

Topics on Fundamental Science

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M.Sc

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Part I

Prologue

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But still try, for who knows what is possible . . .

*(Michael Faraday, in The Life and Letters of Faraday (1870) Vol. II, edited by Henry
Bence Jones, p. 483)*

Chapter 1

The Purpose of Electromagnetism

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Newtonian mechanics is usually enough for most **everyday** applications ranging from calculating the motion of an object falling from a building to predicting the motion of planets. When objects start to move at high speeds,¹ however, the Newtonian model fails to predict accurately and was replaced by special relativity, which was introduced by Einstein in 1905 [1]. For objects which are **extremely small**, sizes comparable to that of atoms, then the theory of relativity fails for different reasons, and is superseded by quantum mechanics.²

Finally, for objects which are both very fast and very small,³ a mechanics which combines both relativity and quantum principles is needed:

This relativistic quantum mechanics is known as quantum field theory.

Even today this new model cannot claim to be a completely satisfactory system.

Luckily for us, in this **Electrodynamics Lecture**, we will **NOT** look at these models in detail but look at a specific subset of it called **electrodynamics** which is generally in the domain of classical mechanics.⁴

Information: Electromagnetism and Relativity

Interestingly, electromagnetism was one of the main catalyst for developing general relativity.

Historically, the *Maxwell equations* were discovered before the theory of special relativity. It was thought light waves we derived above must be oscillations of some substance which fills all of space. This was dubbed the aether.⁵ The idea was that Maxwell's equations only hold in the frame in which the aether is at rest; light should then travel at speed c relative to the aether.

¹as in speeds comparable to the speed of light.

²Developed by Bohr, Schrodinger, Heisenberg, and many others, in the 1920's, mostly.

³As is common in modern particle physics.

⁴which can be extended to other domains if needed.

⁵the postulated medium for the propagation of light [2]. It was originally used to explain the ability of the apparently wave-based light to propagate through empty space, something that waves should not be able to do. The assumption of a spatial plenum of luminiferous aether, rather than a spatial vacuum, provided the theoretical medium that was required by wave theories of light.

We now know that the concept of the aether is unnecessary baggage. Instead, Maxwell's equations hold in all inertial frames and are the first equations of physics which are consistent with the laws of special relativity. Ultimately, it was by studying the Maxwell equations that Lorentz was able to determine the form of the Lorentz transformations which subsequently laid the foundation for Einstein's vision of space and time [3].

1.0.1 Four Kinds of Forces

Classical Mechanics tells us how behaviour is when subjected to a given force. There are four (4) basic forces known to physics. Listing them in the order of decreasing strength:

Strong

Holds protons and neutrons together in the atomic nucleus, have extremely short range, so we do not “feel” them, in spite of the fact that they are a hundred times more powerful than electrical forces.

Weak

Accounts for certain kinds of radioactive decay, are also of short range, and they are far weaker than electromagnetic forces.

Gravitational

it is very weak⁶ that it is only in scale of huge mass concentrations, for example the earth and the sun, that we ever notice it at all.

⁶compared to all of the others

Electromagnetic

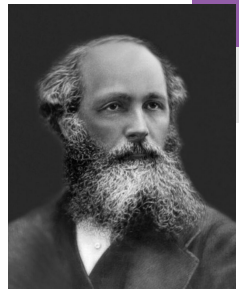
It is the interaction that occurs between particles with electric charge via electromagnetic fields.

For reference, the electrical repulsion between two (2) electrons is 1×10^{42} times as large as their gravitational attraction.

Not only are electromagnetic forces overwhelmingly dominant in everyday life, they are also, the only ones which are completely understood.

There is, of course, a classical theory of gravity and a relativistic one, but no entirely satisfactory quantum mechanical theory of gravity has been constructed. At the present time there is a very successful theory for the weak interactions, and a strikingly attractive candidate, called chromodynamics. for the strong interactions.

All these theories draw their inspiration from electrodynamics. None can claim conclusive experimental verification at this stage. So electrodynamics, a complete and successful theory, has become a pedestal for physicists:



⁷ James Clerk Maxwell
FRS FRSE
(1831 - 1879)

A Scottish physicist and mathematician who was responsible for the classical theory of electromagnetic radiation, which was the first theory to describe electricity, magnetism and light as different manifestations of the same phenomenon. Maxwell's equations for electromagnetism achieved the second great unification in physics, where the first one had been realised by Isaac Newton. Maxwell was also key in the creation of statistical mechanics.

With the publication of *A Dynamical Theory of the Electromagnetic Field* in 1865, Maxwell demonstrated electric and magnetic fields travel through space as waves moving at the speed of light. He proposed light is an undulation in the same medium that is the cause of electric and magnetic phenomena. The unification of light and electrical phenomena led to his prediction of radio waves. As a result of his equations, and other contributions such as introducing an effective method to deal with network problems and linear conductors, he is regarded as a founder of the modern field of electrical engineering.

Table 1.1: The four fundamental forces and their respective properties.

Traits	Strong Force	Electromagnetism	Weak Force	Gravity
Affected Particles	quarks ¹ and gluons ²	electrically charged	quarks and leptons ³	all particles with mass
Force Carrying Particle	gluon (g)	photon ⁴ (γ)	W and Z bosons (W^+ , W^- , Z^0)	graviton (unobserved)
Acting Range	short ($\sim 1 \text{ fm}$) ⁵	∞	short	∞
Strength	1	$1/137$ ⁶	1×10^{-6}	6×10^{-39}

¹ Elementary particle responsible for making **protons** and **neutron**.

² Elementary particle acting as exchange particle for the **strong force** between **quarks**.

³ Elementary particle affected by the **weak force** but not by the **strong force**.

⁴ Elementary particle that is a **quantum** of the **electromagnetic** field, such as light and radio waves.

⁵ 1 femtometre ($1 \text{ fm} = 1 \times 10^{-15} \text{ m}$). i.e., the gold nucleus radius is approx. 8.45 fm.

⁶ Known as the **fine structure constant**.

An ideal model that other theories emulate.

The laws of classical electrodynamics were discovered in bits and pieces by Franklin, Coulomb, Ampere, Faraday, and others, but the person who completed the job, and packaged it all in the compact and consistent form it has today, was James Clerk Maxwell.⁷

The theory is now about 150 years old.

1.0.2 Unifying Physical Theories

In the beginning, electricity and magnetism were entirely separate subjects. The one dealt with glass rods and cat's fur, batteries, currents, electrolysis, and lightning, whereas the other with bar magnets, iron filings, compass needles, and the North Pole. However, in 1820 Ørsted noticed that an electric current could deflect a magnetic compass needle. Soon afterwards, Ampère correctly postulated all magnetic phenomena are due to electric charges in motion. Then, in 1831, Faraday discovered that a moving magnet generates an electric current. By the time Maxwell and Lorentz put the finishing touches on the theory, electricity and magnetism were inextricably intertwined.

They could no longer be regarded as separate subjects, but rather as two aspects of a single subject: electromagnetism.

Faraday speculated that light, too, is electrical in nature. Maxwell's theory provided justification for this hypothesis, and soon optics, which is the study of lenses, mirrors, prisms, interference, and diffraction, was incorporated into electromagnetism. Hertz, who presented the decisive experimental confirmation for Maxwell's theory in 1888, said:

The connection between light and electricity is now established ... In every flame, in

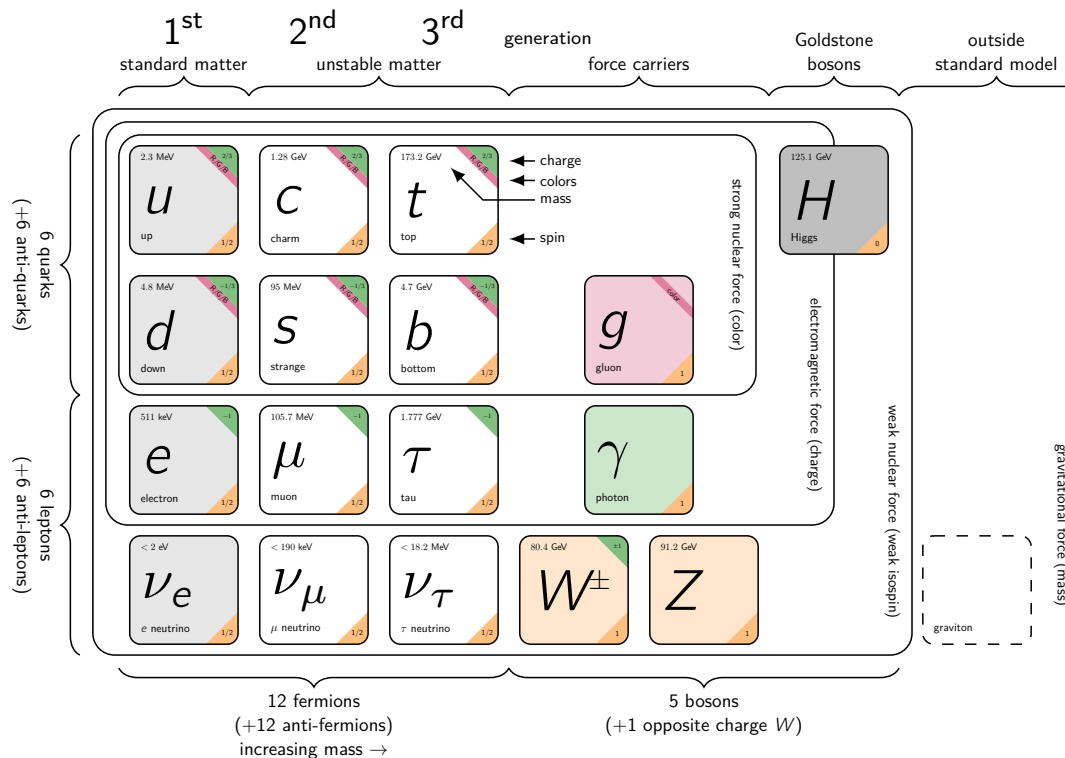


Figure 1.1: The Standard Model of particle physics is the theory describing three of the four known fundamental forces (electromagnetic, weak and strong interactions - excluding gravity) in the universe and classifying all known elementary particles. It was developed in stages throughout the latter half of the 20th century, through the work of many scientists worldwide, with the current formulation being finalized in the mid-1970s upon experimental confirmation of the existence of quarks.

every luminous particle, we see an electrical process . . . Thus, the domain of electricity extends over the whole of nature. It even affects ourselves intimately: we perceive that we possess . . . an electrical organ-the eye.

By 1900, three (3) great branches of physics: **electricity**, **magnetism**, and **optics** had merged into a single unified theory.

It became apparent that visible light represents only a tiny “window” in the vast spectrum of electromagnetic radiation, from radio through microwaves, infrared and ultraviolet . . .

Einstein worked on a further unification, which would combine gravity and electrodynamics, in much the same way as electricity and magnetism had been combined a century earlier. His unified field theory was not particularly successful, but in recent years the same impulse has spawned a hierarchy of increasingly ambitious unification schemes, beginning in the 1960s with the electroweak theory of Glashow, Weinberg, and Salam, which joins the weak and electromagnetic forces, and culminating in the 1980s with the superstring theory.⁸

At each step in this hierarchy, the mathematical difficulties mount, and the gap between inspired conjecture and experimental test widens; nevertheless, it is clear the unification of forces initiated by

⁸an attempt to explain all of the particles and fundamental forces of nature in one theory by modeling them as vibrations of tiny supersymmetric strings.

electrodynamics has become a major theme in the progress of physics.

1.0.3 The Fields of Electrodynamics

The essential problem the theory of electromagnetism hopes to solve is:

If there exists a bunch of electric charges here, what happens to some other charge, over another place?

The classical solution takes the form of a **field theory**:

We say that the space around an electric charge is permeated by **electric** and **magnetic** fields. A second charge, in the presence of these fields, experiences a force. The fields, then, transmit the influence from one charge to the other—they “mediate” the interaction.

When a charge undergoes acceleration, a portion of the field **detaches** itself, in sense, and travels off at the speed of light, carrying with it energy, momentum, and angular momentum. We call this **electromagnetic radiation**. Its existence invites⁹ us to regard the fields as independent dynamical entities in their own right, every bit as **real** as atoms or baseballs.

⁹if not compels

Our interest accordingly shifts from the study of forces between charges to the theory of the fields themselves. But it takes a charge to produce an electromagnetic field, and it takes another charge to detect one, so we had best begin by reviewing the essential properties of electric charge.

1.0.4 Electric Charge

Charge comes in two varieties

We call **plus** and **minus**, as their effects tend to cancel.¹⁰ The interesting point here is that plus and minus charges occur in exactly equal amounts, to high precision, in bulk matter, so that their effects are almost completely neutralised.

¹⁰If we have $+q$ and q at the same point, electrically it is the same as having no charge there at all.

Charge is conserved

Charge cannot be created or destroyed [4]. What there is now has always been. A plus charge can **annihilate** an equal minus charge, but a plus charge cannot simply disappear by itself, something must pick up that electric charge. So the total charge of the universe is fixed for all time.

This is called **global conservation of charge**. Charges also cannot disappear and then reappear in some other place, the charge, if moved must follow a continuous path. This is called **local conservation of charge**.

Charge is quantised

While there is no restrictions in classical electrodynamics which require it to be, electric charge

comes only in **integer multiples** of the basic unit of charge. If we call the charge on the proton $+e$, then the electron carries charge e .¹¹

¹¹It is never $7.392e$, or $1/2e$.

Information: Interpreting Conservation of Charge

Charge conservation does **NOT** mean that individual positive and negative charges cannot be created or destroyed. Electric charge is carried by subatomic particles such as **electrons** and **protons**. Charged particles can be created and destroyed in elementary particle reactions.

In particle physics, charge conservation means, in reactions that create charged particles, equal numbers of positive and negative particles are always created, keeping the net amount of charge unchanged. Similarly, when particles are destroyed, equal numbers of positive and negative charges are destroyed.

This property is supported without exception by all empirical observations so far [4].

Even though the conservation of charge requires that the total quantity of charge in the universe is constant, it leaves open the question of what that quantity is. Most evidence indicates that the net charge in the universe is zero [5]. That is, there are equal quantities of positive and negative charge.

Chapter 2

Vector Calculus

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2.1 Vector Algebra

2.1.1 Vector Operations

Walking 5 km north and then 12 km east, we will have gone a total of 17 km, but if we carefully observe our current position, we will realise that we're not 17 km from where we set out, but only 13. To describe these quantities, We need a set of mathematics principles, which evidently do **NOT** add in the ordinary way.

The reason they don't, is **displacements** have *direction* as well as *magnitude*, and it is essential to take both into account when we combine them. Such objects are called **vectors**.

Examples include: velocity, acceleration, force, momentum ...

By contrast, quantities that have magnitude but no direction are called **scalars**.

Examples include: mass, charge, density, temperature, ...



We shall use **boldface** ($\mathbf{A}$, $\mathbf{B}$, and so on) for vectors and **normal-font** for scalars. The magnitude of a vector $\mathbf{A}$ is written $|\mathbf{A}|$ or, more simply, A . In diagrams, vectors are denoted by arrows: the length of the arrow is proportional to the magnitude of the vector, and the arrow indicates its direction.

Minus $\mathbf{A}$ ($-\mathbf{A}$) is a vector with the same magnitude as $\mathbf{A}$ but of opposite direction.

Vectors have magnitude and direction but *not location*.

Here we will define four (4) vector operations: addition and three kinds of multiplication.

Addition of Two Vectors Place the tail of $\mathbf{B}$ at the head of $\mathbf{A}$. Their sum, $\mathbf{A} + \mathbf{B}$, is the vector from the tail of $\mathbf{A}$ to the head of $\mathbf{B}$. Addition is *commutative*:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A},$$

5 kilometers east followed by 12 kilometers north gets us to the same place as 12 kilometers north followed by 5 kilometers east. Addition is also *associative*:

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}).$$

To subtract a vector, add its opposite

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}).$$

Multiplication by a Scalar Value Multiplication of a vector by a positive scalar a multiplies the *magnitude* but leaves the direction **unchanged**. This means if a is negative, the direction is reversed. Scalar multiplication is *distributive*:

$$a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}.$$

Dot Product of Two Vectors The dot product of two vectors is defined by:

$$\mathbf{A} \cdot \mathbf{B} \equiv AB \cos \theta \quad (2.1)$$

where θ is the angle they form when placed tail-to-tail.

$\mathbf{A} \cdot \mathbf{B}$ is itself a scalar.¹

¹which is why its alternative name is **scalar product**.

The dot product is *commutative*,

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A},$$

and *distributive*,

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}, \quad (2.2)$$

Geometrically, $\mathbf{A} \cdot \mathbf{B}$ is the product of A times the projection of $\mathbf{B}$ along $\mathbf{A}$ ². If the two vectors are parallel, then $\mathbf{A} \cdot \mathbf{B} = AB$. In particular, for any vector $\mathbf{A}$,

$$\mathbf{A} \cdot \mathbf{A} = A^2 \quad (2.3)$$

If $\mathbf{A}$ and $\mathbf{B}$ are perpendicular, then $\mathbf{A} \cdot \mathbf{B} = 0$.

²Or the product of B times the projection of $\mathbf{A}$ along $\mathbf{B}$.

Cross Product of Two Vectors The cross product of two (2) vectors is defined by:

$$\mathbf{A} \times \mathbf{B} \equiv AB \sin \theta \hat{\mathbf{n}} \quad (2.4)$$

where $\hat{\mathbf{n}}$ is a **unit vector** (vector of magnitude 1) pointing perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$. Of course, there are *two* directions perpendicular to any plane: **in** and **out**.

The ambiguity is resolved by the **right-hand rule**: let our fingers point in the direction of the first vector and curl around (via the smaller angle) toward the second; then our thumb indicates the direction of $\hat{\mathbf{n}}$.

$\mathbf{A} \times \mathbf{B}$ is itself a *vector* and it is also known as **vector product**.

The cross product is *distributive*,

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C}) \quad (2.5)$$

but **NOT** commutative:

$$(\mathbf{B} \times \mathbf{A}) = -(\mathbf{A} \times \mathbf{B}) \quad (2.6)$$

If two (2) vectors are parallel, their cross product is zero. In particular,

$$\mathbf{A} \times \mathbf{A} = 0,$$

for any vector $\mathbf{A}$.

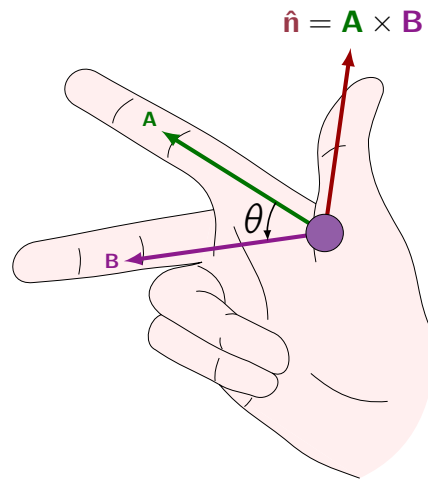


Figure 2.1: A mnemonic, used to define the orientation of axes in three-dimensional space and to determine the direction of the cross product of two vectors, as well as to establish the direction of the force on a current-carrying conductor in a magnetic field.

2.1.2 Vector Component Forms

In the previous section, we defined the four (4) vector operations in abstract form, without reference to any particular coordinate system.

In practice, it is often easier to set up Cartesian coordinates (x, y, z) and work with vector **components**. Let $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ be unit vectors parallel to the x, y , and z axes, respectively. An arbitrary vector $\mathbf{A}$ can be expanded in terms of these **basis vectors**:

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$



The symbols A_x , A_y , A_z , are the **components** of $\mathbf{A}$. In geometrical terms they are the **projections** of $\mathbf{A}$ along the three (3) coordinate axes (i.e., $A_x = \mathbf{A} \cdot \hat{\mathbf{x}}$, $A_y = \mathbf{A} \cdot \hat{\mathbf{y}}$, $A_z = \mathbf{A} \cdot \hat{\mathbf{z}}$).

We can now reformulate each of the four (4) vector operations as a rule for manipulating components:

$$\begin{aligned}\mathbf{A} + \mathbf{B} &= (A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}) + (B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}) \\ &= (A_x + B_x) \hat{\mathbf{x}} + (A_y + B_y) \hat{\mathbf{y}} + (A_z + B_z) \hat{\mathbf{z}}\end{aligned}$$

The operation rules are as follows:

- i. To add vectors, add like components.

$$a\mathbf{A} = (aA_x) \hat{\mathbf{x}} + (aA_y) \hat{\mathbf{y}} + (aA_z) \hat{\mathbf{z}}$$

- ii. To multiply by a scalar, multiply each component. As $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ are mutually perpendicular unit vectors, the following properties are valid:

$$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1 \quad \hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{x}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = 0$$

Accordingly,

$$\mathbf{A} \cdot \mathbf{B} = (A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}) \cdot (B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}})$$

- iii. To calculate the dot product, multiply like components, and add. In particular,

$$\mathbf{A} \cdot \mathbf{A} = A_x^2 + A_y^2 + A_z^2, \quad \text{so} \quad A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Similarly the following relations can be derived:

$$\begin{aligned}\hat{\mathbf{x}} \times \hat{\mathbf{x}} &= \hat{\mathbf{y}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0, \\ \hat{\mathbf{x}} \times \hat{\mathbf{y}} &= -\hat{\mathbf{y}} \times \hat{\mathbf{x}} = \hat{\mathbf{z}}, \\ \hat{\mathbf{y}} \times \hat{\mathbf{z}} &= -\hat{\mathbf{z}} \times \hat{\mathbf{y}} = \hat{\mathbf{x}}, \\ \hat{\mathbf{z}} \times \hat{\mathbf{x}} &= -\hat{\mathbf{x}} \times \hat{\mathbf{z}} = \hat{\mathbf{y}}.\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= (A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}) \times (B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}) \\ &= (A_y B_z - A_z B_y) \hat{\mathbf{x}} + (A_z B_x - A_x B_z) \hat{\mathbf{y}} + (A_x B_y - A_y B_x) \hat{\mathbf{z}}.\end{aligned}$$

This expression look unruly but we can tidy it up and write it more neatly as a **determinant**:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \quad (2.7)$$

- iv. To calculate the cross product, form the determinant whose first row is $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$, whose second row is $\mathbf{A}$ (in component form), and whose third row is $\mathbf{B}$.



2.1.3 Triple Products

As the cross product of two (2) vectors is itself a vector, it can be dotted or crossed with a 3rd vector to form a **triple** product.

Scalar Triple Product Evidently,

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}),$$

for they all correspond to the same value.

The **alphabetical** order is preserved.

Alternatively, the **non-alphabetical** triple products,

$$\mathbf{A} \cdot (\mathbf{C} \times \mathbf{B}) = \mathbf{B} \cdot (\mathbf{A} \times \mathbf{C}) = \mathbf{C} \cdot (\mathbf{B} \times \mathbf{A}),$$

have the **opposite** sign. In component form,

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}.$$

A final point worth mentioning is the dot and cross can be interchanged:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C},$$

however, it is important to stress it out, the placement of the parentheses is **critical**:

Information: Cross Product of Two Scalars

$(\mathbf{A} \cdot \mathbf{B}) \times \mathbf{C}$ is a meaningless expression. We can't make a cross product from a scalar and a vector.

Vector Triple Product The vector triple product can be simplified by the **BAC-CAB** rule:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}).$$

Please observe that the following triple product

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = -\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = -\mathbf{A}(\mathbf{B} \cdot \mathbf{C}) + \mathbf{B}(\mathbf{A} \cdot \mathbf{C})$$

is an entirely different vector.³ All *higher* vector products can be similarly reduced, often by repeated application, so it is never necessary for an expression to contain more than one cross product in any term. For instance,

$$\begin{aligned} (\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) &= (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C}), \\ \mathbf{A} \times [\mathbf{B} \times (\mathbf{C} \times \mathbf{D})] &= \mathbf{B}[\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D})] - (\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \times \mathbf{D}). \end{aligned}$$

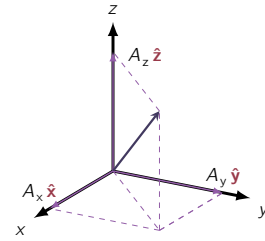


Figure 2.2: The decomposition of the vector $\mathbf{A}$ into its components.

³To reiterate cross-products are not associative.

2.1.4 Position, Displacement, and Separation Vectors

The location of a point in three dimensions can be described by listing its Cartesian coordinates (x, y, z) . The vector to that point from the origin ($\mathcal{O}$) is called the **position vector**:

$$\mathbf{r} \equiv x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}$$

Throughout this course, $\mathbf{r}$ will be used to measure **distance**. Its magnitude:

$$r = \sqrt{x^2 + y^2 + z^2}$$

is the distance from the origin, and

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{(x) \hat{\mathbf{x}} + (y) \hat{\mathbf{y}} + (z) \hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}}$$

is a unit vector pointing **radially outward**. The **infinitesimal displacement vector**, from (x, y, z) to $(x + dx, y + dy, z + dz)$, is

$$d\mathbf{l} = (dx) \hat{\mathbf{x}} + (dy) \hat{\mathbf{y}} + (dz) \hat{\mathbf{z}}.$$

In electrodynamics, one frequently encounters problems involving two (2) points:

source point ($\mathbf{r}'$), where an electric charge is located

field point ($\mathbf{r}$), at which we are calculating the electric or magnetic field

To make these redundant calculations easier to handle, let's use the following short-hand notation:

$$\mathbf{z} \equiv \mathbf{r} - \mathbf{r}'$$

Its magnitude is:

$$z = |\mathbf{r} - \mathbf{r}'|$$

and a unit vector in the direction from $\mathbf{r}'$ to $\mathbf{r}$ is

$$\hat{\mathbf{z}} = \frac{\mathbf{z}}{z} = \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$

In Cartesian coordinates our new short-notations would be as following:

$$\begin{aligned} \mathbf{z} &= (x - x') \hat{\mathbf{x}} + (y - y') \hat{\mathbf{y}} + (z - z') \hat{\mathbf{z}} \\ z &= \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2} \\ \hat{\mathbf{z}} &= \frac{(x - x') \hat{\mathbf{x}} + (y - y') \hat{\mathbf{y}} + (z - z') \hat{\mathbf{z}}}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}} \end{aligned}$$

2.2 Differential Calculus

2.2.1 Ordinary Derivatives

Assume a function of just one (1) variable: $f(x)$. Therefore, the question we need to answer is what does the derivative, df/dx , do and convey?

It tells us how rapidly the function $f(x)$ **varies** when we change the argument x by an infinitesimal amount, dx :

$$df = \left(\frac{df}{dx} \right) dx$$

If we increment x by an infinitesimal amount dx , then f changes by an amount df ⁴.

⁴Here we can think the derivative as the proportionality factor.

Geometrically, the derivative df/dx is the *slope* of the graph of f versus x .

2.2.2 Gradient

Assume a function which accepts three (3) variables. As an example, let's take the temperature $T(x, y, z)$ in the lecture room. Start out in one corner, and set up a system of cardinal directions. Then for each point (x, y, z) in the room, T gives the temperature at that spot. We want to generalise the notion of **derivative** to functions like T , which depend **NOT** on one but on three variables.

A derivative tells us **how fast the function varies**, if we move a little distance. But this time the situation is more complicated, because it depends on what **direction** we move:

- Going straight up, the temperature will probably increase fairly rapidly,
- Moving horizontally, it may not change much at all.

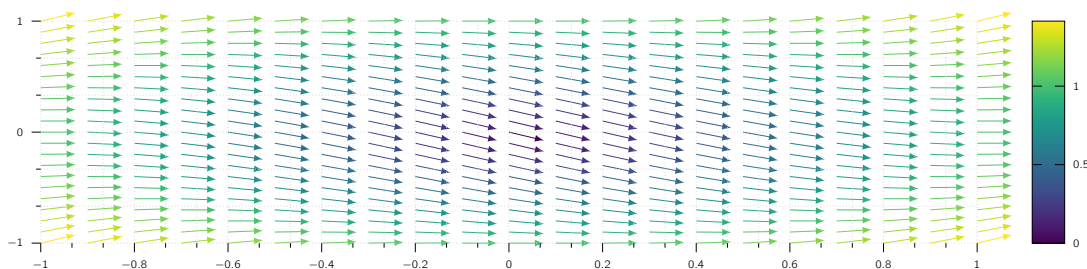


Figure 2.3: An example of a gradient field. Here the field itself is plotted by using arrows to designate the direction of the gradient. To explain the magnitude of the gradient it is either shown by the length of the arrow or by imposing colour on to the plot, which the latter has been used here. Extending this plot to our example, we can see that most of the high temperature resides on the edges of the room whereas the centre remains cool.

David Wilkins suggests Hamilton may have used the nabla as a general purpose symbol or abbreviation for whatever operator he wanted to introduce at any time. In 1837 Hamilton used the nabla, in its modern orientation, as a symbol for any arbitrary function in *Trans. R. Irish Acad. XVII. 236. (OED.)* [2] He used the nabla to signify a permutation operator in "*On the Argument of Abel, respecting the Impossibility of expressing a Root of any General Equation above the Fourth Degree, by any* ⁵*This Cajori generalised Hamilton's nabla and gave it the name of the nabla.* *Mathematical Dictionary* (1839).

is the **gradient** of T . Note that ∇T is a **vector quantity**, with three (3) components.⁵

Hamilton used the rotated nabla (∇), for the vector differential operator in the *Proceedings of the Royal Irish Academy* (1846). Hamilton also used the rotated nabla as the vector differential operator in *On Quaternions; or on a new System of Imaginaries in Algebra*. For more information, please visit [here](#).

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \quad (2.8)$$

1. By a scalar a : $\mathbf{A}a$;
2. By a vector $\mathbf{B}$, via the dot product: $\mathbf{A} \cdot \mathbf{B}$;
3. By a vector $\mathbf{B}$ via the cross product: $\mathbf{A} \times \mathbf{B}$.

1. On a scalar function $T : \nabla T$ (the gradient);

2. On a vector function $\mathbf{v}$, via the dot product: $\nabla \cdot \mathbf{v}$ (divergence)
3. On a vector function $\mathbf{v}$, via the cross product: $\nabla \times \mathbf{v}$ (curl).

It is time to examine the other two (2) vector derivatives: divergence and curl.

Divergence From the definition of ∇ we construct the divergence as follows:

$$\begin{aligned}\nabla \cdot \mathbf{v} &= \left(\left(\frac{\partial}{\partial x} \right) \hat{\mathbf{x}} + \left(\frac{\partial}{\partial y} \right) \hat{\mathbf{y}} + \left(\frac{\partial}{\partial z} \right) \hat{\mathbf{z}} \right) \cdot \left((v_x) \hat{\mathbf{x}} + (v_y) \hat{\mathbf{y}} + (v_z) \hat{\mathbf{z}} \right) \\ &= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}.\end{aligned}$$

Observe that the divergence of a vector function $\mathbf{v}$ is itself a scalar $\nabla \cdot \mathbf{v}$.

Let's try to visualise this concept. The name divergence is well chosen, for $\nabla \cdot \mathbf{v}$ is a measure of how much $\mathbf{v}$ spreads out⁷ from the initial point.

⁷i.e., diverges

For example, a vector function in **Fig. 2.4** has a large (positive) divergence,⁸ two functions in **Fig. 2.4** has zero divergence, and one function has a negative divergence.

⁸if the arrows pointed in, it would be a negative divergence

As an example, imagine standing at the edge of a pond. Sprinkle some sawdust on the surface.

1. If the material spreads out, then we dropped it at a point of positive divergence;
2. If it collects together, we dropped it at a point of negative divergence.

Curl From the definition of ∇ we construct the curl:

$$\begin{aligned}\mathbf{v} \times \mathbf{v} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix} \\ &= \hat{\mathbf{x}} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right).\end{aligned}$$

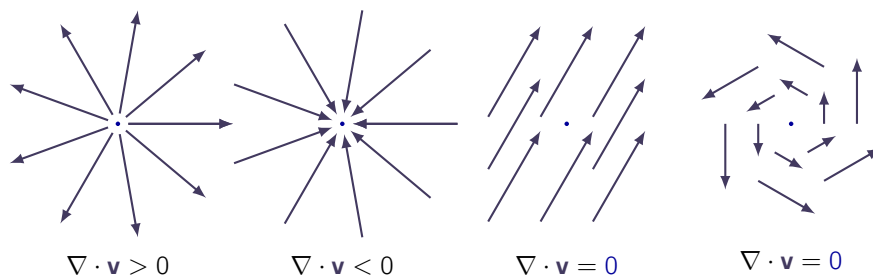


Figure 2.4: Visual description of the divergence operation.

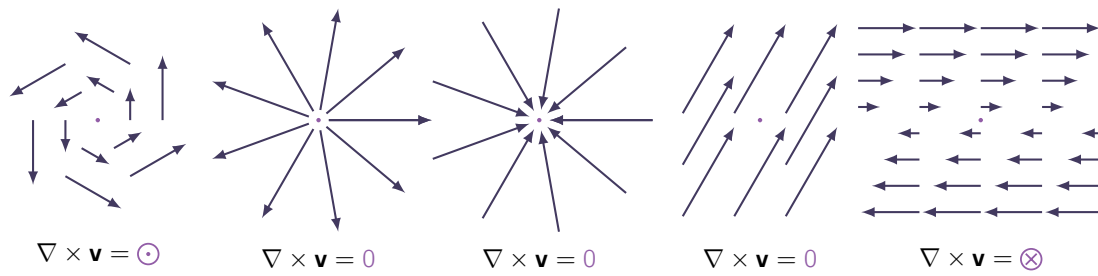


Figure 2.5: Different behaviours of the curl operation.

As with divergence, the name curl is also well chosen, for $\nabla \times \mathbf{v}$ is a measure of how much $\mathbf{v}$ swirls around the point in question. Therefore the functions in Fig. 1.18 all have zero curl, whereas the functions in **Fig. 2.5** all have a substantial curl sans one, pointing in the $\hat{z}$ direction, as the natural right-hand rule would suggest.

To finish these two (2) important operations, let's again imagine we are standing at the edge of a pond. Float a small flower, if it starts to rotate, then we placed it at a point of nonzero curl. A whirlpool would be a region of large curl.

2.2.4 Product Rules

The calculation of ordinary derivatives is facilitated by a number of rules which are as follows:

Table 2.1: Product rules of ordinary derivatives

Sum Rule	$\frac{d}{dx}(f + g) = \frac{df}{dx} + \frac{dg}{dx}$
Constant Multiplication	$\frac{d}{dx}(kf) = k \frac{df}{dx}$
Product Rule	$\frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx}$
Quotient Rule	$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g \frac{df}{dx} - f \frac{dg}{dx}}{g^2}$

Similar relations hold for the vector derivatives. Thus,

$$\begin{aligned}\nabla(f + g) &= \nabla f + \nabla g, \\ \nabla \cdot (\mathbf{A} + \mathbf{B}) &= (\nabla \cdot \mathbf{A}) + (\nabla \cdot \mathbf{B}), \\ \nabla \times (\mathbf{A} + \mathbf{B}) &= (\nabla \times \mathbf{A}) + (\nabla \times \mathbf{B}),\end{aligned}$$

and

$$\nabla(kf) = k\nabla f, \quad \nabla \cdot (k\mathbf{A}) = k(\nabla \cdot \mathbf{A}), \quad \nabla \times (k\mathbf{A}) = k(\nabla \times \mathbf{A}), \quad (2.9)$$

The product rules, on the other hand, are not quite so simple. There are two (2) ways to construct

a scalar as the product of two functions:

$$\begin{aligned} fg & \quad (\text{product of two scalar functions}), \\ \mathbf{A} \cdot \mathbf{B} & \quad (\text{dot product of two vector functions}), \end{aligned}$$

and two ways to make a vector:

$$\begin{aligned} f\mathbf{A} & \quad (\text{scalar times vector}), \\ \mathbf{A} \times \mathbf{B} & \quad (\text{cross product of two vectors}). \end{aligned}$$

Accordingly, there are six (6) product rules, which are given in **Tbl. 2.2**.

Table 2.2: Product rules of vector operations

Gradient I	$\nabla(fg) = f\nabla g + g\nabla f$
Gradient II	$\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$
Divergence I	$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$
Divergence II	$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
Curl I	$\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f),$
Curl II	$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$

The proofs for rules given in **Tbl. 2.2** come straight from the product rule for ordinary derivatives. As an example:

$$\begin{aligned} \nabla \cdot (f\mathbf{A}) &= \frac{\partial}{\partial x}(fA_x) + \frac{\partial}{\partial y}(fA_y) + \frac{\partial}{\partial z}(fA_z) \\ &= \left(\frac{\partial f}{\partial x}A_x + f \frac{\partial A_x}{\partial x} \right) + \left(\frac{\partial f}{\partial y}A_y + f \frac{\partial A_y}{\partial y} \right) + \left(\frac{\partial f}{\partial z}A_z + f \frac{\partial A_z}{\partial z} \right) \\ &= (\nabla f) \cdot \mathbf{A} + f(\nabla \cdot \mathbf{A}). \end{aligned}$$

It is also possible to formulate three quotient rules:

$$\begin{aligned} \nabla \left(\frac{f}{g} \right) &= \frac{g\nabla f - f\nabla g}{g^2}, \\ \nabla \cdot \left(\frac{\mathbf{A}}{g} \right) &= \frac{g(\nabla \cdot \mathbf{A}) - \mathbf{A} \cdot (\nabla g)}{g^2}, \\ \nabla \times \left(\frac{\mathbf{A}}{g} \right) &= \frac{g(\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla g)}{g^2}. \end{aligned}$$

However, given these can be obtained quickly from the previously mentioned product rules, there is no point in listing them separately and is left for the reader as exercise.

2.2.5 Second Derivatives

The gradient, the divergence, and the curl are the only first derivatives we can make with $\nabla \cdot \mathbf{v}$ by applying ∇ twice, we can construct five (5) types of 2nd derivatives.

The gradient ∇T is a vector, so we can take the divergence and curl of it:

1. Divergence of gradient: $\nabla \cdot (\nabla T)$.
2. Curl of gradient: $\nabla \times (\nabla T)$.

The divergence $\nabla \cdot \mathbf{v}$ is a scalar, therefore all we can do is take its gradient:

3. Gradient of divergence: $\nabla(\nabla \cdot \mathbf{v})$.

The curl $\nabla \times \mathbf{v}$ is a vector, so we can take its divergence and curl:

4. Divergence of curl: $\nabla \cdot (\nabla \times \mathbf{v})$.
5. Curl of curl: $\nabla \times (\nabla \times \mathbf{v})$.

This exhausts the possible combinations, and in fact **NOT** all of them give anything new. Let's consider them one at a time:

Divergence of a Gradient

$$\begin{aligned}\nabla \cdot (\nabla T) &= \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \cdot \left(\frac{\partial T}{\partial x} \hat{x} + \frac{\partial T}{\partial y} \hat{y} + \frac{\partial T}{\partial z} \hat{z} \right) \\ &= \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}.\end{aligned}$$

This object, which we write as $\nabla^2 T$ for short, is called the Laplacian⁹ of T , which will be our focus later.

The Laplacian of a scalar T is a scalar value.

Occasionally, we will use the Laplacian of a vector, $\nabla^2 \mathbf{v}$. By this we mean a **vector** quantity whose x-component is the Laplacian of v_x , and so on.

$$\nabla^2 \mathbf{v} \equiv \left(\nabla^2 v_x \right) \hat{x} + \left(\nabla^2 v_y \right) \hat{y} + \left(\nabla^2 v_z \right) \hat{z}$$

This is nothing more than a convenient extension of the meaning of ∇^2 .

⁹A differential operator given by the divergence of the gradient of a scalar function on Euclidean space. It is usually denoted by the symbols $\nabla \cdot \nabla$, ∇^2 or Δ . In a Cartesian coordinate system, the Laplacian is given by the sum of second partial derivatives of the function with respect to each independent variable.

The Laplace operator is named after the French mathematician *Pierre-Simon de Laplace* (1749-1827), who first applied the operator to the study of celestial mechanics: the Laplacian of the gravitational potential due to a given mass density distribution is a constant multiple of that density distribution. Solutions of Laplace's equation $\Delta f = 0$ are called harmonic functions and represent the possible gravitational potentials in regions of vacuum.

The Laplacian occurs in many differential equations describing physical phenomena. Poisson's equation describes electric and gravitational potentials; the diffusion equation describes heat and fluid flow; the wave equation describes wave propagation; and the Schrödinger equation describes the wave function in quantum mechanics. In image processing and computer vision, the Laplacian operator has



Curl of a Gradient

The curl of a gradient is **always** zero:

$$\nabla \times (\nabla T)$$

This is an **important fact**, which will be used repeatedly. Without going into too much detail into the proof, it relies on the following relation:

$$\frac{\partial}{\partial x} \left(\frac{\partial T}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial x} \right)$$

Gradient of Divergence

This operation rarely occurs in physical applications, and it has not been given any special name of its own. Notice that $\nabla (\nabla \cdot \mathbf{v})$ is **NOT** the same as the Laplacian of a vector:

$$\nabla^2 = (\nabla \cdot \nabla) \neq \nabla (\nabla \cdot \mathbf{v})$$

Divergence of a Curl

Similar to the curl of a gradient, it is always zero:

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0.$$

Curl of a Curl

As we can check from the definition of ∇ :

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla (\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}. \quad (2.10)$$

So curl-of-curl gives nothing new as the first term is just the divergence of a curl, and the second is the Laplacian. To put it short, then, there are just two kinds of second derivatives:¹⁰

1. Laplacian,
2. Gradient of divergence.

¹⁰It is possible to work out 3rd derivatives, but fortunately second derivatives suffice for practically all physical applications.

2.2.6 Line, Surface, and Volume Integrals

In electrodynamics, we encounter several different kinds of integrals, among which the most important are **line** (or **path**) **integrals**, **surface integrals** (or **flux**), and **volume integrals**, which will be the focus of this section.



Line Integrals Has an expression of the form:

$$\int_a^b \mathbf{v} \cdot d\mathbf{l}$$

where $\mathbf{v}$ is a vector function, $d\mathbf{l}$ is the infinitesimal displacement vector, and the integral is to be carried out along a prescribed path $\mathcal{P}$ from point a to point b . If the path forms a closed loop,¹¹

¹¹i.e., if $b = a$.

We put a circle on the integral sign:

$$\oint \mathbf{v} \cdot d\mathbf{l}$$

At each point on the path, we take the dot product of $\mathbf{v}$ (evaluated at that point) with the displacement $d\mathbf{l}$ to the next point on the path.

A good example of a line integral is the work done by a force $\mathbf{F}$:

$$W = \int \mathbf{F} \cdot d\mathbf{l}$$

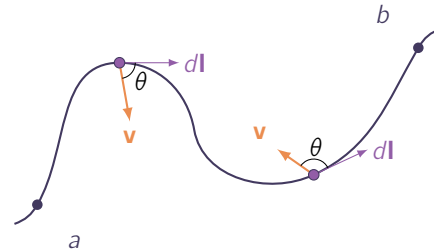


Figure 2.6: The method in which line integral is calculated. At each point the dot product of the vector ($\mathbf{v}$) is taken with the length vector ($d\mathbf{l}$) which is always tangential to the point in which the integration is taken.

Ordinarily, the value of a line integral depends critically on the path taken from a to b , but there is an important special class of vector functions for which the line integral is independent of path and is determined entirely by the end points.

It will be our business in due course to characterise this special class of vectors.

A force which has this property is called **conservative**.

Exercise 2.1 Fluid Flow

A fluid's velocity field is $\mathbf{F} = (x) \hat{x} + (z) \hat{y} + (y) \hat{z}$. Find the flow along the helix $\mathbf{l}(t) = (\cos t) \hat{x} + (\sin t) \hat{y} + (t) \hat{z}$ with a range of $0 \leq t \leq \pi/2$.

SOLUTION

We first evaluate $\mathbf{F}$ on the curve:

$$\mathbf{F} = (x) \hat{x} + (z) \hat{y} + (y) \hat{z} = (\cos t) \hat{x} + (t) \hat{y} + (\sin t) \hat{z}$$

Substitute $x = \cos t$, $z = t$, $y = \sin t$.

and then find $d\mathbf{l}/dt$:

$$\frac{d\mathbf{l}}{dt} = (-\sin t) \hat{x} + (\cos t) \hat{y} + (1) \hat{z}.$$

Then we integrate $\mathbf{F} \cdot (d\mathbf{l}/dt)$ from $t = 0$ to $t = \pi/2$:

$$\begin{aligned} \mathbf{F} \cdot \frac{d\mathbf{l}}{dt} &= (\cos t) (-\sin t) + (t) (\cos t) + (\sin t) (1), \\ &= -\sin t \cos t + t \cos t + \sin t. \end{aligned}$$

Which makes,

$$\text{Flow} = \int_{t=a}^{t=b} \mathbf{F} \cdot \frac{d\mathbf{l}}{dt} dt$$

$$\begin{aligned}
 &= \int_0^{\pi/2} (-\sin t \cos t + t \cos t + \sin t) dt, \\
 &= \left[\frac{\cos^2 t}{2} + t \sin t \right]_0^{\pi/2} \\
 &= \left(0 + \frac{\pi}{2} \right) - \left(\frac{1}{2} + 0 \right) = \frac{\pi}{2} - \frac{1}{2} \blacksquare
 \end{aligned}$$

Exercise 2.2 Field Circulation

Find the circulation of the field $\mathbf{F} = (x - y) \hat{\mathbf{x}} + x \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}$ around the circle $\ell(t) = (\cos t) \hat{\mathbf{x}} + (\sin t) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}$ with a range of $0 \leq t \leq 2\pi$.

SOLUTION On the circle, $\mathbf{F} = (x - y) \hat{\mathbf{x}} + (x) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}} = (\cos t - \sin t) \hat{\mathbf{x}} + (\cos t) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}$ and

$$\frac{d\ell}{dt} = (-\sin t) \hat{\mathbf{x}} + (\cos t) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}.$$

Then

$$\mathbf{F} \cdot \frac{d\ell}{dt} = -\sin t \cos t + \underbrace{\sin^2 t + \cos^2 t}_1,$$

Gives.

$$\begin{aligned}
 \text{Circulation} &= \int_0^{2\pi} \mathbf{F} \cdot \frac{d\ell}{dt} dt \\
 &= \int_0^{2\pi} (1 - \sin t \cos t) dt \\
 &= \left[t - \frac{\sin^2 t}{2} \right]_0^{2\pi} = 2\pi \blacksquare
 \end{aligned}$$

Surface Integrals A surface integral is an expression of the form:

$$\int_S \mathbf{v} \cdot d\mathbf{a}$$

where $\mathbf{v}$ is a vector function, and the integral is over a specified surface S . Here $d\mathbf{a}$ is an infinitesimal patch of area, with direction **perpendicular to the surface**. There are, two (2) directions perpendicular to any surface, so the **sign** of a surface integral is intrinsically ambiguous.

If the surface is closed,¹² we put a circle on the integral sign:

$$\oint \mathbf{v} \cdot d\mathbf{a}$$

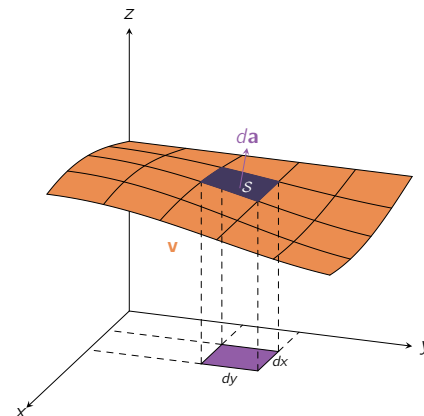


Figure 2.7: An visual description of the surface integral.

¹²Imagine it forming a balloon.

Tradition dictates that **outward** is positive, but for open surfaces it's arbitrary.

As an example, if $\mathbf{v}$ describes the flow of a fluid,¹³ then $\int \mathbf{v} \cdot d\mathbf{a}$ represents the total mass per unit time passing through the surface.

¹³Measured in mass per unit area per unit time.

Ordinarily, the value of a surface integral depends on the particular surface chosen, but there is a special class of vector functions for which it is **independent** of the surface and is determined entirely by the boundary line. An important task will be to characterise this special class of functions.

Volume Integrals A volume integral is an expression of the form:

$$\int_V T d\tau$$

where T is a scalar function and $d\tau$ is an infinitesimal volume element. In Cartesian coordinates,

$$d\tau = dx dy dz$$

As an example, if T is the density of a substance,¹⁴ then the volume integral would give the total mass.

¹⁴This might vary from point to point.

Occasionally we shall encounter volume integrals of vector functions:

$$\int \mathbf{v} d\tau = \int (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}) d\tau = \hat{\mathbf{x}} \int v_x d\tau + \hat{\mathbf{y}} \int v_y d\tau + \hat{\mathbf{z}} \int v_z d\tau.$$

As the unit vectors ($\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$) are constants, they can be taken outside the integral.

Exercise 2.3 Double Integrals

Find the following double integrals:

$$\begin{aligned} \int_0^1 \int_x^{2x} (x+y)^2 dy dx, & \quad \int_0^1 \int_y^{\sqrt{y}} (1-2xy) dx dy, \\ \int_0^3 \int_x^3 \cosh(x+y) dy dx, & \quad \int_0^1 \int_0^{y^3} \exp y^4 dx dy. \end{aligned}$$

SOLUTION

The solution to integrations are as follows:

$$\begin{aligned} & \int_0^1 \int_x^{2x} (x+y)^2 dy dx \\ &= \int_0^1 \int_x^{2x} x^2 + 2xy + y^2 dy dx, \\ &= \int_0^1 \left[yx^2 + xy^2 + \frac{y^3}{3} \right]_x^{2x} dx, \\ &= \int_0^1 \left(4x^3 + \frac{7x^3}{3} \right) dx, \\ &= \left[4x^3 + \frac{7x^4}{12} \right]_0^1 = \frac{19}{12} \quad \blacksquare \\ & \int_0^1 \int_y^{\sqrt{y}} (1-2xy) dx dy \\ &= \int_0^1 \left[x - x^2y \right]_y^{\sqrt{y}} dy, \\ &= \int_0^1 \left[(\sqrt{y} - y^2) - (y - y^3) \right] dy \\ &= \int_0^1 \left[y^3 + \sqrt{y} - y^2 - y \right] dy, \end{aligned}$$

$$\begin{aligned}
&= \left[\frac{y^4}{4} + \frac{2}{3} y^{3/2} - \frac{y^3}{3} - \frac{y^2}{2} \right] \bigg|_0^1, \\
&= \left(\frac{1}{4} + \frac{2}{3} - \frac{1}{3} - \frac{1}{2} \right) - (0) = \frac{1}{12} \quad \blacksquare \\
&\int_0^3 \int_x^3 \cosh(x+y) \, dy \, dx \\
&= \int_0^1 \left[\sinh(x+y) \right] \bigg|_x^3 \, dx \\
&= \int_0^1 [\sinh(3+x) - \sinh(2x)] \, dx
\end{aligned}$$

2.2.7 The Fundamental Theorems of Vector Calculus

Assume $f(x)$ is a function of one (1) variable. Based on this, the fundamental theorem of calculus says:

Theory 2.1: Calculus Theorem

The **integral** of a **derivative** over some **region** is given by the **value of the function** at the end points (**boundaries**)

$$\int_a^b \left(\frac{df}{dx} \right) dx = f(x) - f(a) \quad \text{or} \quad \int_a^b F(x) dx = f(x) - f(a)$$

In vector calculus there are three (3) species of derivative¹⁵ and each has its own “fundamental theorem” with essentially the same format. Our purpose here is to not prove these theorems here, but rather, understand what they **mean**.

¹⁵These are gradient, divergence, and curl

The Fundamental Theorem for Gradients Suppose we have a scalar function of three (3) variables $T(x, y, z)$. Starting at point **a**, move a small distance $d\mathbf{l}_1$. The function T will change by an amount:

$$dT = (\nabla T) \cdot d\mathbf{l}_1$$

Now let's move an additional small displacement $d\mathbf{l}_2$. The incremental change in T will be now:

$$dT = (\nabla T) \cdot d\mathbf{l}_2$$

In this manner, proceeding by infinitesimal steps, we make the journey to point **b**. At each step we compute the gradient of T , at that point, and dot it into the displacement $d\mathbf{l}$... this gives us the change in T .

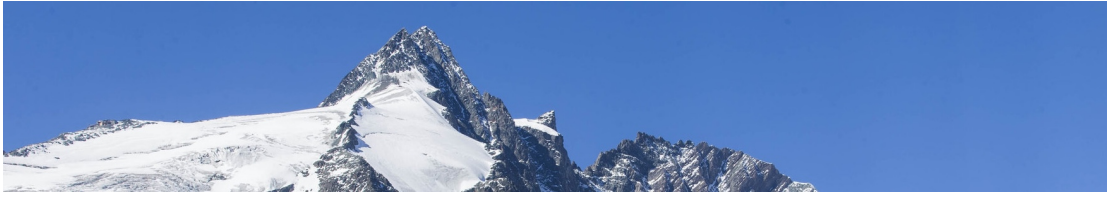


Figure 2.8: To measure the height of a mountain, it doesn't matter what way we take, as long as we know the base and the top, we will know the height.

Theory 2.2: Gradient Theorem

The total change in T in going from $\mathbf{a}$ to $\mathbf{b}$ (along the path selected) is:

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

Similar to “ordinary” fundamental theorem, this theorem says the integral¹⁶ of a derivative, which here the gradient, is given by the value of the function at the boundaries which are $\mathbf{a}$ and $\mathbf{b}$ respectively.

¹⁶In this case it is a line integral

As an example, assume we want to measure the height of GrossGlockner. We could climb the mountain from base, or take the high alpine road, or take a helicopter ride all the way up to top. Regardless of the options we take, we should get the same answer either way.¹⁷

¹⁷This is the essence of the fundamental theorem

Theory 2.3: Line Independence of Gradient

Gradients have the special property that their line integrals are path independent:

- $\int_a^b (\nabla T) \cdot d\mathbf{l}$ is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$.
- $\oint (\nabla T) \cdot d\mathbf{l} = 0$, since the beginning and end points are identical, and hence $T(\mathbf{b}) - T(\mathbf{a}) = 0$.

The Fundamental Theorem for Divergences

This theorem has at least three (3) special names:

1. Gauss's theorem,
2. Green's theorem,
3. Divergence theorem.

The fundamental theorem for divergences states:

Theory 2.4: Divergence Theorem

the **integral** of a **derivative** (the **divergence**) over a **region** (in this case the **volume**, $\mathcal{V}$) is equal to

the value of the function at the **boundary** (in this case the **surface** S bounding the volume).

$$\int_V (\nabla \cdot \mathbf{v}) \, d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}.$$

The boundary term is itself an integral, more specifically, a surface integral. This is reasonable: the “boundary” of a line is just two end points, but the boundary of a volume is a closed surface. To create an analogy, if $\mathbf{v}$ represents the flow of an incompressible fluid, then the flux $\mathbf{v}$ is the total amount of fluid passing out through the surface, per unit time. Now, the divergence measures the *spreading out* of the vectors from a point, a place of high divergence is like a tap, pouring out liquid. If we have a bunch of tap in a region filled with incompressible fluid, an equal amount of liquid will be forced out through the boundaries of the region. In fact, there are two (2) ways we could determine how much is being produced:

- i. we could count up all the faucets, recording how much each puts out
- ii. we could go around the boundary, measuring the flow at each point, and add it all up

We get the same answer either way:

$$\int (\text{faucets within the volume}) = \oint (\text{flow out through the surface})$$

Exercise 2.4 An Example of Divergence Theorem

Evaluate both sides of the Divergence theorem for the expanding vector field:

$$\mathbf{F} = (x) \hat{\mathbf{x}} + (y) \hat{\mathbf{y}} + (z) \hat{\mathbf{z}},$$

over the sphere:

$$x^2 + y^2 + z^2 = a^2$$

SOLUTION

The outer unit normal to S , calculated from the gradient of $f(x, y, z) = x^2 + y^2 + z^2 - a^2$, is:

$$\begin{aligned} \hat{\mathbf{n}} &= \frac{\nabla S}{|\nabla S|} \\ &= \frac{(2x) \hat{\mathbf{x}} + (2y) \hat{\mathbf{y}} + (2z) \hat{\mathbf{z}}}{\sqrt{4x^2 + 4y^2 + 4z^2}} \\ &= \frac{(x) \hat{\mathbf{x}} + (y) \hat{\mathbf{y}} + (z) \hat{\mathbf{z}}}{a}, \end{aligned}$$

where $x^2 + y^2 + z^2 = a^2$ on S

Therefore:

$$(\mathbf{F} \cdot \hat{\mathbf{n}}) \, da = \frac{x^2 + y^2 + z^2}{a} \, da = \frac{a^2}{a} \, da = a \, da.$$

This in turn gives us:

$$\begin{aligned} &\iint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) \, da \\ &= \iint_S a \, da = a \iint_S da \end{aligned}$$

$$=a(4\pi a^2) = 4\pi a^3.$$

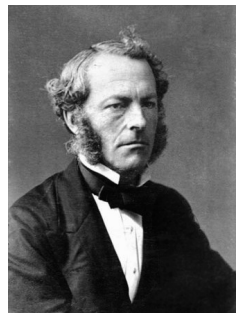
where Area of S is $4\pi a^2$

The divergence of F is:

$$\nabla \cdot F = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 3,$$

So,

$$\begin{aligned} \iiint_V (\nabla \cdot \mathbf{v}) \, d\tau &= \iiint_V 3 \, d\tau \\ &= 3 \left(\frac{4}{3} \pi a^3 \right) = 4\pi a^3 \quad \blacksquare \end{aligned}$$



¹⁸Sir George Gabriel Stokes, 1st Baronet, (1819 - 1903)

was an Irish mathematician and physicist. Born in County Sligo, Ireland, Stokes spent his entire career at the University of Cambridge, where he served as the Lucasian Professor of Mathematics for 54 years, from 1849 until his death in 1903, the longest tenure held by the Lucasian Professor. As a physicist, Stokes made seminal contributions to fluid mechanics, including the Navier-Stokes equations; and to physical optics, with notable works on polarisation and fluorescence. As a mathematician, he popularised "Stokes' theorem" in vector calculus and contributed to the theory of asymptotic expansions. Stokes, along with Felix Hoppe-Seyler, first demonstrated the oxygen transport function of haemoglobin, and showed colour changes produced by the aeration of haemoglobin solutions.

2.2.8 The Fundamental Theorem for Curls

The fundamental theorem for curls, also known as **Stokes' theorem**, states:¹⁸

Theory 2.5: Stokes' Theorem

Integral of a **derivative** over a **region** (S) is equal to the value of the function at **boundary** ($\mathcal{P}$).

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l}.$$

Similar to the divergence theorem, the boundary term is itself an integral. Specifically, a *closed line integral*.

Remember the curl measures the *twist* of the vectors $\mathbf{v}$. Think of a region of high curl as a whirlpool, where if we put a wheel there, it will rotate. Now, the integral of the curl over some surface (or, more precisely, the *flux* of the curl through the surface) represents the *total amount of swirl*, and we can determine that just as well by going around the edge and finding how much the flow is following the boundary.

$\oint \mathbf{v} \cdot d\mathbf{l}$ is sometimes called the **circulation** of $\mathbf{v}$.

There seems to be an ambiguity in Stokes' theorem: concerning the boundary line integral:

Which way are we supposed to go around?¹⁹

¹⁹clockwise or counterclockwise.

The answer is that it doesn't matter which way we go **as long as we are consistent**, for there is an additional sign ambiguity in the surface integral:

Which way does $d\mathbf{a}$ point?

For a closed surface,²⁰ $d\mathbf{a}$ points in the direction of the outward normal. But for an open surface, which way would be defined as out? Consistency in Stokes' theorem is given by the right-hand rule. If our rings point in the direction of the line integral, then our thumb fixes the direction of $d\mathbf{a}$.

²⁰i.e., the divergence theorem.

Ordinary, a flux integral depends critically on what surface we integrate over, but this is **NOT** the case with curls. For Stokes' theorem says that $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ is equal to the line integral of $\mathbf{v}$ around the boundary, and the latter makes no reference to the specific surface we choose.

Proposition I $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ depends only on the boundary line, not on the particular surface used.

Proposition II $\oint (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$ for any closed surface.

Exercise 2.5 Surface Area of an Implicit Surface

Find the area of the surface cut from the bottom of the paraboloid $x^2 + y^2 - z = 0$ by the plane $z = 4$.

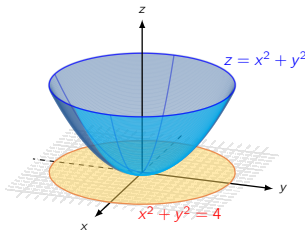


Figure 2.9: The paraboloid of “Surface Area of an Implicit Surface”.

SOLUTION We sketch the surface S and the region R below it in the xy -plane (Fig. 2.9). The surface S is part of the level surface $F(x, y, z) = x^2 + y^2 - z = 0$, and R is the disk $x^2 + y^2 \leq 4$ in the xy -plane.

To get a unit vector normal (i.e., $\hat{\mathbf{n}}$) to the plane R , we can take $\hat{\mathbf{n}} = \hat{\mathbf{z}}$. At any point (x, y, z) on the surface, we have:

$$F(x, y, z) = x^2 + y^2 - z$$

$$\nabla F = (2x) \hat{\mathbf{x}} + (2y) \hat{\mathbf{y}} + (-1) \hat{\mathbf{z}}$$

$$|\nabla F| = \sqrt{(2x)^2 + (2y)^2 + (-1)^2}$$

$$= \sqrt{4x^2 + 4y^2 + 1}$$

$$|\nabla F \cdot \hat{\mathbf{n}}| = |\nabla F \cdot \hat{\mathbf{z}}| = |-1| = 1.$$

In the region R , the area is defined to be $dA = dx dy$. Therefore:

$$\begin{aligned} \text{Surface Area} &= \iint_R \frac{|\nabla F|}{|\nabla F \cdot \hat{\mathbf{n}}|} dA \\ &= \iint_{x^2+y^2 \leq 4} \sqrt{4x^2 + 4y^2 + 1} dx dy \\ &= \int_0^{2\pi} \int_0^2 \sqrt{4r^2 + 1} r dr d\theta \\ &= \int_0^{2\pi} \left[\frac{1}{12} (4r^2 + 1)^{3/2} \right]_0^2 d\theta \\ &= \int_0^{2\pi} \frac{1}{12} (17^{3/2} - 1) d\theta \\ &= \frac{\pi}{6} (17\sqrt{17} - 1) \quad \blacksquare \end{aligned}$$

Exercise 2.6 Stokes' Theorem Over a Hemisphere

Evaluate Stokes's theorem for the hemisphere S : $x^2 + y^2 + z^2 = 9, z \geq 0$, with its bounding circle C : $x^2 + y^2 = 9, z = 0$ and the field is:

$$\mathbf{F} = (y) \hat{\mathbf{x}} + (-x) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}.$$

Tip: Parametrisation of a circle is: $x = r \cos \theta, y = r \sin \theta$ and $da = r d\theta$

SOLUTION

We start by calculating the counter-clockwise circulation around C using the following parametrisation:

$$\ell(\theta) = (3 \cos \theta) \hat{\mathbf{x}} + (3 \sin \theta) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}},$$

where $0 \leq \theta \leq 2\pi$.

Using this we can calculate the counter-clockwise circulation.

$$d\ell = (-3 \sin \theta d\theta) \hat{\mathbf{x}} + (3 \cos \theta d\theta) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}},$$

$$\mathbf{F} = (y) \hat{\mathbf{x}} + (-x) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}$$

$$= (3 \sin \theta) \hat{\mathbf{x}} + (-3 \cos \theta) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}},$$

$$\mathbf{F} \cdot d\mathbf{l} = -9 \sin^2 \theta d\theta - 9 \cos^2 \theta d\theta = -9 d\theta,$$

$$\oint_C \mathbf{F} \cdot d\mathbf{l} = \int_0^{2\pi} -9 d\theta = -18\pi.$$

For the curl of integral we have:

$$\begin{aligned} \nabla \times \mathbf{F} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_x & F_y & F_z \end{vmatrix} \\ &= (0-0) \hat{\mathbf{x}} + (0-0) \hat{\mathbf{y}} + (-1-1) \hat{\mathbf{z}} \\ &= -2 \hat{\mathbf{z}} \end{aligned}$$

$$\begin{aligned} \hat{\mathbf{n}} &= \frac{\nabla S}{|\nabla S|} \\ &= \frac{(x) \hat{\mathbf{x}} + (y) \hat{\mathbf{y}} + (z) \hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}} \\ &= \frac{(x) \hat{\mathbf{x}} + (y) \hat{\mathbf{y}} + (z) \hat{\mathbf{z}}}{3} \end{aligned} \quad \text{Unit normal}$$

Now it is time to define the area of integration (da):

$$\begin{aligned} da &= \frac{|\nabla S|}{|\nabla S \cdot \hat{\mathbf{z}}|} dA \\ &= \frac{|(2x) \hat{\mathbf{x}} + (2y) \hat{\mathbf{y}} + (2z) \hat{\mathbf{z}}|}{2z} \\ &= \frac{2 \sqrt{x^2 + y^2 + z^2}}{2z} \\ &= \frac{3}{z} dA, \end{aligned}$$

$$\nabla \times \mathbf{F} \cdot \mathbf{n} da = -\frac{2z}{3} \frac{3}{z} dA = -2 dA$$

The cardinal direction $\hat{\mathbf{z}}$ comes from being the direction **perpendicular** to the surface (S).

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} da = \iint_{x^2+y^2 \leq 9} -2 dA = -18\pi$$

The circulation around the circle equals the integral of the curl over the hemisphere ■

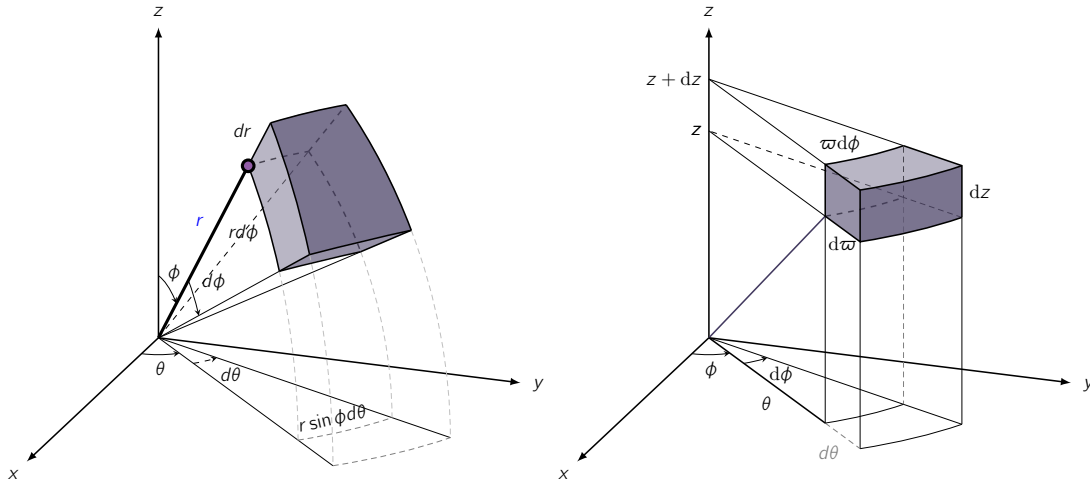


Figure 2.10: The two types of coordinate systems in question. (a) Spherical coordinate system (b) Spherical coordinate system.

2.3 Curvilinear Coordinates

2.3.1 Spherical Coordinate System

It is possible to label a point P in Cartesian coordinates (x, y, z) , but sometimes it is more convenient to use **spherical** coordinates (r, θ, ϕ) ; r is the distance from the origin, θ is called the **polar angle**, and ϕ is the **azimuthal angle**. Their relation to Cartesian coordinates can be read from Fig. 2.10a.

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$$

Fig. 2.10a also shows three unit vectors, $\hat{r}$, $\hat{\theta}$, $\hat{\phi}$, pointing in the direction of increase of the corresponding coordinates.

They constitute an **orthogonal** basis set,²¹ similar to $\hat{x}$, $\hat{y}$, $\hat{z}$, and any vector $\mathbf{A}$ can be expressed in terms of them, in the usual way:

$$\mathbf{A} = (A_r) \hat{r} + (A_\theta) \hat{\theta} + (A_\phi) \hat{\phi}$$

Here, A_r , A_θ , A_ϕ are the radial, polar, and azimuthal components of vector $\mathbf{A}$. In terms of the Cartesian unit vectors:

$$\begin{aligned} \hat{r} &= (\sin \theta \cos \phi) \hat{x} + (\sin \theta \sin \phi) \hat{y} + (\cos \theta) \hat{z}, \\ \hat{\theta} &= (\cos \theta \cos \phi) \hat{x} + (\cos \theta \sin \phi) \hat{y} + (-\sin \theta) \hat{z}, \\ \hat{\phi} &= (-\sin \phi) \hat{x} + (\cos \phi) \hat{y} + (0) \hat{z}. \end{aligned}$$

An infinitesimal displacement in the $\hat{r}$ direction is simply dr , just as an infinitesimal element of length in the $\hat{x}$ direction is dx :

$$dl_r = dr$$

²¹meaning mutually perpendicular.

On the other hand, an infinitesimal element of length in the $\hat{\theta}$ direction is **NOT** just $d\theta$ but rather,

$$dl_{\theta} = r d\theta$$

Similarly, an infinitesimal element of length in the $\hat{\phi}$ direction is

$$dl_{\phi} = r \sin \theta d\phi$$

Thus the general infinitesimal displacement $d\mathbf{l}$ is:

$$d\mathbf{l} = (dr) \hat{r} + (r d\theta) \hat{\theta} + (r \sin \theta) \hat{\phi}$$

This plays the role $d\mathbf{l} = (dx) \hat{x} + (dy) \hat{y} + (dz) \hat{z}$ plays in Cartesian coordinates. The infinitesimal volume element $d\tau$, in spherical coordinates, is the product of the three (3) infinitesimal displacements:

$$d\tau = dl_r dl_{\theta} dl_{\phi} = r^2 \sin \theta dr d\theta d\phi.$$

It is not possible to give a general expression for surface elements $d\mathbf{a}$, since these depend on the orientation of the surface. We simply have to analyse the geometry for any given case, which goes for Cartesian and curvilinear coordinates.

Integrating over the surface of a sphere, for instance, makes r constant, whereas θ and ϕ change:

$$d\mathbf{a}_1 = dl_{\theta} dl_{\phi} \hat{r} = r^2 \sin \theta d\theta d\phi \hat{r}$$

On the other hand, if the surface lies in the xy plane, making θ is constant, while r and ϕ vary:

$$d\mathbf{a}_2 = dl_r dl_{\phi} \hat{\theta} = r dr d\phi \hat{\theta}$$

Finally: r ranges from 0 to ∞ , ϕ from 0 to 2π , and θ from 0 to π .

Exercise 2.7 The Volume of a Sphere

Find the volume of a sphere of radius R .

SOLUTION

The derivation is as follows:

$$V = \int d\tau$$

$$\int_{r=0}^R \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} r^2 \sin \theta dr d\theta d\phi,$$

$$= \left(\int_0^R r^2 dr \right) \left(\int_0^{\pi} \sin \theta d\theta \right) \left(\int_0^{2\pi} d\phi \right)$$

$$= \left(\frac{R^3}{3} \right) (2) (2\pi) = \frac{4}{3} \pi R^3 \quad \blacksquare$$

2.3.2 Cylindrical Coordinates

The cylindrical coordinates (s, ϕ, z) of a point P are defined in **Fig. 2.10b**. Observe that ϕ has the same meaning as in spherical coordinates, and z is the same as Cartesian; s is the distance to P

from the z axis, whereas the spherical coordinate r is the distance from the origin. The relation to Cartesian coordinates is:

$$x = s \cos \phi \quad y = s \sin \phi \quad z = z.$$

The unit vectors are:

$$\begin{aligned}\hat{s} &= \cos \phi \hat{x} + \sin \phi \hat{y} \\ \hat{\phi} &= -\sin \phi \hat{x} + \cos \phi \hat{y} \\ \hat{z} &= \hat{z}\end{aligned}$$

The infinitesimal displacements are

$$dl_s = ds \quad dl_\phi = s d\phi, \quad dl_z = dz$$

which makes:

$$d\mathbf{l} = ds \hat{s} + s d\phi \hat{\phi} + dz \hat{z}.$$

and the volume element is

$$d\tau = s ds d\phi dz$$

The range of s is $(0, \infty)$, ϕ is from 0 to 2π and z is from $-\infty$ to $+\infty$



2.4 Dirac Delta Function

2.4.1 A Mathematical Anomaly

Consider the following vector function:

$$\mathbf{v} = \frac{1}{r^2} \hat{\mathbf{r}}$$

At every location, $\mathbf{v}$ is directed **radially outward** which we can see in **Fig. 2.11**. Let's calculate its divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r^2} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} (1) = 0$$

This is interesting as this calculation gives us an unforeseen solution. Let's look at it closer. Suppose we integrate over a sphere of radius R , centred at the origin. The surface integral therefore is

$$\oint \mathbf{v} \cdot d\mathbf{a} = \int \left(\frac{1}{R^2} \hat{\mathbf{r}} \right) \cdot (R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}) = \left(\int_0^\pi \sin \theta d\theta \right) \left(\int_0^{2\pi} d\phi \right) = 4\pi$$

But the volume integral ($\int \nabla \cdot \mathbf{v} d\tau$), should be zero (0) if we assume the aforementioned calculation to be true.

Does this mean that the divergence theorem is false?

The source of the problem lies at the point $r = 0$, where $\mathbf{v}$ **blows up**. It is true that $\nabla \cdot \mathbf{v} = 0$ everywhere **except** the origin, but right at the origin is the situation gets a little bit complicated.

Observe, the surface integral is **independent** of R . If the divergence theorem is to be true, we should expect

$$\int \nabla \cdot \mathbf{v} d\tau = 4\pi,$$

for any non-zero vector and the origin.

This means the value of 4π must be coming from the point $r = 0$. Therefore, $\nabla \cdot \mathbf{v}$ has the unique property that it vanishes everywhere except at one point, and yet its **integral** is 4π .

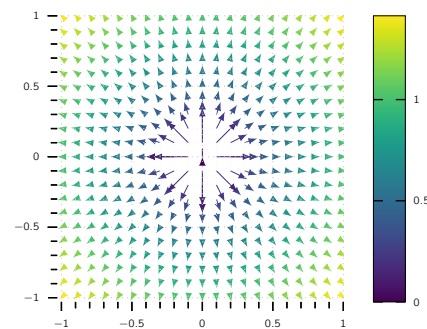


Figure 2.11: The vector plot of the “divergence problem”.

No normal function behaves like that.

Information: An Analogy with Density

To wrap our heads around this property think of **density**.

The density of a point particle is zero except at the exact location of the particle, and yet its **integral** is finite. Namely, the mass of the particle.

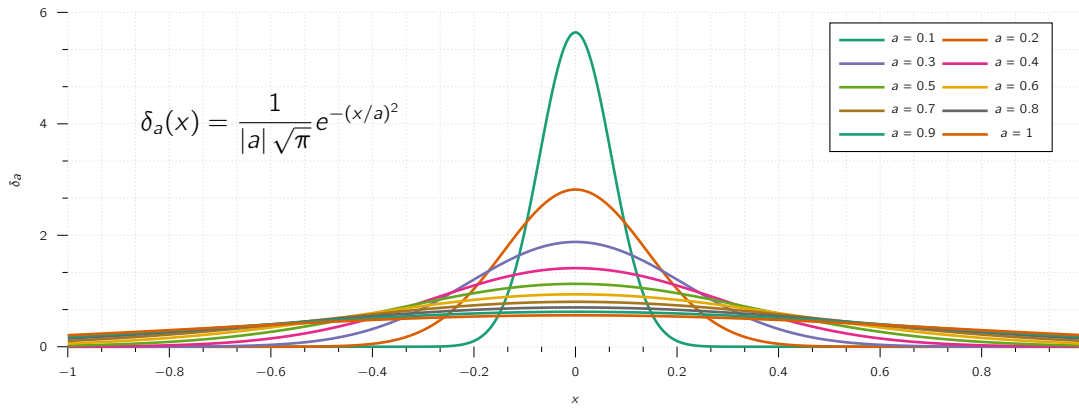


Figure 2.12: A visual representation of a 1D Dirac Delta Function. Think of it as a distribution function being squeezed to an infinitely small width.

What we have stumbled upon is called the **Dirac delta function**. It arises in numerous branches of theoretical physics and plays a central role in the theory of electrodynamics.

2.4.2 The 1D Dirac Delta Function

The one-dimensional Dirac delta function ($\delta(x)$), can be pictured as an infinitely high, infinitesimally narrow “spike”, with area 1.

That is to say:

$$\delta(x) = \begin{cases} 0, & \text{if } x \neq 0 \\ \infty, & \text{if } x = 0 \end{cases}$$

and,

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

In a strict sense of definition, $\delta(x)$ is **NOT** a function at all, as its value is not finite at $x = 0$. In literature it is known as a generalised function.²²

If $f(x)$ is some “ordinary” function, then the product ($f(x)\delta(x)$) is zero everywhere except at $x = 0$. It follows that:

$$f(x)\delta(x) = f(0)\delta(x). \quad (2.11)$$

The product is zero anyway except at $x = 0$. Based on this property, we may as well replace $f(x)$ by the value it assumes at the origin.

Particularly

²²Objects extending the notion of functions on real or complex numbers. There is more than one recognised theory, for example the theory of distributions. Generalised functions are especially useful for treating discontinuous functions more like smooth functions, and describing discrete physical phenomena such as point charges.



$$\int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0) \int_{-\infty}^{\infty} \delta(x) dx = f(0). \quad (2.12)$$

Under an integral, then, the delta function “picks out” the value of $f(x)$ at $x = 0$.²³ Of course, we can shift the spike from $x = 0$ to some other point, $x = a$:

$$\delta(x - a) = \begin{cases} 0, & \text{if } x \neq a \\ \infty, & \text{if } x = a \end{cases} \quad \text{with} \quad \int_{-\infty}^{\infty} \delta(x - a) dx = 1. \quad (2.13)$$

²³Here and below, the integral need not run from $-\infty$ to $+\infty$; it is sufficient that the domain extend across the delta function, and $-\epsilon$ to $+\epsilon$ would do as well.

which turns Eq. (2.11) into:

$$f(x) \delta(x - a) = f(a) \delta(x - a), \quad (2.14)$$

and Eq. (2.12) generalises to:

$$\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a). \quad (2.15)$$

While δ itself is not a proper function, integrals over δ are perfectly acceptable. In fact, it's best to think of the delta function as something that is always intended for use under an integral sign. In particular, two expressions involving delta functions, say $D_1(x)$ and $D_2(x)$, are considered equal if:

$$\int_{-\infty}^{\infty} f(x) D_1(x) dx = \int_{-\infty}^{\infty} f(x) D_2(x) dx, \quad (2.16)$$

for all **ordinary** functions $f(x)$.

Exercise 2.8 A Simple Dirac Integral

Evaluate the following integral:

$$\int_0^3 x^3 \delta(x - 2) dx.$$

The delta function picks out the value of x^3 at the point $x = 2$, so the integral is $2^3 = 8$. Notice, however, that if the upper limit had been 1, instead of 3, the answer would be 0, because the spike would then be outside the domain of integration ■

SOLUTION

2.4.3 The 3D Dirac Delta Function

Once we have defined the 1D Dirac, it is simple to generalise it to 3D with the following:

$$\delta^3(\mathbf{r}) = \delta(x) \delta(y) \delta(z), \quad (2.17)$$

As we can see, it is similar to 1D, where 3D Dirac is zero everywhere except at $(0, 0, 0)$, where it blows up. Its volume integral is 1:

$$\int_{\text{all space}} \delta^3(\mathbf{r}) d\tau = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(x) \delta(y) \delta(z) dx dy dz = 1$$

And, the general form is:

$$\int_{\text{all space}} f(\mathbf{r}) \delta^3(\mathbf{r} - \mathbf{a}) d\tau = f(\mathbf{a}). \quad (2.18)$$

As in the 1D case, integration with δ picks out the value of the function f at the location of the spike.

We can fix the paradox introduced in Section 2.4. Remember, the divergence of $\hat{\mathbf{r}}/r^2$ is zero everywhere except at the origin, however, its integral over any volume containing the origin is a constant.

These are precisely the defining conditions for the Dirac delta function; evidently

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^3(\mathbf{r})$$

Or in a more general fashion:

$$\nabla \cdot \left(\frac{\hat{\mathbf{z}}}{z^2} \right) = 4\pi \delta^3(\hat{\mathbf{z}}) \quad (2.19)$$

Differentiation here is with respect to $\mathbf{r}$, while $\mathbf{r}'$ is held constant.

2.5 Vector Field Theory

2.5.1 Helmholtz Theorem

Electricity and magnetism are generally expressed as **electric and magnetic fields**, $\mathbf{E}$ and $\mathbf{B}$ and like many physical laws, these are most compactly expressed as **differential equations**.

As $\mathbf{E}$ and $\mathbf{B}$ are **vectors**, the differential equations naturally involve vector derivatives: *divergence* and *curl*. Maxwell reduced the entire theory to four (4) fundamental equations, specifying respectively the divergence and the curl of $\mathbf{E}$ and $\mathbf{B}$.

This formulation raises an interesting question:

To what extent is a vector function determined by its divergence and curl?

To study this case let's assume a vector of $\mathbf{F}$. If the divergence of $\mathbf{F}$ is a specified scalar function D ,

$$\nabla \cdot \mathbf{F} = D,$$

and the curl of $\mathbf{F}$ is a specified vector function $\mathbf{C}$,

$$\nabla \times \mathbf{F} = \mathbf{C},$$

and for consistency, we assume $\mathbf{C}$ to have **NO** divergence,

$$\nabla \cdot \mathbf{C} = 0,$$

Remember, the divergence of a curl is **ALWAYS** zero.

Using this knowledge, is it possible to determine the function $\mathbf{F}$?

Without knowing more information, it is not really possible. There are many functions whose divergence and curl are both zero everywhere.

Some examples are:

$$\mathbf{F} = 0,$$

$$\mathbf{F} = (y) \hat{x} + (zx) \hat{y} + (xy) \hat{z},$$

$$\mathbf{F} = (\sin x \cosh y) \hat{x} + (-\cos x \sinh y) \hat{y} + (.) \hat{z}$$

If we recall **Higher Mathematics I**, to solve a differential equation with a particular solution, we must also be supplied with appropriate **boundary conditions**.

In electrodynamics we typically require the fields go to zero at infinity. With that extra information, the **Helmholtz theorem**²⁴ guarantees the field is uniquely determined by its divergence and curl.



²⁴Hermann Ludwig Ferdinand von Helmholtz (1821 - 1894)

was a German physicist and physician who made significant contributions in several scientific fields, particularly hydrodynamic stability. The Helmholtz Association, the largest German association of research institutions, was named in his honour.

In physics, he is known for his theories on the conservation of energy and on the electrical double layer, work in electrodynamics, chemical thermodynamics, and on a mechanical foundation of thermodynamics. Although credit is shared with Julius von Mayer, James Joule, and Daniel Bernoulli-among others-for the energy conservation principles that eventually led to the first law of thermodynamics, he is credited with the first formulation of the energy conservation principle in its maximally general form.

2.5.2 Potentials

If the curl of a vector field ($\mathbf{F}$) vanishes everywhere, then $\mathbf{F}$ can be written as the **gradient of a scalar potential** (V):

$$\nabla \times \mathbf{F} = 0 \iff \mathbf{F} = -\nabla V$$

The minus sign is purely conventional.

That's the essential idea of the following theorem:

Theory 2.6: Zero Curl Fields

The following conditions are **equivalent**.

- i. $\nabla \times \mathbf{F} = 0$ everywhere,
- ii. $\int_a^b \mathbf{F} \cdot d\mathbf{l}$ is independent of path, for any given end points,
- iii. $\oint \mathbf{F} \cdot d\mathbf{l} = 0$ for any closed loop,
- iv. $\mathbf{F}$ is the gradient of some scalar function: $\mathbf{F} = -\nabla V$.

The potential is **NOT** unique as any constant can be added to V , since this will not affect its gradient.

If the divergence of a vector field ($\mathbf{F}$) vanishes everywhere, then $\mathbf{F}$ can be expressed as the curl of a **vector potential** ($\mathbf{A}$):

$$\nabla \cdot \mathbf{F} = 0 \iff \mathbf{F} = \nabla \times \mathbf{A}$$

That's the main conclusion of the following theorem:

Theory 2.7: Zero Divergence Fields

The following conditions are **equivalent**:

- i. $\nabla \cdot \mathbf{F} = 0$ everywhere.
- ii. $\oint \mathbf{F} \cdot d\mathbf{a}$ is independent of surface, for any given boundary line.
- iii. $\oint \mathbf{F} \cdot d\mathbf{a} = 0$ for any closed surface.
- iv. $\mathbf{F}$ is the curl of some vector function: $\mathbf{F} = \nabla \times \mathbf{A}$.

The vector potential is **NOT** unique as the gradient of any scalar function can be added to $\mathbf{A}$ without affecting the curl, given the curl of a gradient is zero.

Incidentally, in all cases, a vector field $\mathbf{F}$ can be written as the gradient of a scalar plus the curl of a vector.

$$\mathbf{F} = -\nabla V + \nabla \times \mathbf{A}$$

Chapter Bibliography

1. Tait, P. G. *An elementary treatise on quaternions* (CUP Archive, 2018).
2. Hamilton, W. R. On the Argument of Abel. *Transactions of the Royal Irish Academy* **18**, 171–259 (1839).



Part II

Electric Fields

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Wahrlich es ist nicht das Wissen, sondern das Lernen, nicht das Besitzen sondern das Erwerben, nicht das Da-Seyn, sondern das Hinkommen, was den grössten Genuss gewährt.

It is not knowledge, but the act of learning, not possession but the act of getting there, which grants the greatest enjoyment.

(Carl Friedrich Gauss in Letter to Farkas Bolyai (2 September 1808))

Chapter 3

Electrostatics

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3.1 The Electric Field

In essence, problem of electrodynamics is to solve:

We have some electric charges, $q_1, q_2, q_3, \dots$, let's call them **source charges**. What force do they exert on another charge (i.e., **test charge**), Q ?

The positions of the source charges are given¹ and the trajectory of the test particle is to be calculated. For most realistic cases, both the **source charges** and the **test charge** are in motion.

¹usually as functions of time

The solution to this problem is solved with the **principle of superposition**, which states:

The interaction between any two (2) charges is **unaffected** by the presence of others.

This means to determine the force on Q , we can first calculate the force $\mathbf{F}_1$, caused by q_1 alone, then we calculate the force $\mathbf{F}_2$, and then q_2 alone and so on.

Finally, we take the vector sum of all these individual forces:

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots$$

Therefore, if we can find the force on Q caused by a single source charge q , we are, in principle, done².

It seems easy, right? Why don't we just write down the formula for the force on Q directed **towards** q , and calculate it?

We could, but some technicalities limit us, for not only does the force on Q depend on the separation distance z between the charges, but also depends on *both* their velocities and on the *acceleration* of q .

Moreover, it is not the position, velocity, and acceleration of q **right now** that matter:

electromagnetic particles/waves travels at the speed of light, so what concerns Q is the position, velocity, and acceleration q *had* at some earlier time, when the message left.

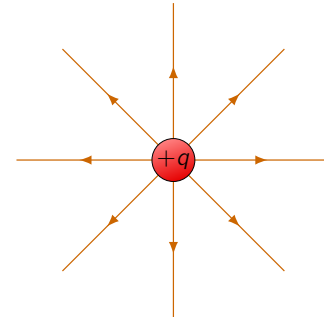


Figure 3.1: An example of a singular charge in space. As can be seen a charge with a plus value will generate a repulsive force.

²The principle of superposition may seem "obvious" to us, but it did **NOT** have to be simple. For example, if the electromagnetic force were proportional to the square of the total source charge, the principle of superposition would not hold, as $(q_1 + q_2)^2 \neq q_1^2 + q_2^2$, as there would be "cross terms" to consider. Superposition is **NOT** a logical necessity, but an experimental fact.

Therefore, in spite of the fact that the basic question is easy to state³, therefore we shall not answer this with a **direct** answer now but go at it by stages.

To simplify this problem, we assume time to be **irrelevant**, making the special case of **electrostatics** in which all the *source charges are stationary* whereas the test charge may be in motion.

³Most complicated problem are funny enough relatively easy to state (see Collatz conjecture)

3.1.1 Coulomb's Law

What is the force on a test charge Q due to a single point charge q , that is at *rest* a distance z away? The answer is given by Coulomb's law⁴:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{z^2} \hat{z} \quad (3.1)$$

The constant ϵ_0 is the **permittivity of free space**. In SI units, where force is in newtons (N) distance in meters (m), and charge in coulombs (C).

$$\epsilon_0 8.85 \times 10^{-12} \text{ C}^2 \cdot \text{N}^{-1} \cdot \text{m}^2$$

In other words,

the force is proportional to the product of the charges and inversely proportional to the square of the separation distance.

Remember, z is the separation vector from r_i (the location of q) to r (the location of Q):

$$\mathbf{z} = \mathbf{r} - \mathbf{r}_i$$



⁴Charles-Augustin de Coulomb (1736 - 1806) was a French officer, engineer, physicist and the discoverer of what is now called Coulomb's law, the description of the electrostatic force of attraction and repulsion. He also did important work on friction, and his work on earth pressure formed the basis for the later development of much of the science of soil mechanics. The SI unit of electric charge, the **coulomb**, was named in his honor in 1880.

$\hat{z}$ is its magnitude, and $\hat{z}$ is its direction. The force points **along the line** from q to Q ;

■ it is repulsive if q and Q have the same sign,

■ it is attractive if their signs are opposite

Coulomb's law and the principle of superposition form the physical foundation for electrostatics with the rest, except for some special properties of matter, is **mathematical derivation** of these fundamental rules.

3.1.2 The Electric Field

If we have *several* point charges $q_1, q_2, \dots, q_n$, at distances $z_1, z_2, \dots, z_n$ from Q , the total force on Q is evidently

$$\begin{aligned} \mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \dots &= \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 Q}{z_1^2} \hat{z}_1 + \frac{q_2 Q}{z_2^2} \hat{z}_2 + \dots \right) \\ &= Q \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{z_1^2} \hat{z}_1 + \frac{q_2}{z_2^2} \hat{z}_2 + \dots \right) \end{aligned}$$

or in a more cleaner way:

$$\mathbf{F} = Q\mathbf{E} \quad (3.2)$$

where

$$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{z_i^2} \hat{z}_i \quad (3.3)$$

$\mathbf{E}$ is called the **electric field** of the source charges. It is important to mention, it is a function of the position ($\mathbf{r}$), as the separation vectors $\hat{z}_i$ depend on the location of the **field point** P .

But it makes no reference to the test charge Q .

The electric field is a **vector quantity** that varies from point to point and is determined by the configuration of source charges. Physically, $\mathbf{E}(\mathbf{r})$ is the force per unit charge that would be exerted on a test charge, if you were to place one at P .

Exercise 3.1 Electric Field at a Distance

Find the $\mathbf{E}$ field a distance z above the midpoint between two (2) equal charges (q), a distance d apart:

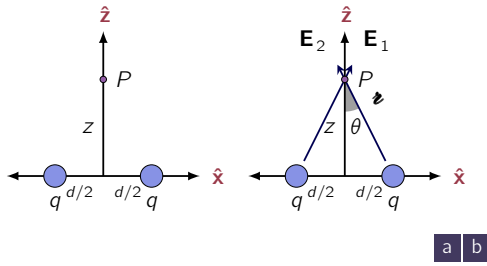


Figure 3.2: The diagram for the question “Electric Field at a Distance” (a) The geometrical description (b) Additional annotations added.

SOLUTION Let E_1 be the field of the left charge, and E_2 that of the right charge alone. Adding them, the horizontal components cancel and the vertical components add to:

$$E_z = 2 \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cos\theta.$$

Here $r = \sqrt{z^2 + (d/2)^2}$ and $\cos\theta = z/r$, therefore:

$$E = \frac{1}{4\pi\epsilon_0} \frac{2qz}{[z^2 + (d/2)^2]^{3/2}} \hat{z} \quad \blacksquare$$

3.1.3 Continuous Charge Distributions

Our definition of the electric field Eq. (3.3) assumes the field source is a **collection** of discrete point charges q_i . If, the charge is distributed continuously over some region, the sum becomes an integral:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{\mathbf{r}} dq$$

1. If charge is spread out along a **line**, with charge-per-unit-length λ , then $dq = \lambda dl'$, where dl' is an element of length along the line.
2. If charge is smeared out over a **surface**, with charge-per-unit-area σ , then $dq = \sigma da'$, where da' is an element of area on the surface.
3. If charge fills a **volume** with charge-per-unit-volume ρ , then $dq = \rho d\tau'$, where $d\tau'$ is an element of volume.

These aforementioned volumetric definitions can be rewritten as:

$$dq \rightarrow \lambda dl' \sim \sigma da' \sim \rho d\tau'. \quad (3.4)$$

Therefore the electric field of a line charge is,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{r^2} \hat{\mathbf{r}} dl',$$

for a surface charge,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{r^2} \hat{\mathbf{r}} da',$$

and for a volume charge,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}} d\tau'. \quad (3.5)$$

Eq. (3.5) is often referred to as **Coulomb's law**, as it is such a short step from the original shown in Eq. (3.1), and because a volume charge is in a sense the most general and realistic case.

3.2 Divergence and Curl of Electrostatic Fields

As we have seen by now, the integrals involved in computing $\mathbf{E}$ can be complicated and often laborious, even for reasonably simple charge distributions.

Therefore, rest of electrostatics from this point onwards is devoted to assembling methods for **avoiding these integrals**.

It all begins with the **divergence** and **curl** of $\mathbf{E}$. We shall calculate the divergence of $\mathbf{E}$ directly from Eq. (3.5), but first we should develop an intuitive view of $\mathbf{E}$.

Let's begin with the simplest possible case:

A singular point charge (q), situated at the **origin**:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

As the equation tells, the field falls off with a factor of $1/r^2$, the vectors get shorter as you go farther away from the origin and **they always point radially outward**. But there is a nicer way to represent this field, and that's to connect up the arrows, to form **field lines**. The magnitude of the field is indicated by the **density** of the field lines:

It's strong near the center where the field lines are close together, and weak farther out, where they are relatively far apart.

In truth, the field-line diagram can be deceptive, as when we draw it on a two-dimensional surface, for the density of lines passing through a circle of radius r is the total number divided by the circumference ($n/2\pi r$), which goes at a rate of $(1/r)$, and not $1/r^2$.

But if were to imagine the model in three dimensions then the density of lines is the total number divided by the area of the sphere ($n/4\pi r^2$), which **does** change with a factor of $1/r^2$.

Such diagrams are also convenient for representing more complicated fields. Field lines begin on positive charges and end on negative ones and they cannot simply terminate in midair, though they may extend out to infinity.

If they were to terminate, the divergence of $\mathbf{E}$ would not be zero, and that cannot happen in empty space. In addition, field lines can never cross at the intersection,

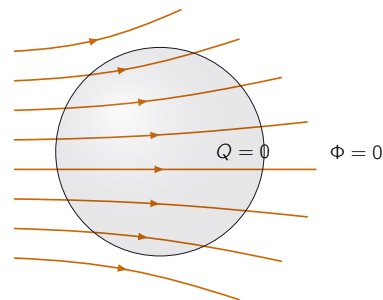


Figure 3.3
An example of imagining electric field as a collection of lines passing through. Here we can see the flux lines (i.e., the arrows) passing through an object.

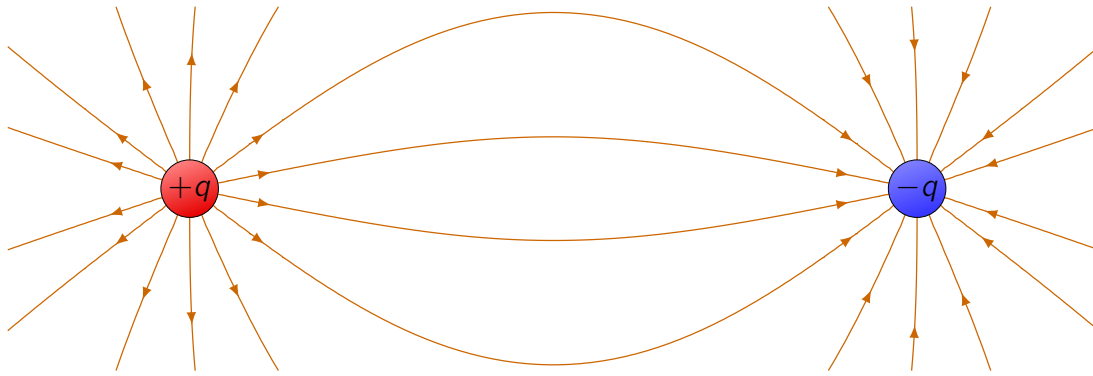


Figure 3.4

An example interaction of two charges interacting with one another with their field lines visible. Please observe that the field line spread away from the plus charge and are collected by the minus charge.

With all this in mind, it is easy to sketch the field of any simple configuration of point charges.

In this model, the **flux** of $\mathbf{E}$ through a surface S ,

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{a}. \quad (3.6)$$

is a measure of the **number of field lines** passing through S . It is worth mentioning here that it is only a representation of the field lines⁵, as the **total** number would be infinite. But **for a given sampling rate** the flux is *proportional* to the number of lines drawn, because the field strength, is proportional to the density of field lines, and therefore $\mathbf{E} \cdot d\mathbf{a}$, is proportional to the number of lines passing through the infinitesimal area $d\mathbf{a}$.

This means the flux through any **closed** surface is a measure of the total mass inside the surface. For the field lines that originate on a positive charge must either pass through the surface or else terminate on a negative charge inside. On the other hand, a charge *outside* the surface will contribute nothing to the total flux, since its field lines pass in one side and out the other.

This is the basic idea behind Gauss' law.

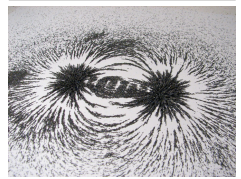
3.2.1 Quantification

In the case of a point charge q at the origin, the flux of $\mathbf{E}$ through a spherical surface of radius (r) is:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \int \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2} \hat{\mathbf{r}} \right) \cdot (r^2 \sin\theta d\theta d\phi \hat{\mathbf{r}}) = \frac{1}{\epsilon_0} q$$

Observe that the radius of the sphere **cancels out**, as the surface area goes **up** as r^2 , the field goes **down** as $1/r^2$, so the product is **constant**. In terms of the field-line picture, this makes sense, given the same number of field lines pass through any sphere centered at the origin, regardless of its size.

In fact, it doesn't have to be a sphere. **Any** closed surface, whatever is shape, would be pierced by the same number of field lines. Evidently the flux through any surface enclosing the charge is q/ϵ_0 .



⁵In the history of physics, a line of force in Michael Faraday's extended sense is synonymous with James Clerk Maxwell's line of induction. According to J.J. Thomson, Faraday usually discusses lines of force as chains of polarized particles in a dielectric, yet sometimes Faraday discusses them as having an existence all their own as in stretching across a vacuum. From the 20th century perspective, lines of force are energy linkages embedded in a 19th-century field theory that led to more mathematically and experimentally sophisticated concepts and theories, including Maxwell's equations and Albert Einstein's theory of relativity.

Let's extend this idea a bit further and assume instead of a single charge at the origin, we have a bunch of charges scattered about.

According to the principle of superposition, the total field is the vector sum of all the individual fields:

$$\mathbf{E} = \sum_{i=1}^n \mathbf{E}_i$$

The flux through a surface that encloses them all is:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \sum_{i=1}^n \left(\oint \mathbf{E}_i \cdot d\mathbf{a} \right) = \sum_{i=1}^n \left(\frac{1}{\epsilon_0} q_i \right)$$

For any closed surface, then,

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}} \quad (3.7)$$

where Q_{enc} is the total charge enclosed within the surface. This is the quantitative statement of Gauss's law. As it stands, Gauss's law is an *integral* equation, but we can easily turn it into a *differential* one, by applying the divergence theorem:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{E}) d\tau$$

Rewriting Q_{enc} in terms of the charge density ρ , we have

$$Q_{\text{enc}} = \int_V \rho d\tau$$

So Gauss's law becomes

$$\int_V (\nabla \cdot \mathbf{E}) d\tau = \int_V \left(\frac{\rho}{\epsilon_0} \right) d\tau$$

And since this holds for *any* volume, the integrands must be equal:

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho \quad (3.8)$$

Eq. (3.8) carries the same message as Eq. (3.7) as it is **Gauss's law in differential form**. The differential version is idler, but the integral form has the advantage in that it accommodates point, line, and surface charges more naturally.

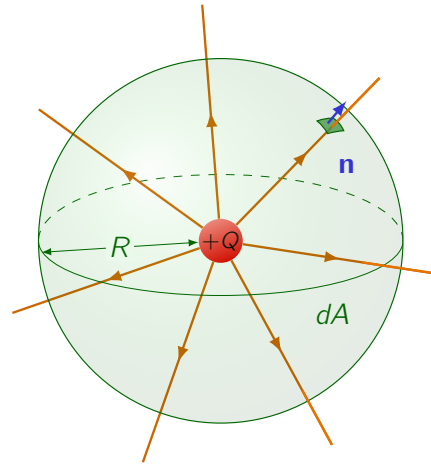


Figure 3.5: A visual representation of the gauss' law.

3.2.2 Divergence of $\mathbf{E}$

Let's go back, now, and calculate the divergence of $\mathbf{E}$ directly from Eq. (3.5):

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_{\text{all space}} \frac{\mathbf{z}}{z^2} \rho(\mathbf{r}') d\tau'$$

While the interaction is originally defined over the volume occupied by the charge, it is much easier to encompass all the space as $\rho = 0$ outside the volume regardless.

Noting that the $\mathbf{r}$ -dependence of the aforementioned equation is contained in the following:

$$\mathbf{z} = \mathbf{r} - \mathbf{r}', \quad \text{where we have} \quad \nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\mathbf{z}}{z^2} \right) \rho(\mathbf{r}') d\tau'$$

This is precisely the divergence we calculated in the previous chapter regarding the divergence anomaly.

$$\left(\frac{\mathbf{z}}{z^2} \right) = 4\pi\delta^2(\mathbf{z})$$

Using this relation we can arrive at the following:

$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int 4\pi\delta^3(\mathbf{r} - \mathbf{r}') \rho(\mathbf{r}') d\tau' = \frac{1}{\epsilon_0} \rho(\mathbf{r})$$

which is Gauss's law in differential form given in Eq. (3.8). To recover the integral form in Eq. (3.7), we run the previous argument **in reverse**, integrate over a volume and apply the divergence theorem:

$$\int_V \nabla \cdot \mathbf{E} d\tau = \oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} \int_V \rho d\tau = \frac{1}{\epsilon_0} Q_{\text{enc}}. \quad (3.9)$$

3.2.3 Applications of Gauss's Law

Let us look at some of the use cases of Gauss's law, in integral form. When the problem allows symmetry, it allows us the **quickest** and **easiest** way of computing electric fields. Let's understand this statement with a series of examples.

Exercise 3.2 A Gaussian Sphere

Find the field outside a uniformly charged solid sphere of radius R and total charge q .

SOLUTION Imagine a spherical surface at radius $r > R$, shown below. This is called a Gaussian surface.

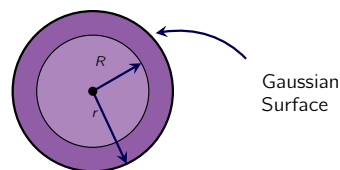


Figure 3.6: An imaginary Gaussian sphere.

Gauss's law says that:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}},$$

For this case it is $Q_{\text{enc}} = q$. At first glance this doesn't seem to get us very far, as the quantity we want ($\mathbf{E}$) is buried inside the surface integral. Luckily, symmetry allows us to extract $\mathbf{E}$ from under the integral sign: $\mathbf{E}$ certainly points radially outward,⁵ as does $d\mathbf{a}$, so we can drop the dot product,

$$\int_S \mathbf{E} \cdot d\mathbf{a} = \int_S |\mathbf{E}| da$$

and the magnitude of $\mathbf{E}$ is constant over the Gaussian surface, so it comes outside the integral

$$\int_S |\mathbf{E}| da = |\mathbf{E}| \int_S da = |\mathbf{E}| 4\pi r^2$$

Therefore:

$$|\mathbf{E}| 4\pi r^2 = \frac{1}{\epsilon_0} q \quad \text{or} \quad \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \quad \blacksquare$$

Gauss's law is always **true**, however it may **NOT** always be useful. If ρ had not been uniform,⁶ or if we had chosen some other shape for our Gaussian surface, it would still have been true that the flux of $\mathbf{E}$ is q/ϵ_0 , but $\mathbf{E}$ would not have pointed in the same direction as $d\mathbf{a}$, and its magnitude would not have been constant over the surface, and without that we cannot get $|\mathbf{E}|$ outside of the integral.

⁶or spherically symmetrical.

Symmetry is crucial to this application of Gauss's law.

There are three (3) kinds of symmetry that work:

Spherical it is a concentric sphere,

Cylindrical it is a coaxial cylinder,

Plane A pillbox which straddles the surface.

While cylindrical and plane symmetries technically require infinitely long cylinders, and planes are assumed to extend to infinity, we shall often use them to get approximate answers for "long" cylinders or "large" planes, at points far from the edges.

Although the direct use of Gauss's law to compute electric fields is limited to cases of spherical, cylindrical, and planar symmetry, we can put together **combinations** of objects possessing such symmetry, even though the arrangement as a whole is not symmetrical. For example, invoking the principle of superposition, we could find the field in the vicinity of two uniformly charged parallel cylinders, or a sphere near an infinite charged plane.

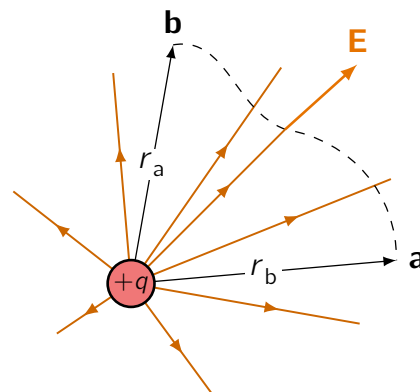


Figure 3.7

3.2.4 Curl of $\mathbf{E}$

It is time to calculate the curl of $\mathbf{E}$, as we did the divergence. As usual, to derive this behaviour, we shall



use the simplest case:

a point charge at the origin.

In this case we write down:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$$

To get a feel of the curve of $\mathbf{E}$, let us calculate the line integral of this field from some point $\mathbf{a}$ to some other point $\mathbf{b}$:

$$\int_a^b \mathbf{E} \cdot d\mathbf{l}$$

As we are working in spherical coordinates, $(dr) \hat{\mathbf{r}} + (r d\theta) \hat{\boldsymbol{\theta}} + (r \sin \theta d\phi) \hat{\boldsymbol{\phi}}$, so

$$\mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr$$

Therefore

$$\int_a^b \mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \int_a^b \frac{q}{r^2} dr = -\frac{1}{4\pi\epsilon_0} \frac{q}{r} \Big|_{r_a}^{r_b} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_a} - \frac{q}{r_b} \right)$$

where r_a is the distance from the origin to the point $\mathbf{a}$ (i.e., $r_a = |\mathbf{a} - \mathcal{O}|$) and r_b is the distance to $\mathbf{b}$ (i.e., $r_b = |\mathbf{b} - \mathcal{O}|$). The integral around a closed path is evidently zero (for then $r_a = r_b$):

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0 \quad (3.10)$$

And through the use of **Stokes' Theorem** one could arrive its differential form:

$$\nabla \times \mathbf{E} = 0 \quad (3.11)$$

Now, we proved Eq. (3.10) and Eq. (3.11) only for the field of a single point charge at the origin, but these results make no reference to what is, a perfectly arbitrary choice of coordinates.

They hold no matter where the charge is located. Moreover, if we have many charges, the principle of superposition states that the total field is a vector sum of their individual fields:

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \dots$$

which means

$$\nabla \times \mathbf{E} = \nabla \times (\mathbf{E}_1 + \mathbf{E}_2 + \dots)$$

Therefore, Eq. (3.10) and Eq. (3.11) hold for any static charge distribution.



3.3 Electric Potential

Now we look at another important concept in electric fields which is generally used as a mathematical concept but has significant use nonetheless.

3.3.1 Defining the Potential

The electric field $\mathbf{E}$ is a **special kind** of vector function, one whose curl is zero.

For example, $\mathbf{E} = y \hat{\mathbf{x}}$, could **NOT** possibly be an electrostatic field as **NO** set of charges, regardless of their sizes and positions, could ever produce such a field.

We're going to exploit this special property of electric fields to reduce a **vector** problem (finding $\mathbf{E}$) to a **scalar** problem. We know from the previous chapter that any vector whose curl is zero is equal to the gradient of some scalar. What we are going to do now amounts to a proof in the context of electrostatics.

As $\nabla \times \mathbf{E} = 0$, the line integral of $\mathbf{E}$ around any closed loop is zero.⁷ As $\oint \mathbf{E} \cdot d\mathbf{l} = 0$, the line integral of $\mathbf{E}$ from point $\mathbf{a}$ to point $\mathbf{b}$ is the same for all paths.

⁷This is a consequence of the Stokes' Theorem.

Because the line integral is independent of path, we can define a function

$$V(\mathbf{r}) \equiv - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} \quad (3.12)$$

Here $\mathcal{O}$ is some standard reference point on which we have agreed beforehand. V then depends only on the point $\mathbf{r}$. We will call this the **electric potential**.

The potential difference between two (2) points $\mathbf{a}$ and $\mathbf{b}$ is

$$V(\mathbf{b}) - V(\mathbf{a}) = - \int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} + \int_{\mathcal{O}}^{\mathbf{a}} \mathbf{E} \cdot d\mathbf{l} \quad (3.13)$$

$$= - \int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} - \int_{\mathbf{a}}^{\mathcal{O}} \mathbf{E} \cdot d\mathbf{l} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}. \quad (3.14)$$

Now, the fundamental theorem for gradients states the following:

$$V(\mathbf{b}) - V(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} (\nabla V) \cdot d\mathbf{l}, \quad \text{and} \quad \int_{\mathbf{a}}^{\mathbf{b}} (\nabla V) \cdot d\mathbf{l} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}.$$

Since, finally, this is true for any points $\mathbf{a}$ and $\mathbf{b}$, the integrands must be equal:

$$\mathbf{E} = -\nabla V \quad (3.15)$$



Eq. (3.15) is the differential version of Eq. (3.12) which says that the electric field is the gradient of a scalar potential, which is what we set out to prove.

Notice the subtle but crucial role played by path independence⁸ in this argument. If the line integral of $\mathbf{E}$ depended on the path taken, then the “definition” of V , Eq. (3.12), would **NOT** be useful as it simply would **NOT** define a function, since changing the path would alter the value of $V(\mathbf{r})$.

⁸or, equivalently, the fact that $\nabla \times \mathbf{E} = 0$

The minus sign in $\mathbf{E} = -\nabla V$ is there due to historical conventions.

Information: An Important Distinction

It is easy to be confused about the term **electrical potential** and a **voltage source** we encounter when we analyse a normal circuit. However there is a significant difference between them.

Electrical potential is the energy per unit charge gained or lost when a charge is moved from some reference point at which the potential is defined to be zero. **Voltage**, on the other hand, is the difference in potential between two (2) arbitrary points at which the potential is **NOT** necessarily zero.

For example, the potential at point **A** relative to a defined reference point⁹ might be 100 V, and the potential at point **B** might be 10 V volts. Then the voltage between **B** and **A** is 90 V volts.

⁹sometimes an infinite distance away

3.3.2 Some Digressions on Potential

Naming

The naming of V as **potential** is unfortunate as it inevitably reminds us of potential *energy*. This is particularly problematic, as there is a connection between **potential** and **potential energy**. Incidentally, a surface over which the potential is constant is called an **equipotential**.

Advantages of the Potential

If we know V , we can easily get $\mathbf{E}$. It's simple, just take the gradient: $\mathbf{E} = -\nabla V$. This is due to the three (3) components of $\mathbf{E}$ not being independent. In fact, they are **explicitly** interrelated by the very condition we started with, $\nabla \times \mathbf{E} = 0$.

In terms of components,

$$\frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}, \quad \frac{\partial E_z}{\partial y} = \frac{\partial E_y}{\partial z}, \quad \frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}.$$

The Reference Point

There is a **necessary** ambiguity in the definition of potential, as the choice of reference point $\mathcal{O}$ was arbitrary. Changing reference points amounts to adding a constant K to the potential:

$$V'(r) = - \int_{\mathcal{O}}^r \mathbf{E} \cdot d\mathbf{l} = - \int_{\mathcal{O}'}^{\mathcal{O}} \mathbf{E} \cdot d\mathbf{l} - \int_{\mathcal{O}}^r \mathbf{E} \cdot d\mathbf{l} = K + V(r)$$

where K is the line integral of $\mathbf{E}$ from the old reference point $\mathcal{O}$ to the new one $\mathcal{O}'$. Of course, adding a constant to V will **NOT** affect the potential **difference** between two (2) points:

$$V'(\mathbf{b}) - V'(\mathbf{a}) = V(\mathbf{b}) - V(\mathbf{a}),$$

since the K 's cancel out.¹⁰ Nor does the ambiguity affect the gradient of V :

$$\nabla V' = \nabla V,$$

since the derivative of a constant is zero (0). That's why all such V 's, differing only in their choice of reference point, correspond to the same field $\mathbf{E}$.

Potential as such carries no real physical significance, for at any given point we can adjust its value at will by a suitable relocation of $\mathcal{O}$.

¹⁰Actually, it was already clear from Eq. (3.13) that the potential difference is independent of $\mathcal{O}$, as it can be written as the line integral of $\mathbf{E}$ from $\mathbf{a}$ to $\mathbf{b}$, with no reference to $\mathcal{O}$.

However, therefore V 's a "reference level to use for $\mathcal{O}$ in electrostatics—almost-one-like equations to sea level for altitude—and that is a point infinitely far from the charge. Ordinarily, then, we "set the zero of potential at infinity." (Since $V(\mathcal{O}) = 0$, choosing a reference point is equivalent to selecting a place where V is to be zero.) But I must warn you that there is one special circumstance in which this convention fails: when the charge distribution itself extends to infinity. The symptom of trouble, in such cases, is that the potential blows up. For instance, the field of a uniformly charged plane is $(\sigma/2\epsilon_0)$

$$V(z) = - \int_{\infty}^z \frac{1}{2\epsilon_0} \sigma dz = - \frac{1}{2\epsilon_0} \sigma (z - \infty).$$

The remedy is simply to choose some other reference point (in this example you might use a point on the plane). Notice that the difficulty occurs only in textbook problems; in "real life" there is no such thing as a charge distribution that goes on forever, and we can always use infinity as our reference point.

Superposition Principle

The original superposition principle pertains to the force on a test charge (Q). It says that the total force on Q is the **vector sum** of the forces attributable to the source charges individually:

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 \dots$$

Dividing through by Q , we see that the electric field, too, obeys the superposition principle:

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \dots$$

Integrating from the common reference point to $\mathbf{r}$, it follows that the potential also satisfies such a principle:

$$V = V_1 + V_2 + \dots$$

That is, the potential at any given point is the sum of the potentials due to all the source charges separately. Only this time it is an ordinary sum, not a vector sum, which makes it a lot easier to work with.

Units of Potential

For our system of units, SI, force is measured in newtons (N) and charge in coulombs (C), so electric fields are in newtons per coulomb ($\text{N} \cdot \text{C}^{-1}$).

Accordingly, potential is known-meters per coulomb ($\text{m} \cdot \text{C}^{-1}$), or joules per coulomb ($\text{J} \cdot \text{C}^{-1}$).

A joule per coulomb is a volt.

3.3.3 Poisson's Equation and Laplace's Equation

We now know the electric field ($\mathbf{E}$) can be written as the **gradient of a scalar potential**:

$$\mathbf{E} = -\nabla V$$

The question arises:

How are divergence and curl of $\mathbf{E}$ related to our new definition of electric potential, V ?

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \text{and} \quad \nabla \times \mathbf{E} = 0$$

Well, we can write the aforementioned equations in the following form:

$$\nabla \cdot \mathbf{E} = \nabla \cdot (-\nabla V) = -\nabla^2 V$$

so, apart from that persistent minus sign, the divergence of $\mathbf{E}$ is the **Laplacian** of V . Gauss's law, then, says:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad (3.16)$$

This is known as **Poisson's equation**. In regions where there is no charge, so $\rho = 0$, Poisson's equation reduces to **Laplace's equation**:

$$\nabla^2 V = 0$$

We'll explore this in Chapter 3 when we work with on **Advanced Mathematical Methods**.

What about the curl? Let's do the math and write the following form:

$$\nabla \times \mathbf{E} = \nabla \times (-\nabla V) = 0$$

But that's no condition on V as curl of gradient is always zero.

Of course, here, we used curl to show $\mathbf{E}$ could be expressed as the gradient of a scalar, so it's not really surprising that this works out. It takes only one (1) Partial Differential Equation (PDE) (Poisson's) to determine V , as V is a scalar; for $\mathbf{E}$ we needed two (2), the divergence and the curl.¹¹

¹¹This comes from *Helmholtz theorem*, which we discussed previously.

3.3.4 The Potential of a Localised Charge Distribution

Previously, we have defined V in terms of $\mathbf{E}$ as shown in Eq. (3.12).

Ordinarily, though, it's $\mathbf{E}$ that we're looking for.¹² The idea is that it might be easier to get V first, and then calculate $\mathbf{E}$ by taking the gradient. Typically, then, we know where the charge is,¹³ and we want to find V .

¹²if we already knew $\mathbf{E}$, there wouldn't be much point in calculating V

¹³that is, we know where the charge density (ρ) is.

Now, Poisson's equation relates V and ρ , but unfortunately it's "the wrong way around":

it would give us ρ , if we knew V , whereas we want V , knowing ρ .

What we need to do is **invert** Poisson's equation. That's the idea for this section, although we shall do it by roundabout means, beginning, as always, with a point charge at the origin.

The electric field is:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{r^2} \right) \hat{\mathbf{r}},$$

so

$$\mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr \quad \text{where} \quad d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}.$$

Setting the reference point at a place in infinity,¹⁴ the potential of a point charge q at the origin is

$$V(r) = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

¹⁴We will see here the advantage of using infinity for the reference point as it kills the lower limit on the integral.

Please observe the sign of V ; presumably the conventional minus sign in the definition Eq. (3.12) was chosen to make the potential of a positive charge come out positive.

Information: Think Topographically

It is useful to remember that regions of positive charge are potential "hills", regions of negative charge are potential "valleys", and the electric field points "downhill", from plus toward minus.



In general, the potential of a point charge q is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{z}, \quad (3.17)$$

where z , as always, is the distance from q to $\mathbf{r}$. Invoking the superposition principle, then, the potential of a collection of charges is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{z_i}, \quad (3.18)$$

or, for a continuous distribution,

$$\oint_V V[\mathbf{r}] = \frac{1}{4\pi\epsilon_0} \int \frac{1}{z} dq. \quad (3.19)$$

In particular, for a volume charge, it's

$$\oint_V V[\mathbf{r}] = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z} d\tau'. \quad (3.20)$$

This is the equation we were looking for, telling us how to compute V when we know ρ . It is useful to think of this equation as the “solution” to Poisson’s equation, for a localized charge distribution.

Let’s compare Eq. (3.20) with the corresponding formula for the electric field in terms of ρ Eq. (3.5):

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z^2} \hat{\mathbf{z}} d\tau'$$

The main point to notice is that the unit vector ($\hat{\mathbf{z}}$) is gone, so there is no need to fuss with components. The potentials of line and surface charges are

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{z} dl' \quad \text{and} \quad V = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{z} da' \quad \text{and} \quad (3.21)$$

It is worth stressing that everything in this section is predicated on the assumption that the reference point is at **infinity**. This is hardly apparent in Eq. (3.20), but remember that we got that equation from the potential of a point charge at the origin, $(1/4\pi\epsilon_0)(q/r)$, which is valid only when $\mathcal{O} = \infty$.

If we try to apply these formulas to one of those artificial problems in which the charge itself extends to infinity, the integral will diverge.

3.3.5 Boundary Conditions

In the typical electrostatic problem we are given a source charge distribution (ρ), and we want to find the electric field ($\mathbf{E}$) it produces.

Unless the symmetry of the problem allows a solution by Gauss’s law, it is generally to our advantage to calculate the potential first, as an intermediate step.

These are the three fundamental quantities of electrostatics:

ρ , $\mathbf{E}$, and V .

We have, in the course of our discussion, derived all six (6) formulas interrelating them. These equations are neatly summarized in **Fig. 3.8**. We began with just two (2) experimental observations:

1. The principle of superposition—a broad general rule applying to **all** electromagnetic forces, and
2. Coulomb's law—the fundamental law of electrostatics.

From these, all else are derived. You may have noticed that the electric field always undergoes a discontinuity when you cross a surface charge σ . In fact, it is a simple matter to find the **amount** by which $\mathbf{E}$ changes at such a boundary. Suppose we draw a wafer-thin Gaussian pillbox, extending just barely over the edge in each direction as seen in **Fig. 3.9**. Gauss's law says that

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \sigma A,$$

where A is the area of the pillbox lid (m^2).¹⁵ Now, the sides of the pillbox contribute **nothing** to the flux, in the limit as the thickness ϵ goes to zero, so we are left with:

$$E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \frac{1}{\epsilon_0} \sigma, \quad (3.22)$$

where $E_{\text{above}}^{\perp}$ denotes the component of $\mathbf{E}$ **perpendicular** to the surface immediately above, and $E_{\text{below}}^{\perp}$ is the same, only just below the surface. For consistency, we let “upward” be the positive direction for both.

Based on this model we can say normal component of $\mathbf{E}$ (for clarity it is, $\mathbf{E} \cdot \hat{n}$) is discontinuous by an amount σ/ϵ_0 at any boundary. In particular, where there is **no** surface charge, $E^{\perp}$ is continuous, as for instance at the surface of a uniformly charged solid sphere. The tangential component of $\mathbf{E}$, by contrast, is always continuous.

For if we apply:

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

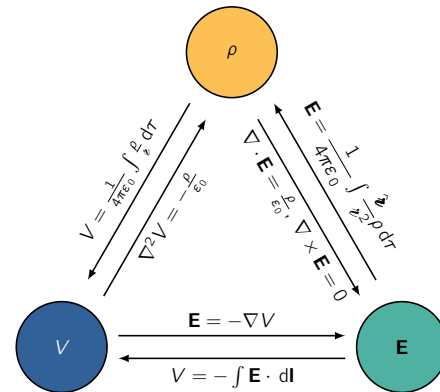


Figure 3.8

The relation between three (3) major variables used in the study of electrostatics.

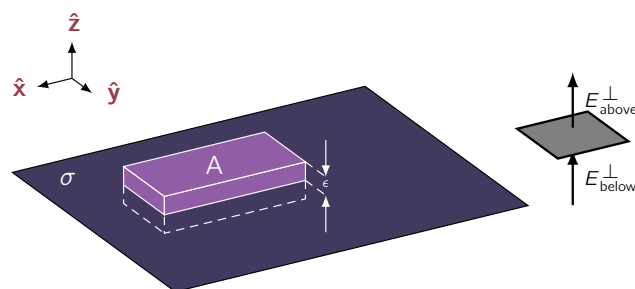


Figure 3.9

A thought experiment on how to figure out the discrepancy when a boundary is changed.



to the thin rectangular loop of Fig. 2.37, the ends give nothing,¹⁶ and the sides give $(E_{\text{above}}^{\parallel} l - E_{\text{below}}^{\parallel} l)$,¹⁶ as $\epsilon \rightarrow 0$.
so

$$E_{\text{above}}^{\parallel} = E_{\text{below}}^{\parallel} \quad (3.23)$$

where $E^{\parallel}$ stands for the components of E parallel to the surface. The boundary conditions on E (Eqs. 2.31 and 2.32) can be combined into a single formula:

$$E_{\text{above}} - E_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{n}, \quad (3.24)$$

where $\hat{n}$ is a unit vector perpendicular to the surface, pointing from "below" to "above."⁸ The potential, meanwhile, is continuous across any boundary (Fig. 2.38), since $V_{\text{above}} - V_{\text{below}} = -\int_0^b E \cdot d\mathbf{l}$; as the path length shrinks to zero, so too does the integral:

$$V_{\text{above}} = V_{\text{below}}. \quad (3.25)$$

However, the **gradient** of V inherits the discontinuity in E given and following this $E = -\nabla V$, which Eq. (3.24) implies that:

$$\nabla V_{\text{above}} - \nabla V_{\text{below}} = -\frac{1}{\epsilon_0} \sigma \hat{n}, \quad (3.26)$$

or, more conveniently,

$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\epsilon_0} \sigma, \quad (3.27)$$

where

$$\frac{\partial V}{\partial n} = \nabla V \cdot \hat{n} \quad (3.28)$$

denotes the **normal derivative** of V .¹⁷ Please note that these boundary conditions relate the fields and potentials just above and **just** below the surface. For example, the derivatives in Eq. (3.27) are the **limiting** values as we approach the surface from either side.¹⁸

¹⁷That is, the rate of change in the direction perpendicular to the surface

¹⁸Observe that it doesn't matter which side you call **above** and which **below**, given the reversal would switch the direction of $\hat{n}$.

Incidentally, if we're only interested in the field due to the essentially flat local patch of surface charge itself, the answer is $(\sigma/2\epsilon_0) \hat{n}$ immediately above the surface, and $-(\sigma/2\epsilon_0) \hat{n}$ immediately below. Evidently the entire discontinuity in E is attributable to this local patch of surface charge.



3.4 Work and Energy

3.4.1 To Move a Charge

To understand how to define work, Let us assume that we have a stationary configuration of source charges, and we want to move a test charge (Q) from point $\mathbf{a}$ to point $\mathbf{b}$. Here we need to ask the following question:

How much work will we have to do?

At any point along the path, the electric force on Q is $\mathbf{F} = Q\mathbf{E}$; the force we must exert, in opposition to this electrical force, is $-\mathbf{F}$. The work we do is therefore:

$$W = \int_0^b \mathbf{F} \cdot d\mathbf{l} = -Q \int_0^b \mathbf{E} \cdot d\mathbf{l} = Q [V(\mathbf{b}) - V(\mathbf{a})] .$$

Please observe the answer is **independent of the path** we take from $\mathbf{a}$ to $\mathbf{b}$;

In mechanics, we call the electrostatic force “conservative”.¹⁹

Dividing through by Q , we have:

$$V(\mathbf{b}) - V(\mathbf{a}) = \frac{W}{Q} . \quad (3.29)$$

In other words, the potential difference between points $\mathbf{a}$ and $\mathbf{b}$ is equal to the work per unit charge required to carry a particle from $\mathbf{a}$ to $\mathbf{b}$. In particular, if we want to bring Q in from far away and stick it at point $\mathbf{r}$, the work we must do is

$$W = Q [V(\mathbf{r}) - V(\infty)] ,$$

so, if we have set the reference point at infinity,

$$W = QV(\mathbf{r}) \quad (3.30)$$

Here, **potential** is potential **energy**, the work it takes to create the system, per unit charge, just as the field is the force per unit charge.

3.4.2 The Energy of a Point Charge Distribution

How much work would it take to assemble an entire **collection** of point charges? Imagine bringing in the charges, one by one, from far away (Fig. 2.40). The first charge, q_1 , takes no work, since there is no field yet to fight against. Now bring in q_2 . According to Eq. (3.30), this will cost you $q_2 V_1(\mathbf{r}_2)$, where V_1 is the potential due to q_1 , and $\mathbf{r}_2$ is the place we're putting q_2 :

$$W = \frac{1}{4\pi\epsilon_0} q_2 \left(\frac{q_1}{r_{12}} \right)$$

¹⁹A force with the property which the total work done by the force in moving a particle between two (2) points is independent of the path taken. Equivalently, if a particle travels in a closed loop, the total work done by a conservative force is zero.



(λ_{12}) is the distance between q_1 and q_2 once they are in position). As you bring in each charge, nail it down in its final location, so it doesn't move when you bring in the next charge. Now bring in q_3 ; this requires work $q_3 V_{1,2}(\mathbf{r}_3)$, where $V_{1,2}$ is the potential due to charges q_1 and q_2 , namely, $(1/4\pi\epsilon_0)(q_1/\lambda_{13} + q_2/\lambda_{23})$. Thus

$$W_3 = \frac{1}{4\pi\epsilon_0} q_3 \left(\frac{q_1}{\lambda_{13}} + \frac{q_2}{\lambda_{23}} \right).$$

Similarly, the extra work to bring in q_4 will be

$$W_4 = \frac{1}{4\pi\epsilon_0} q_4 \left(\frac{q_1}{\lambda_{14}} + \frac{q_2}{\lambda_{24}} + \frac{q_3}{\lambda_{34}} \right).$$

The total work necessary to assemble the first four charges, then, is

$$W = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{\lambda_{12}} + \frac{q_1 q_3}{\lambda_{13}} + \frac{q_1 q_4}{\lambda_{14}} + \frac{q_2 q_3}{\lambda_{23}} + \frac{q_2 q_4}{\lambda_{24}} + \frac{q_3 q_4}{\lambda_{34}} \right).$$

You see the general rule: Take the product of each pair of charges, divide by their separation distance, and add it all up:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{\lambda_{ij}}. \quad (3.31)$$

The stipulation $j > i$ is to remind you not to count the same pair twice. A nicer way to accomplish this is **intentionally** to count each pair twice, and then divide by 2:

$$W = \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{j=i}^n \frac{q_i q_j}{\lambda_{ij}} \quad (3.32)$$

(we must still avoid $i = j$, of course). Notice that in this form the answer plainly does not depend on the **order** in which you assemble the charges, since every pair occurs in the sum. Finally, let's pull out the factor q_i :

$$W = \frac{1}{2} \sum_{i=1}^n q_i \left(\sum_{j \neq i}^n \frac{1}{4\pi\epsilon_0 \lambda_{ij}} q_j \right).$$

The term in parentheses is the potential at point $\mathbf{r}_i$ (the position of q_i) due to all the **other** charges—all of them, now, not just the ones that were present at some stage during the assembly. Thus,

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i). \quad (3.33)$$

That's how much work it takes to assemble a configuration of point charges; it's also the amount of work you'd get back if you dismantled the system. In the meantime, it represents energy stored in the configuration ("potential" energy, if you insist, though for obvious reasons I prefer to avoid that word in this context).

3.4.3 The Energy of a Continuous Charge Distribution

For a volume charge density (ρ), Eq. (3.33) becomes.²⁰

$$W = \frac{1}{2} \int \rho V d\tau \quad (3.34)$$

²⁰The corresponding integrals for line and surface charges would be $\int \lambda V dl$ and $\int \sigma V da$.



There is a tidy way to rewrite this result, in which ρ and V are eliminated in favor of $\mathbf{E}$. First use Gauss's law to express ρ in terms of $\mathbf{E}$:

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E} \quad \text{so} \quad W = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) V \, d\tau$$

Now use integration by parts, to transfer the derivative from $\mathbf{E}$ to V :

$$W = \frac{\epsilon_0}{2} \left[- \int \mathbf{E} \cdot (\nabla V) \, d\tau + \oint V \mathbf{E} \cdot d\mathbf{a} \right].$$

But we know $\nabla V = -\mathbf{E}$, so

$$W = \frac{\epsilon_0}{2} \left(\int_V E^2 \, d\tau + \oint_S V \mathbf{E} \cdot d\mathbf{a} \right) \quad (3.35)$$

But what volume is this we're integrating over? Let's go back to the formula we started with, Eq. (3.34). From its derivation, it is clear that we should integrate over the region where the charge is located. But actually, any larger volume would do just as well:

The "extra" territory we throw in will contribute nothing to the integral, since $\rho = 0$ out there.

With this in mind, we return to Eq. (3.35). What happens here, as we enlarge the volume beyond the minimum necessary to trap all the charge? Well, the integral of E^2 can only increase;²¹ evidently the surface integral must decrease correspondingly to leave the sum intact.²²

²¹the integrand being positive.

It is now worth stressing out that Eq. (3.35) gives us the correct energy W , whatever volume we use;²³ but the contribution from the volume integral goes up, and that of the surface integral goes down, as you take larger and larger volumes.

²²In fact, at large distances from the charge, $\mathbf{E}$ goes $1/r^2$ and V $1/r$, while the surface area grows like r^2 ; roughly speaking, then, the surface integral goes down like $1/r$.

In particular, why not integrate over all space?

Then the surface integral goes to zero, and we are left with:

$$W = \frac{\epsilon_0}{2} \int E^2 \, d\tau \quad (3.36)$$

²³as long as it encloses all the charge

3.4.4 Comments on Electrostatic Energy

An Interesting Inconsistency

We can see Eq. (3.36) implies the energy of a stationary charge distribution is always **positive**. On the other hand, Eq. (3.33) which we derived Eq. (3.36), can be either **positive** or **negative**.

For instance, according to Eq. (3.33), the energy of two (2) equal but **opposite** charges a distance z apart is $-(1/4\pi\epsilon_0)(q^2/z)$.

What's gone wrong? Which equation is correct?



The answer is that **both** are correct, but they speak to slightly different questions. Eq. (3.33) does **NOT** take into account the work necessary to make the point charges in the first place; we **started** with point charges and simply found the work required to bring them together. This is wise strategy, given Eq. (3.36) indicates that the energy of a point charge is in fact **infinite**:

$$W = \frac{\epsilon_0}{2(4\pi\epsilon_0)^2} \int \left(\frac{q^2}{r^4} \right) (r^2 \sin \theta dr d\theta d\phi) = \frac{q^2}{8\pi\epsilon_0} \int_0^\infty \frac{1}{r^2} dr = \infty.$$

Eq. (3.36) is more **complete**, in the sense that it tells you the **total** energy stored in a charge configuration, but Eq. (3.33) is more appropriate when we're dealing with point charges, because we prefer to leave out that portion of the total energy that is attributable to the fabrication of the point charges themselves.

In practice, after all, the point charges are given to us ready-made; all we do is move them around. Since we did not put them together, and we cannot take them apart, it is immaterial how much work the process would involve.²⁴

Now, it is a valid question to ask when the inconsistency crept into an apparently watertight derivation. The "flaw" lies between Eq. (3.33) and Eq. (3.34):

In the former, $V(\mathbf{r}_i)$ represents the potential due to all the **other** charges **but not** q_i , whereas in the latter, $V(\mathbf{r})$ is the **full** potential. For a **continuous** distribution, there is no distinction, given the amount of charge **right at the point** $\mathbf{r}$ is vanishingly small, and its contribution to the potential is zero.

²⁴Still, the infinite energy of a point charge is a recurring source of embarrassment for electromagnetic theory, afflicting the quantum version as well as the classical.

In the presence of point charges we should better stick with Eq. (3.33).

The Place where Energy is Stored

Eq. (3.34) and Eq. (3.36) offer two (2) different ways of calculating the same thing.

1. The first is an integral over the charge distribution;
2. The second is an integral over the field.

These can involve completely different regions. For instance, in the case of the spherical shell, the charge is confined to the surface, whereas the electric field is everywhere **outside** this surface.

Where is the energy, then?

Is it stored in the field, as Eq. (3.36) seems to suggest, or is it stored in the charge, as Eq. (3.34) implies? At the present stage this is simply an unanswerable question: I can tell you what the total energy is, and I can provide you with several different ways to compute it, but it is impertinent to worry about where the energy is located. In the context of radiation theory it is useful to regard the energy as stored in the field, with a density:

$$\frac{\epsilon_0}{2} E^2 = \text{energy per unit volume.}$$

But in electrostatics one could just as well say it is stored in the charge, with a density $\frac{1}{2}\rho V$.

The difference is purely a matter of bookkeeping.

Principle of Superposition

Because electrostatic energy is quadratic in the fields, it does **NOT** obey a superposition principle. The energy of a compound system is **NOT** the sum of the energies of its parts considered separately as there are also “cross terms” to take into account:

$$W_{\text{tot}} = \frac{\epsilon_0}{2} \int E^2 d\tau' = \frac{\epsilon_0}{2} \int (\mathbf{E}_1 + \mathbf{E}_2)^2 d\tau' \quad (3.37)$$

$$= \frac{\epsilon_0}{2} \int (E_1^2 + E_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2) d\tau \quad (3.38)$$

$$= W_1 + W_2 + \epsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 d\tau. \quad (3.39)$$

For example, if you double the charge everywhere, you **quadruple** the total energy.

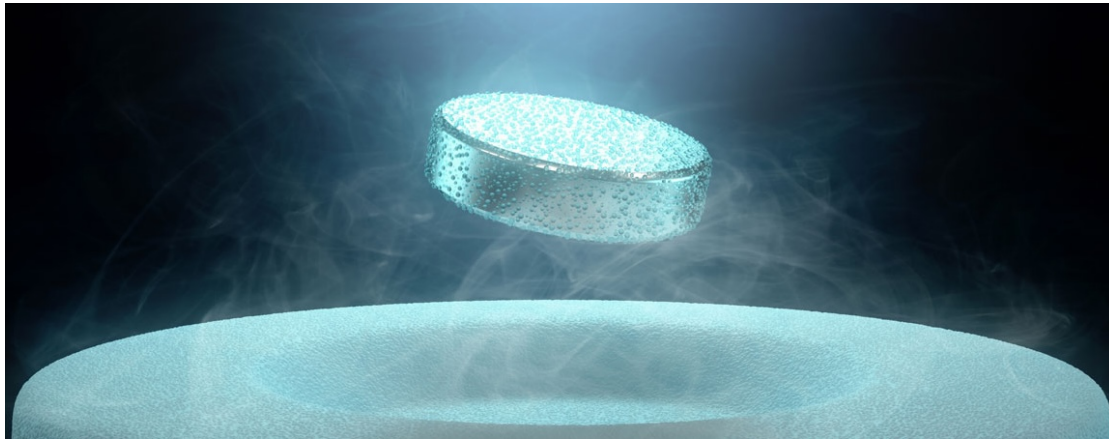


Figure 3.10

Some may say that superconductors can be considered as “perfect conductors”, however, they exhibit quantum effects such as the Meissner effect and quantization of magnetic flux. In perfect conductors, the interior magnetic field must remain fixed but can have a zero or nonzero value [1]. In real superconductors, all magnetic flux is expelled during the phase transition to superconductivity (the Meissner effect), and the magnetic field is always zero within the bulk of the superconductor [2].

3.5 Conductors

3.5.1 Properties

In an insulator, for example glass or rubber, each electron is on a short leash, as it is attached to a particular atom. In a metallic **conductor**, by contrast, one or more electrons per atom are free to roam.²⁵ A **perfect** conductor would contain an **unlimited** supply of free charges.

In real life there are no such things perfect conductors, but metals come pretty close, for most purposes and therefore we can assume them to be perfect.

²⁵In liquid conductors such as salt water, it is ions that do the moving. However, for now we shall only focus on solid materials.

From this aforementioned definition, the basic electrostatic properties of ideal conductors immediately follow:

i. No Electric Field Inside a Conductor

Let’s have a think about this statement. If there **were** any field, those free charges would **move**, and it wouldn’t be electrostatics any more. To give a more concrete explanation, we better examine what happens when we put a conductor into an external electric field $\mathbf{E}_0$, seen in **Fig. 3.12**.

Initially, the field will drive any free positive charges to the right, and negative ones to the left.²⁶ When they come to the edge of the material, the charges pile up: plus on the right side, minus on the left. Now, these induced charges produce a field of their own ($\mathbf{E}_1$), which, as we can see from the figure, is in the **opposite direction** to $\mathbf{E}_0$.

That’s the crucial point, for it means the field of the induced charges tends to cancel the original field. Charge will continue to flow until this cancellation is complete, and the resultant field inside the conductor is precisely zero.²⁷ The whole process is practically instantaneous.

²⁶In practice, it’s the electrons that do the moving, but when they depart, the right side is left with a net positive charge, so it doesn’t really matter which charges move; the effect is the same.

²⁷Outside the conductor the field is not zero, for here $\mathbf{E}_0$ and $\mathbf{E}_1$ do **NOT** tend to cancel.

ii. Charge Density is Zero inside a Conductor

This follows from Gauss's law:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

If $\mathbf{E}$ is zero, then by extension, so is ρ . There is still charge around, but exactly as much plus as minus, so the **net** charge density in the interior is zero.

iii. Any Net Charge is on the Surface

That's the only place left.

iv. A Conductor is an Equipotential

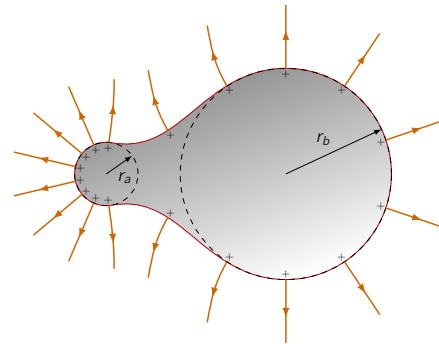
For if $\mathbf{a}$ and $\mathbf{b}$ are any two (2) points within (or at the surface of) a given conductor,

$$V(\mathbf{b}) - V(\mathbf{a}) = - \int_a^b \mathbf{E} \cdot d\mathbf{l} = 0, \quad \text{therefore} \quad V(\mathbf{b}) = V(\mathbf{a}).$$

v. $\mathbf{E}$ is Perpendicular to the Surface

Otherwise, charge will immediately flow around the surface until it kills off the tangential component which are shown in **Fig. 3.11** ²⁸.

It is an astonishing fact the charge on a conductor flows to the surface. Given their mutual repulsion, the charges naturally **spread out** as much as possible, but for all of them to go to the surface seems like a waste of the interior space. Surely we could do better, from the point of view of making each charge as far as possible from its neighbors, to sprinkle **some** of them throughout the volume. Well, it simply is not so. You do best to put all the charge on the surface, and this is true regardless of the size or shape of the conductor.



²⁸Perpendicular to the surface, charge cannot flow, of course, since it is confined to the conducting object.

Figure 3.11

A visual representation of treating a non-standard body and the generated electric field. Please observe that all lines flowing from the body is perpendicular.

The problem can also be phrased in terms of **energy**.

Like any other free dynamical system, the charge on a conductor will seek the configuration that minimizes its potential energy. What property (iii) asserts is that the electrostatic energy of a solid object²⁹ is a minimum when charge is spread over the surface.

For instance, the energy of a sphere is

$$(1/8\pi\epsilon_0)(q^2/R)$$

if the charge is uniformly distributed over the surface, but it is,

$$(3/20\pi\epsilon_0)(q^2/R)$$

if the charge is uniformly distributed throughout the volume

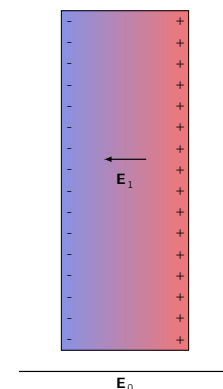


Figure 3.12

A visual representation of the electric field inside a conductor.

²⁹with specified shape and total charge

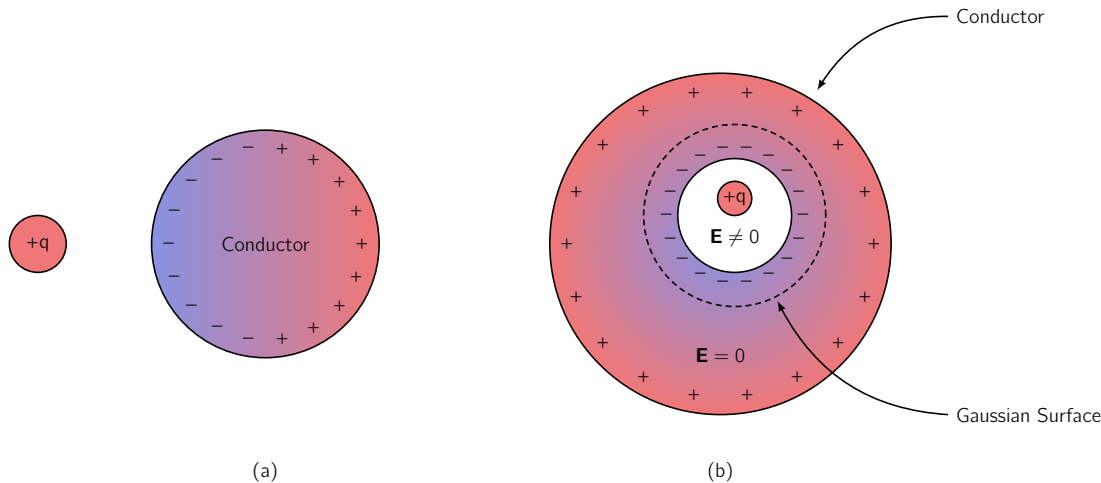


Figure 3.13

Charges induced by a point charge in a conductor (a) When a charge approaches a conductor the charges will start to move where opposite charges will move towards the point charge and same charges will move away. While there is an imbalance now, the overall charge of the conductor remains the same (b) When a charge is given inside the cavity, again similar behaviour is to be expected inside the conductor.

3.5.2 Induced Charges

If we were to hold a charge ($+q$) near an **uncharged** conductor seen in **Fig. 3.13a**, the two (2) will attract one another. The reason for this is that $+q$ will pull minus charges ($-q$) over to the near side and repel plus charges to the far side.³⁰ Since the negative induced charge is closer to q , there is a net force of attraction.

When we speak of the field, charge, or potential “inside” a conductor, we should think of the “meat” of the conductor; if there is some hollow **cavity** in the conductor, and within that cavity we put some charge, then the field in the cavity will **NOT** be zero. But in a remarkable way the cavity and its contents are electrically isolated from the outside world by the surrounding conductor, shown in **Fig. 3.13b**.

No external fields penetrate the conductor; they are cancelled at the outer surface by the induced charge there. Similarly, the field due to charges within the cavity is cancelled, for all exterior points, by the induced charge on the inner surface. However, the compensating charge left over on the outer surface of the conductor effectively “communicates” the presence of q to the outside world.

The total charge induced on the cavity wall is equal and opposite to the charge inside, for if we surround the cavity with a Gaussian surface, all points of which are in the conductor (**Fig. 3.13b**), $\oint \mathbf{E} \cdot d\mathbf{a} = 0$, and therefore with Gauss’s law, the net enclosed charge must be zero (0). But $Q_{\text{enc}} = q + q_{\text{induced}}$, so $q_{\text{induced}} = -q$. Then if the conductor as a whole is electrically neutral, there must be a charge $+q$ on its outer surface.

If a cavity surrounded by conducting material is itself empty of charge, then the field within the cavity is zero. For any field line would have to begin and end on the cavity wall, going from a plus charge to a minus charge, shown in **Fig. 3.14**.

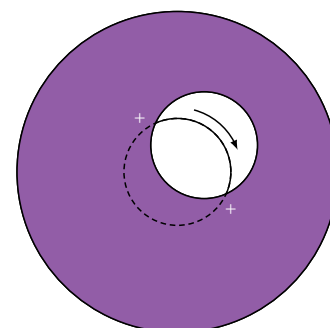


Figure 3.14

Visual description of an electric field inside a cavity. As can be seen all the lines generated by the positive side of the edge is cancelled by the negative side with it absorbs making the inside of the cavity have 0 field.

³⁰Another way to think of it is that the charge moves around in such a way as to kill off the field of q for points inside the conductor, where the total field must be zero.

Letting that field line be part of a closed loop, the rest of which is entirely inside the conductor (where $\mathbf{E} = 0$), the integral $\oint \mathbf{E} \cdot d\mathbf{l}$ is distinctly **positive**, in violation of $\oint \mathbf{E} \cdot d\mathbf{l} = 0$. It follows that $\mathbf{E} = 0$ within an **empty** cavity, and there is in fact **no** charge on the surface of the cavity.³¹

3.5.3 Surface Charge and the Force on a Conductor

Because the field inside a conductor is zero (**0**), boundary condition 2.33 requires the field immediately **outside** is:

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}, \quad (3.40)$$

consistent with our earlier conclusion that the field is **normal to the surface**. In terms of potential, Eq. (3.27) gives us:

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n} \quad (3.41)$$

These equations enable us to calculate the surface charge on a conductor, if we can determine $\mathbf{E}$ or V .³² In the presence of an electric field, a surface charge will experience a force; the force per unit area ($\mathbf{f}$), is $\sigma \mathbf{E}$.

But there's a problem here:

The electric field is **discontinuous** at a surface charge.

So what are we supposed to use: E_{above} , E_{below} , or something in between? The answer is that we should use the average of the two:

$$\mathbf{f} = \sigma \mathbf{E}_{\text{average}} = \frac{1}{2} \sigma (\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}). \quad (3.42)$$

You may ask:, why the average?

The reason is very simple, though the telling makes it sound complicated: Let's focus our attention on a tiny patch of surface surrounding the point in question (Fig. 2.50).³³ The total field consists of two parts—that attributable to the patch itself, and that due to everything else (other regions of the surface, as well as any external sources that may be present): $\mathbf{E} = \mathbf{E}_{\text{patch}} + \mathbf{E}_{\text{other}}$. Now, the patch cannot exert a force on itself, any more than you can lift yourself by standing in a basket and pulling up on the handles. The force on the patch, then, is due exclusively to $\mathbf{E}_{\text{other}}$, and this suffers no discontinuity (if we removed the patch, the field in the "hole" would be perfectly smooth). The discontinuity is due entirely to the charge on the patch, which puts out a field ($\sigma/2\epsilon_0$) on either side, pointing away from the surface. Thus,

$$\begin{aligned} \mathbf{E}_{\text{above}} &= \mathbf{E}_{\text{other}} + \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}, \\ \mathbf{E}_{\text{below}} &= \mathbf{E}_{\text{other}} - \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}. \end{aligned} \quad \text{therefore,} \quad \mathbf{E}_{\text{other}} = \frac{1}{2} (\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}) = \mathbf{E}_{\text{average}}.$$

Averaging is really just a device for removing the contribution of the patch itself. That argument applies to **any** surface charge; in the particular case of a conductor, the field is zero inside and



³¹This is why we are **relatively safe** inside a metal car during a thunderstorm—you may get cooked, if lightning strikes, but we will **NOT** be **electrocuted**. The same principle applies to the placement of sensitive apparatus inside a grounded Faraday cage, to shield out stray electric fields. In practice, the enclosure doesn't even have to be solid conductor—chicken wire will often suffice [3].

³²we shall use them frequently in the next chapter.

³³Make it small enough so it is essentially flat and the surface charge on it is essentially constant.

(σ/ϵ_0) m outside as shown in Eq. (3.40), so the average is $(\sigma/2\epsilon_0) \hat{n}$, and the force per unit area is

$$\mathbf{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{n}. \quad (3.43)$$

This amounts to an outward **electrostatic pressure** on the surface, tending to draw the conductor into the field, regardless of the sign of σ . Expressing the pressure in terms of the field just outside the surface,

$$P = \frac{\epsilon_0}{2} E^2. \quad (3.44)$$

3.5.4 Capacitors

Suppose we have two (2) conductors, and we put charge $+Q$ on one and $-Q$ on the other as shown in **Fig.** 3.15. Since V is **constant** over a conductor, we can speak unambiguously of the potential difference between them:

$$V = V_+ - V_- = - \int_-^+ \mathbf{E} \cdot d\mathbf{l}.$$

We don't know how the charge distributes itself over the two conductors, and calculating the field would be a nightmare, if their shapes are complicated, but this much we do know:

E is proportional to Q .

For $\mathbf{E}$ is given by Coulomb's law:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{\lambda^2} \hat{\mathbf{r}} d\tau, \quad [\text{V} \cdot \text{m}^{-1}]$$

so if we double ρ , we double $\mathbf{E}$.³⁴ Since $\mathbf{E}$ is proportional to Q , so also is V . The constant of proportionality is called the **capacitance** of the arrangement:

$$C \equiv \frac{Q}{V} \quad [\text{V} \cdot \text{C}^{-1} \text{ or F}] \quad (3.45)$$

Capacitance is a **purely geometrical quantity**, determined by the sizes, shapes, and separation of the two (2) conductors. In SI units, C is measured in farad (F); a farad is a coulomb-per-volt. This turns out to be inconveniently large; more practical units are the microfarad (1×10^{-6} F) and the picofarad (1×10^{-12} F).

Notice V is, by definition, the potential of the **positive** conductor less that of the negative one; likewise, Q is the charge of the **positive** conductor. Accordingly, capacitance is an intrinsically positive quantity.³⁵

Charging a Capacitor

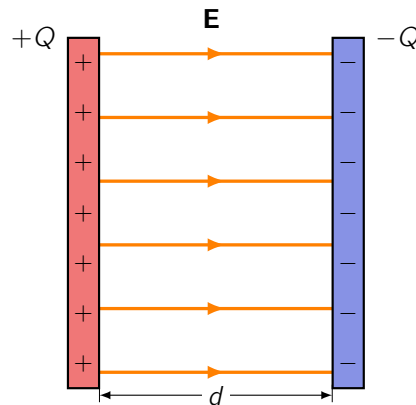


Figure 3.15

A parallel-plate capacitor, as shown in part A, consists of two flat conducting plates. Here one plate is positively charged and the other one is charged negatively. At a distance of d apart, an electric field (E) is generated between the two (2) plates.

³⁴Doubling Q doubles ρ everywhere, it doesn't shift the charge around.

³⁵We may occasionally hear someone speak of the capacitance of a single conductor. In this case the "second conductor", with the negative charge, is an imaginary spherical shell of infinite radius surrounding the one conductor. It contributes **nothing** to the field, so the capacitance is given by Eq. (3.45), where V is the potential with infinity as the reference point.

To “charge up” a capacitor, so to say, we have to remove electrons from the positive plate and carry them to the negative plate. By doing so, we fight against the electric field, which is pulling them back toward the positive conductor and pushing them away from the negative one.

How much work does it take, then, to charge the capacitor up to a final amount Q ?

Suppose that at some intermediate stage in the process the charge on the positive plate is q , so the potential difference is q/C . According to Eq. (3.29), the work we need do to transport the next piece of charge, dq , is:

$$dW = \left(\frac{q}{C} \right) dq. \quad [J]$$

The total work necessary, then, to go from $q = 0$ to $q = Q$, is

$$W = \int_0^Q \left(\frac{q}{C} \right) dq = \frac{1}{2} \frac{Q^2}{C}, \quad [J]$$

or, since $Q = CV$,

$$W = \frac{1}{2} CV^2 \quad [J] \quad (3.46)$$

where V is the final potential of the capacitor.

Chapter 4

Advanced Mathematical Methods

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4.1 Solving Electrostatics

The principle of electrostatics is to find the electric field ($\mathbf{E}$) of a given stationary charge distribution. In terms of mathematical expression, this is accomplished by *Coulomb's law*, in the following form:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{\mathbf{z}}}{z^2} \rho(\mathbf{r}') d\tau'. \quad [\text{V} \cdot \text{m}^{-1}]$$

Where $\mathbf{r}'$ is the position of the source charge. Unfortunately, these types integrals can be difficult to calculate. Occasionally we can get around this by **exploiting symmetry** and using *Gauss's law*, but ordinarily the best strategy is first to calculate the electric potential (V) which is given by the slightly less complicated expression:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{z} \rho(\mathbf{r}') d\tau'. \quad [\text{V}] \quad (4.1)$$

Even this integral is often too tough to handle analytically. Moreover, in problems involving conductors, the charge density (ρ) itself may **NOT** be known in advance as charge is free to move around, the only thing we control directly is the total charge¹ of each conductor. For these cases, it is useful to view the problem in differential form, using **Poisson's** equation:

$$\nabla^2 V = -\frac{1}{\epsilon_0} \rho, \quad [\text{where } \nabla^2 \text{ is the Laplacian.}]$$

¹or perhaps the potential

which, together with appropriate **boundary conditions**, is equivalent to Eq. (4.1). Often we are interested in finding the potential in a region where $\rho = 0^2$.

In this case, Poisson's equation reduces to **Laplace's equation**:

$$\nabla^2 V = 0, \quad \text{in Cartesian coordinates} \quad \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0.$$

The importance of this equation can't be downplayed as the field of electrostatics is the study of Laplace's equation. This equation very common in physics as it appears also in gravitation and magnetism, the theory of heat³, and the study of bubbles [1].

To get an understanding of Laplace's equation and its solutions⁴, let's begin with the 1D and 2D versions, which are easier to picture, and illustrate all the essential properties of the 3D case.

4.1.1 Laplace's Equation in One Dimension

Let us assume V depends on only one (1) variable, x . Then Laplace's equation becomes:

$$\frac{d^2 V}{dx^2} = 0 \quad \text{or} \quad V'' = 0.$$

The general solution becomes:

$$V(x) = mx + b, \quad (4.2)$$

as we can see, the equation geometrically represents a **straight line**. It contains two (2) undetermined constants (m , b), as is appropriate for a 2nd-order Ordinary Differential Equation (ODE). They are fixed, in any particular case, by the boundary conditions of that problem (or initial conditions if we recall **M.Sc Higher Mathematics I.**)

As an example, it might be specified that $V = 4$ at $x = 1$, and $V = 0$ at $x = 5$. In that case, $m = -1$ and $b = 5$, so $V = -x + 5$, shown in **Fig. 4.1**.

Let's try to formulate the two (2) features of this result, while obvious in 1D, where we can write down the general solution **explicitly**, but the analogs in 2D and 3D dimensions are **powerful** and by no means obvious:

1. $V(x)$ is the **average** of $V(x+a)$ and $V(x-a)$, for any a :

$$V(x) = \frac{1}{2} [V(x+a) + V(x-a)]$$

Laplace's equation can be taught of as an **averaging instruction** as it tells to assign to the point x the average of the values to the left and to the right of x .

²If $\rho = 0$ everywhere, then $V = 0$, and there is nothing further to say. This **NOT** the intent. There can be plenty of charge elsewhere, but we're confining our attention to places where there is no charge.

³Which will be our topic of discussion for **M.Sc Higher Mathematics II**.

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⁴known as **harmonic** functions.

A harmonic function is a twice-continuously differentiable function that satisfies Laplace's equation, meaning the sum of its second partial derivatives is zero ($u_{xx} + u_{yy} = 0$) for a two-variable function. This mathematical property is equivalent to the function's value at any point being the average of its values over any circle centered at that point. In physics and engineering, harmonic functions are known as **potential functions** and represent equilibrium conditions, such as steady-state temperature or electrostatic fields.

Figure 4.1: A particular solution to a 1D Laplace.

2. Laplace's equation **tolerates no local maxima or minima**.⁵ Extreme values of V **must occur at the end points**. Actually, this is a consequence of the previous statement, for if there were a local maximum, V would be greater at that point than on either side, and therefore could not be the average.

⁵This is because the second derivative must be greater than or less than zero for maxima or minima but it is equal to zero as per Laplace equation.

Solving in Two Dimensions

When V depends on two (2) variables, Laplace's equation becomes:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \quad \text{or} \quad V_{xx} + V_{yy} = 0. \quad (4.3)$$

While the complexity does **NOT** rise in an exponential manner, nevertheless, we can no longer say we are working with an ODE.⁶ It is a PDE and therefore, some of the simple rules we may be familiar with will not apply.

⁶that is, one involving ordinary derivatives only

For instance, the general solution to the aforementioned equation doesn't contain just two (2) arbitrary constants.⁷ Indeed, one cannot write down a **general solution**.⁸

⁷or, for that matter, any finite number, despite being a second-order equation.

Nevertheless, it is possible to deduce certain properties **common to all solutions**.

⁸at least, **NOT** in a closed form we are familiar with.

It may help to have a physical example in mind. Picture a thin rubber sheet stretched over some support. For definiteness, suppose you take a cardboard box, cut a way line all the way around, and remove the top part, such as the one shown in **Fig. 4.2**.

Now glue a tightly stretched rubber membrane over the box, so it fits like a drum head.⁹ Now, if you lay out coordinates (x, y) on the bottom of the box, the height $V(x, y)$ of the sheet above the point (x, y) will satisfy Laplace's equation.

figures.pdf

Figure 4.2

A visual aid on how a Laplace equation in 2D would be tackled.

⁹It won't be a flat drumhead, of course, unless you chose to cut the edges off straight.

Harmonic functions in 2D have the same properties we noted in one dimension:

1. The value of V at a point (x, y) is the **average** of those around the point. If we were to draw a circle of any radius R about the point (x, y) , the average value of V on the circle is **equal to the value at the centre**:

$$V(x, y) = \frac{1}{2\pi R} \oint_{\text{circle}} V dl.$$

Information: Method of Relaxation

This suggests the method of relaxation¹⁰, on which computer solutions to Laplace's equation are based. Starting with specified values for V at the boundary, and reasonable guesses for V

¹⁰In numerical mathematics, relaxation methods are iterative methods for solving systems of equations, including nonlinear systems.



on a grid of interior points, the first pass reassigns to each point the average of its nearest neighbors. The second pass repeats the process, using the corrected values, and so on. After a few iterations, the numbers begin to settle down, so that subsequent passes produce negligible changes, and a numerical solution to Laplace's equation, with the given boundary values, has been achieved.

2. V has **no local maxima or minima**; all extrema occur at the boundaries. Again, Laplace's equation picks the most featureless function possible, consistent with the boundary conditions: no hills, no valleys, just the smoothest conceivable surface. From a geometrical point of view, just as a straight line is the shortest distance between two points, so a harmonic function in two dimensions minimizes the surface area spanning the given boundary line.

Solving in Three Dimensions

In 3D, unfortunately, we can neither derive an explicit solution nor describe the problem using a physical example as an analogy. However, even with all the additional complexities the third dimension brings, the same two properties remain true:

1. The value of V at some point $\mathbf{r}$ is the **average value** of V over a spherical surface of radius R centered at $\mathbf{r}$:

$$V(\mathbf{r}) = \frac{1}{4\pi R^2} \oint_{\text{sphere}} V d\mathbf{a}.$$

2. As a consequence of having more than one dimension, V can have no local maxima or minima. The extreme values of V must occur at the boundaries.¹¹

¹¹For if V had a local maximum at $\mathbf{r}$, then by the very nature of maximum I could draw a sphere around $\mathbf{r}$ over which all values of V .

4.1.2 Boundary Conditions and Uniqueness Theorems

Laplace's equation does **not by itself** determine V :

suitable boundary conditions must be supplied.

This raises an important question.

What are considered appropriate conditions to determine the answer?

For the case of 1D, it is easy, for here the general solution:

$$V = mx + b$$

contains two (2) arbitrary constants, and we require two (2) boundary conditions. We could:

- specify the value of the function at each end,

[Dirichlet]



■ give the value of the function and its derivative at one end,

[Neumann]

■ give the value at one end and the derivative at the other,

[Robin]

and so on. But we cannot get away with just the value or just the derivative at one end. This is **insufficient** information or specifying the derivatives at both ends as this would either be redundant if the two are equal or inconsistent if they are **NOT**.

In 2D or 3D dimensions we are confronted by a PDE, and it is not so obvious **what would constitute acceptable boundary conditions**.

Is the shape of a number membrane uniquely determined by the frame over which it is stretched, or, like a scanning jar lid, can it snap from one stable configuration to another?

The answer, is that V is **uniquely determined by its value at the boundary**. However, other boundary conditions can also be used. The proof that a proposed set of boundary conditions will suffice is usually presented in the form of a **uniqueness theorem**. There are many such theorems for electrostatics, all sharing the same basic format. Let's look at the two (2) most useful ones.

Theory 4.8: First Uniqueness

The solution to Laplace's equation in some volume $\mathcal{V}$ is **uniquely determined** if V is specified on the boundary surface $\mathcal{S}$.

Proof In **Fig. 4.3** we can see such a region and its boundary.¹² Assume there were two (2) solutions to Laplace's equation:

$$\nabla^2 V_1 = 0, \quad \text{and} \quad \nabla^2 V_2 = 0,$$

both expression assume the specified value on the surface. Let's prove they must be equal. The main idea is look at their **difference**:

$$V_3 \equiv V_1 - V_2.$$

This obeys Laplace's equation,

$$\nabla^2 V_3 = \nabla^2 V_1 - \nabla^2 V_2 = 0,$$

and it takes the value **zero** on all boundaries.¹³ But Laplace's equation allows no local maxima or minima as all extrema occur on the boundaries. Therefore, the maximum and minimum of V_3 are both zero. This implies V_3 must be zero everywhere, which means:

$$V_1 = V_2 \quad \blacksquare$$

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¹²There could also be "islands" inside, so long as V is given on all their surfaces. In addition, the outer boundary could be at infinity, where V is ordinarily taken to be zero.

¹³Given V_1 and V_2 are equal there.

Figure 4.3: A visual aid for understanding the "First Uniqueness Theorem".



The uniqueness theorem allows unique approaches as it doesn't matter how we come to a solution. If it satisfies Laplace's equation and it has the correct value on the boundaries, then it's **right**.

We'll see the power of this argument when we come to the method of images. Incidentally, it is easy to improve on the first uniqueness theorem: I assumed there was no charge inside the region in question, so the potential obeyed Laplace's equation, but we may as well throw in some charge.¹⁴ The argument is the same, only this time.

¹⁴in which case V obeys Poisson's equation.

$$\nabla^2 V_1 = -\frac{1}{\epsilon_0} \rho, \quad \nabla^2 V_2 = -\frac{1}{\epsilon_0} \rho,$$

which makes:

$$\nabla^2 V_3 = \nabla^2 V_1 - \nabla^2 V_2 = -\frac{1}{\epsilon_0} \rho + \frac{1}{\epsilon_0} \rho = 0.$$

Once again the *difference* ($V_3 \equiv V_1 - V_2$) satisfies Laplace's equation and has the value zero on all boundaries, so $V_3 = 0$ and hence $V_1 = V_2$.

The potential in a volume $\mathcal{V}$ is uniquely determined if (a) the charge density throughout the region, and (b) the value of V on all boundaries, are specified.

4.1.3 Conductors and the Second Uniqueness Theorem

The **easiest** way to set the boundary conditions for an electrostatic problem is to specify the value of V on all surfaces surrounding the region of interest. And this situation often occurs in practice:

In the laboratory, we have conductors connected to batteries, which maintain a given potential, or to **ground**, which is the experimental work for $V = 0$.

However, there are other circumstances in which we do not know the potential at the boundary, but rather the charges on various conducting surfaces. Let's say we put charge Q_a on the first conductor, Q_b on the second, and so on.

We are **NOT** telling you how the charge distributes itself over each conducting surface, because as soon as we put it on, it moves around in a way we have no control over.

And for good measure, let's say there is some specified charge density in the region between the conductors. Is the electric field now uniquely determined? Or are there perhaps a number of different ways the charges could arrange themselves on their respective conductors, each leading to a different field?



Theory 4.9: Second Uniqueness

In a volume $\mathcal{V}$ surrounded by conductors and containing a specified charge density (ρ), the electric field is uniquely determined if the *total charge* on each conductor is given,¹⁵ which can be seen in Fig. 4.4.

Proof Suppose there are two (2) fields satisfying the conditions of the given problem. They both obey Gauss's law in differential form in the space between the conductors:

$$\nabla \cdot \mathbf{E}_1 = \frac{1}{\epsilon_0} \rho, \quad \nabla \cdot \mathbf{E}_2 = \frac{1}{\epsilon_0} \rho.$$

And they both obey Gauss's law in integral form for a Gaussian surface enclosing each conductor:

$$\oint_{\text{ith conducting surface}} \mathbf{E}_1 \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_i, \quad \oint_{\text{ith conducting surface}} \mathbf{E}_2 \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_i.$$

Similarly, for the outer boundary,¹⁶

$$\oint_{\text{outer boundary}} \mathbf{E}_1 \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{tot}}, \quad \oint_{\text{outer boundary}} \mathbf{E}_2 \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{tot}}.$$

As before, we examine the difference

$$\mathbf{E}_3 \equiv \mathbf{E}_1 - \mathbf{E}_2, \quad \text{which obeys} \quad \nabla \cdot \mathbf{E}_3 = 0, \quad (4.4)$$

in the region **between** the conductors, and,

$$\oint \mathbf{E}_3 \cdot d\mathbf{a} = 0 \quad (4.5)$$

over each boundary surface. Now there is one final piece of information we must use. While we don't know how the charge Q_i distributes itself over the i^{th} conductor, we **do** know that each conductor is an equipotential, and hence V_3 is a **constant**¹⁷ over each conducting surface.¹⁸ Next comes a **trick**. Invoking product rule number 5, we find that:

$$\nabla \cdot (V_3 \mathbf{E}_3) = V_3 (\nabla \cdot \mathbf{E}_3) + \mathbf{E}_3 \cdot (\nabla V_3) = -(\mathbf{E}_3)^2.$$

Here we have use Eq. (4.4), and $\mathbf{E}_3 = -\nabla V_3$. Integrating this over $\mathcal{V}$, and applying the **divergence theorem** to the Left Hand Side (LHS):

$$\int_{\mathcal{V}} \nabla \cdot (V_3 \mathbf{E}_3) d\tau = \oint_S V_3 \mathbf{E}_3 \cdot d\mathbf{a} = - \int_{\mathcal{V}} (\mathbf{E}_3)^2 d\tau.$$

The surface integral covers all boundaries of the region in question which is the conductors and outer boundary. Now V_3 is a constant over each surface,¹⁹ so it comes outside each integral, and what remains is zero, according to Eq. (4.5). Therefore,

$$\int_{\mathcal{V}} (\mathbf{E}_3)^2 d\tau = 0.$$

¹⁵The region as a whole can be bounded by another conductor, or else unbounded.

¹⁶whether this is just inside an enclosing conductor or at infinity.

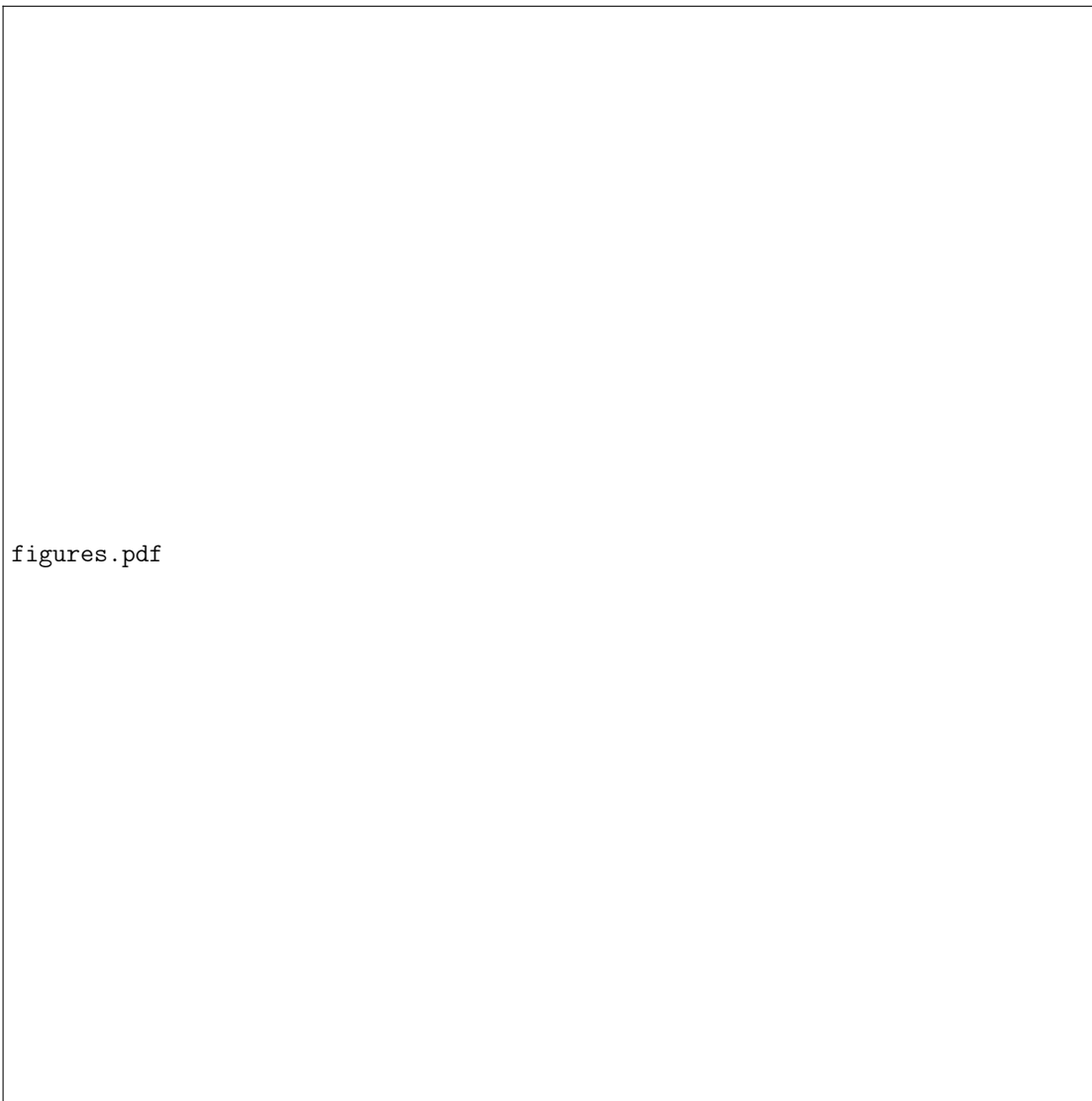
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Figure 4.4: A visual explanation of the second uniqueness theorem. Here the dashed lines represent the surface of integration and solid line describe the actual surface (S) of the charge.

¹⁷This does not necessarily need to be the same constant.

¹⁸It need not be zero, for the potentials V_1 and V_2 may not be equal as all we know for sure is that *both* are constant over any given conductor.

¹⁹if the outer boundary is infinity, $V_3 = 0$ there.



figures.pdf

Figure 4.5

But this integrand is **never negative**. The only way the integral can vanish is if $E_3 = 0$ everywhere. Consequently, $E_1 = E_2$, and the theorem is proved ■

This proof was not easy, and there is a real danger that the theorem itself will seem more plausible to you than the proof. To give a better understanding of the second uniqueness theorem, consider this example of Purcell's:

Fig. 4.5b shows a simple electrostatic configuration, consisting of four (4) conductors with charges Q , situated so that the pluses are near the minuses.

Up to now, it all looks very plausible. Now, what happens if we join them in pairs, by tiny wires, as

indicated in **Fig. 4.5a**?

Since the positive charges are very near negative charges²⁰ we might well guess that **nothing** will happen. But it's wrong. The configuration in **Fig. 4.5b** is **impossible**.

²⁰which is where we think they would be

For there are now effectively two (2) conductors, and the total charge on each is zero. **One** possible way to distribute zero charge over these conductors is to have no accumulation of charge anywhere, and hence zero field everywhere **Fig. 4.5c**.

By the second uniqueness theorem, this must be the solution. The charge will flow down the tiny wires, canceling itself off.

4.2 Method of Images

The method of images²¹ is a mathematical tool for solving differential equations, in which boundary conditions are satisfied by combining a solution not restricted by the boundary conditions with its possibly weighted mirror image. This method has numerous applications in the field of electrical engineering which electrostatics is one of them.

²¹or method of mirror images.

4.2.1 A Classic Problem

To start let's discuss a simple problem. Let us assume a point charge (q) is held a distance (d) above an infinite grounded conducting plane (Fig. 3.10).

Now, we ask the Question:

What is the potential in the region above the plane?

Unfortunately we can't say it is $(1/4\pi\epsilon_0) q/a$, as q will induce a certain amount of negative charge on the nearby surface of the conductor; the total potential is due in part to q directly, and in part to this induced charge.

But how can we possibly calculate the potential (V), when we don't know how much charge is induced or how it is different?

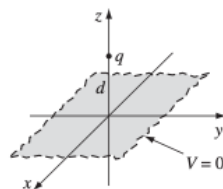


FIGURE 3.10

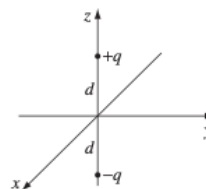


FIGURE 3.11

Figure 4.6

From a mathematical point of view, our problem is to solve Poisson's equation in the region $z > 0$, with a single point charge q at $(0, 0, d)$, subject to the boundary conditions:

1. $V = 0$ when $z = 0$;²² and

$$[V(x, y, 0) = 0] \quad \text{²²since the conducting plane is grounded.}$$

2. $V \rightarrow 0$ far from the charge (that is, for $x^2 + y^2 + z^2 \gg d^2$).

$$[\lim_{\infty} V = 0]$$

The first uniqueness theorem we looked previously guarantees that there is only one (1) function which meets these requirements.



The trick is to forget about the actual problem and study a *completely different* situation.

Let's rethink our problem and set a new configuration. This new configuration consists of two (2) point charges:

$$+q \text{ at } (0, 0, d), \text{ and } -q \text{ at } (0, 0, -d),$$

and we will have no conducting plane in this new setup. For this configuration, we can easily write down the potential:

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right]. \quad (4.6)$$

The denominators²³ are the distances from (x, y, z) to the charges $+q$ and $-q$, respectively.

²³Here, denominator means the value under the fraction sign.

It follows that:

$$1. V = 0 \text{ when } z = 0, \quad [V(x, y, 0) = 0]$$

$$2. V \rightarrow 0 \text{ for } x^2 + y^2 + z^2 \gg d^2, \quad [\lim_{\infty} V = 0]$$

which were defined previously, and the only charge in the region $z > 0$ is the charge $+q$ at $(0, 0, d)$.

These are precisely the conditions of the original problem.

Evidently the second configuration happens to produce **exactly the same potential** as the first configuration, in the “upper” region $z \geq 0$:²⁴

Notice the crucial role played by the **uniqueness theorem** in this argument: without it, no one would believe this solution, since it was obtained for a completely different charge distribution. But the uniqueness theorem certifies it:

If it satisfies Poisson's equation in the region of interest, and assumes the correct value at the boundaries, then it must be right.

²⁴The “lower” region, $z < 0$, is completely different, but as we don't care about the lower region we can disregard that part.

4.2.2 Induced Surface Charge

Now that we know the potential, it is a straightforward matter to compute the surface charge σ induced on the conductor. Looking back at the previous chapter:

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$$

where $\partial V/\partial n$ is the normal derivative of V at the surface. In this case the normal direction is the z direction, so

$$\sigma = -\epsilon_0 \left. \frac{\partial V}{\partial z} \right|_{z=0}$$

And using our previously defined “mirror image” vector potential, in Eq. (4.6), we can write:

$$\frac{\partial V}{\partial z} = \frac{1}{4\pi\epsilon_0} \left[\frac{-q(z-d)}{[x^2 + y^2 + (z-d)^2]^{3/2}} + \frac{q(z+d)}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right],$$

so:

$$\sigma(x, y) = \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}}.$$

As expected, the induced charge is negative²⁵ and greatest at $x = y = 0$. While we’re at it, let’s compute the **total** induced charge:

$$Q = \int \sigma \, da.$$

This integral, over the xy plane, could be done in Cartesian coordinates, with $da = dx \, dy$, but it’s a little easier to use polar coordinates (r, ϕ) , with $r^2 = x^2 + y^2$ and $da = r \, dr \, d\phi$. Then:

$$\sigma(r) = \frac{-qd}{2\pi(r^2 + d^2)^{3/2}},$$

and

$$Q = \int_0^{2\pi} \int_0^\infty \frac{-qd}{2\pi(r^2 + d^2)^{3/2}} r \, dr \, d\phi = \left. \frac{qd}{\sqrt{r^2 + d^2}} \right|_0^\infty = -q.$$

The total charge induced on the plane is $-q$ ■

²⁵Under the assumption, that q is positive.

4.2.3 Force and Energy

The charge q is attracted toward the plane, because of the negative induced charge. Let’s calculate the force of attraction. Since the potential in the vicinity of q is the same as in the analogue problem,²⁶ so also is the field and, therefore, the force:

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{\mathbf{z}}. \quad (4.7)$$

With the two (2) point charges and no conductor:

$$W = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d}.$$

But for a single charge and conducting plane, the energy is *half* of this:

$$W = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}.$$

²⁶the one with $+q$ and $-q$ but no conductor.



Why half? Think of the energy stored in the fields:

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau.$$

In the first case, both the upper region ($z > 0$) and the lower region ($z < 0$) contribute equally. But in the second case, only the upper region contains a nonzero field, and hence the energy is half as great.

Of course, one could also determine the energy by calculating the work required to bring q in from infinity. The force required (to oppose the electrical force in Eq. (4.7)) is

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \left(q^2/4z^2 \right) \hat{\mathbf{z}}$$

so:

$$W = \int_{\infty}^d \mathbf{F} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \int_{\infty}^d \frac{q^2}{4z^2} dz = \frac{1}{4\pi\epsilon_0} \left(-\frac{q^2}{4z} \right) \Big|_{\infty}^d = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d} \quad \blacksquare$$

As we move q towards the conductor, we do work **only on q** . It is true that induced charge is moving in over the conductor, but this costs us nothing, since the whole conductor is at potential zero. By contrast, if I simultaneously bring in *two* point charges without any conductor present, we do work on **both** of them, and the total is twice as much.



4.3 Separation of Variables

In separation of variables²⁷ we will attack Laplace's equation **directly**. The method is applicable in situations where the potential (V) or the charge density (σ) is **specified** on the boundaries of some region, and we are tasked with finding the potential in the **interior**.

²⁷This is the physicist's and engineer's favorite tool for solving PDEs.

We will have a look at this method in more detail in **M.Sc Higher Mathematics II** but for now let's use this opportunity to get a sneak-peek of this method.

The basic strategy is very simple:

We look for solutions that are products of functions, each of which depends on only one (1) of the coordinates.

The algebraic details, however, can be formidable, therefore let's try to understand this through exercises. We'll start with Cartesian coordinates and then do spherical coordinates.

4.3.1 Case Study: 2D Grounded Plates

To begin with let's assume the following problem

Two infinitely long grounded metal plates (■) at $y = 0$ and $y = a$ are connected at $x = \pm b$ by metal strips maintained at a constant potential V_0 (■) as shown. (a thin layer of insulation at each corner prevents the metal plates from shorting out just for pedantry). Find the potential inside the resulting rectangular pipe.

As can be seen, the configuration is **independent** from z -direction. Therefore, our goal here is to solve a 2D Laplace's equation ($\nabla^2 f = 0$) is as follows:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0,$$

which are subject to the following boundary conditions:

- | | |
|---|--|
| <p>(i) $V = 0$ when $y = 0$,</p> <p>(iii) $V = V_0$ when $x = b$,</p> | <p>(ii) $V = 0$ when $y = a$,</p> <p>(iv) $V = V_0$ when $x = -b$.</p> |
|---|--|

To solve this problem we need to use **Separation of Variables** method which is as follows. We will assume a solution with the following description:

$$V(x, y) = X(x)Y(y) \quad \text{putting to Laplace, we get} \quad Y(y)\frac{d^2 X}{dx^2} + X(x)\frac{d^2 Y}{dy^2} = 0.$$

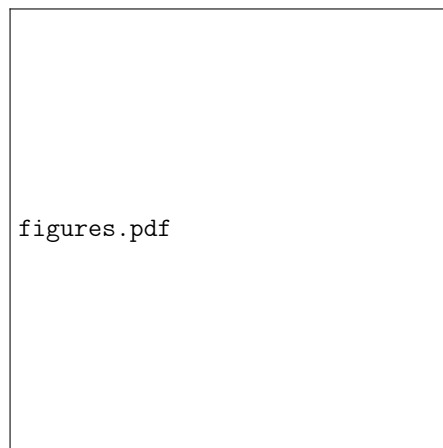


Figure 4.7

It is time to **isolate** the variables,

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0.$$

First term depends on x , and the second term depends on y . From this we can write:

$$f(x) + g(y) = 0.$$

For this to be true both of these functions needs to be constants.

$$\frac{d^2 X}{dx^2} = k^2 X, \quad \frac{d^2 Y}{dy^2} = -k^2 Y.$$

Deriving the ODE Solutions

It is time to solve these equations, lets have a look at the first one:

$$\frac{d^2 X}{dx^2} = k^2 X.$$

As we should be aware by now, this is classified as an homogeneous 2nd-order linear differential equation with the generic form of:²⁸

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + c y(x) = 0.$$

where a , b , c are arbitrary constants to the equation. As this is a homogeneous equation, we can use the **characteristic** equation form to determine the quadratic form. The general solution for these kind of equation is along with the root formula:

$$y(x) = A e^{-\lambda_1 x} + B e^{-\lambda_2 x} \quad \text{where} \quad \lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Based on this form, the solution to the first ODE is;

$$X(x) = A \exp kx + B \exp -kx, \quad \text{where} \quad \lambda_1 = -k \quad \text{and} \quad \lambda_2 = +k.$$

If we were to apply the second ODE to the previously mentioned form, we arrive at the following.

$$Y(y) = C \exp(-j ky) + D \exp(j ky)$$

While the imaginary solution is perfectly valid, we want to use the real only solution as the equation we are trying to solve is real as well. To convert this let's use Euler's identity.

$$\exp(j \theta) = \cos \theta + j \sin \theta \quad \text{which gives} \quad Y(y) = C \sin ky + D \cos ky.$$

The solution to this problem is as follows the same path we have looked at the lecture up to the following part:

$$V(x, y) = (A \exp kx + B \exp -kx) (C \sin ky + D \cos ky)$$

²⁸If you need a light refreshment on this topic, please have a look at the beginning chapters of **M.Sc Higher Mathematics I** LectureBook.

Applying the Boundary Conditions

Now we have the equation let's restrict the equation so we can get a singular solution.

To begin, we have to say, we cannot set $A = 0$ as the region in question does not extend to $x = \infty$, so $\exp kx$ is perfectly acceptable, whereas, the situation is symmetric with respect to x , so $V(-x, y) = V(x, y)$ and it follows $A = B$. Using the hyperbolic cosine function identity:

$$A \exp kx + A \exp -kx = 2A \cosh kx,$$

distributing A to C and D , we have;²⁹

$$V(x, y) = \cosh kx (2AC \sin ky + 2AD \cos ky)$$

Boundary conditions (i) and (ii) require D to be equal to zero and $k = n\pi/a$ so:

$$V(x, y) = 2AC \cosh (n\pi x/a) \sin (n\pi y/a),$$

Because $V(x, y)$ is even in x , it will automatically meet condition (iv) if it fits (iii). It remains, therefore, to construct the linear combination,

$$V(x, y) = \sum_{n=1}^{\infty} G_n \cosh (n\pi x/a) \sin (n\pi y/a), \quad \text{where } G = 2AC.$$

and pick coefficients C_n such that it satisfies the condition on (iii):

$$V(b, y) = \sum_{n=1}^{\infty} G_n \cosh (n\pi b/a) \sin (n\pi y/a) = V_0,$$

Simplifying the Fourier Expression

This is the same problem in *Fourier analysis* we worked in the exercise in part 3.

To solve this equation, we first need to multiply both sides with $\sin \frac{m\pi y}{a}$.

$$\sin \frac{m\pi y}{a} V_0(0) = \sum_1^{\infty} G_n \cosh (n\pi b/a) \sin \frac{n\pi y}{a} \sin \frac{m\pi y}{a},$$

Time to integrate both sides over y , from zero to a :

$$\int_0^a \sin \frac{m\pi y}{a} V_0(0) dy = \sum_1^{\infty} G_n \cosh (n\pi b/a) \int_0^a \sin \frac{n\pi y}{a} \sin \frac{m\pi y}{a} dy,$$

This is basically **Fourier Transform** on both sides. It is time to focus on the integration on the right hand side.

$$\begin{aligned} \int_0^a \sin \frac{n\pi y}{a} \sin \frac{m\pi y}{a} dy &= \left[-\frac{a}{n\pi} \sin \frac{m\pi y}{a} \cos \frac{n\pi y}{a} \right]_0^a + \frac{m}{n} \int_0^a \cos \frac{m\pi y}{a} \cos \frac{n\pi y}{a} dy \\ &= \frac{m}{n} \left[\frac{a}{n\pi} \cos \frac{m\pi y}{a} \sin \frac{n\pi y}{a} \right]_0^a + \frac{m}{n} \int_0^a \sin \frac{m\pi y}{a} \sin \frac{n\pi y}{a} dy, \end{aligned}$$

²⁹We do this distribution as we don't care about individual coefficients, so only their end product. So we can state, here $A \times C \rightarrow C$.

Simplification of this integral yields,

$$\left(1 - \frac{m^2}{n^2}\right) \int_0^a \sin \frac{m\pi y}{a} \sin \frac{n\pi y}{a} dy = 0.$$

There are two possibilities here: either $m = n$ or $m \neq n$. if the latter is the case, then:

$$\int_0^a \sin \frac{m\pi y}{a} \sin \frac{n\pi y}{a} dy = 0 \quad (m \neq n)$$

If on the other hand $m = n$, then:

$$\begin{aligned} \int_0^a \sin^2 \frac{n\pi y}{a} dy &= -\frac{a}{n\pi} \sin \frac{n\pi y}{a} \cos \frac{n\pi y}{a} \Big|_0^a + \int_0^a \cos^2 \frac{n\pi y}{a} dy \\ 2 \int_0^a \sin^2 \frac{n\pi y}{a} dy &= \int_0^a \sin^2 \frac{n\pi y}{a} dy + \int_0^a \cos^2 \frac{n\pi y}{a} dy = \int_0^a dy = a. \end{aligned}$$

Therefore:

$$\int_0^a \sin \frac{n\pi y}{a} \sin \frac{m\pi y}{a} dy = \begin{cases} 0, & \text{if } n \neq m \\ \frac{a}{2} & \text{if } n = m \end{cases}$$

Based on this, we only care about the solutions where $m = n$. If we take our focus to the rest of the equation.

$$\int_0^a \sin \frac{n\pi y}{a} V_0 dy = \sum_1^\infty G_n \cosh(n\pi b/a) \overbrace{\int_0^a \sin \frac{n\pi y}{a} \sin \frac{n\pi y}{a} dy}^{=a/2},$$

If we isolate the G_n on one side:

$$G_n = \frac{2}{a} \frac{1}{\cosh(n\pi b/a)} \int_0^a \sin \frac{n\pi y}{a} V_0 dy$$

Doing the integral on the right hand side yields the following:

$$G_n = \frac{2}{a} \frac{1}{\cosh(n\pi b/a)} \int_0^a \sin \frac{n\pi y}{a} V_0 dy = \frac{1}{\cosh(n\pi b/a)} \frac{2V_0}{n\pi} (1 - \cos n\pi).$$

The results are:

$$C_n \cosh(n\pi b/a) = \begin{cases} 0, & \text{if } n \text{ is even,} \\ \frac{4V_0}{n\pi}, & \text{if } n \text{ is odd.} \end{cases}$$

The potential in this case is given by:

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots} \frac{1}{n} \frac{\cosh(n\pi x/a)}{\cosh(n\pi b/a)} \sin(n\pi y/a) \quad \nabla \quad \blacksquare$$

The success of this method relies on two (2) important properties of the separable solutions (Eqs. 3.28 and 3.29): **completeness** and **orthogonality**.

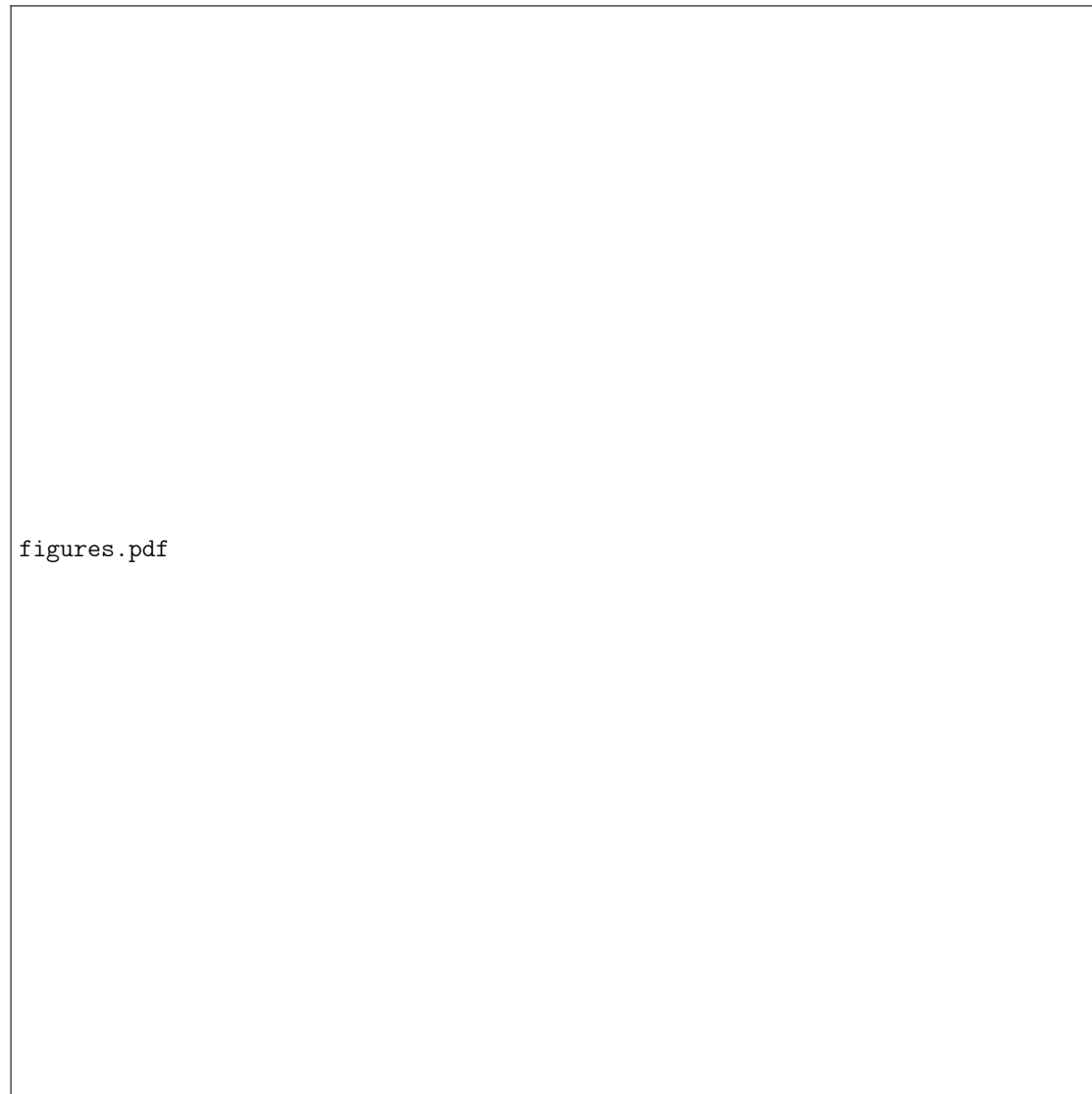


Figure 4.8

The solution to the 2D-Grounded plate. Please observe that the solution is generated by summing up consecutive parts of a Fourier series to achieve a more accurate result.

A set of functions $f_n(y)$ is said to be complete if any other function $f(y)$ can be expressed as a linear combination of them:

$$f(y) = \sum_{n=1}^{\infty} C_n f_n(y)$$

The functions $\sin(n\pi y/a)$ are complete on the interval $0 \leq y \leq a$. It was this fact, guaranteed by Dirichlet's theorem, which assured us Eq. 3.31 could be satisfied, given the proper choice of the coefficients C_n .

A set of functions is **orthogonal** if the integral of the product of any two (2) different members of the set is zero:

$$\int_0^a f_n(y) f_{n'}(y) dy = 0 \quad \text{for} \quad n' \neq n.$$

The sine functions are orthogonal (Eq. 3.33). This is the property on which Fourier's trick is based, allowing us to kill off all terms but one in the infinite series and thereby solve for the coefficients C_n .

³⁰The proof of completeness, for a particular set of functions, is an extremely difficult business, and unfortunately this margin is too narrow to write.

**Example 4.1: Spherical Separation of Variables**

The potential $V_0(\theta)$ is again specified on the surface of a sphere of radius R , but this time we are asked to find the potential **outside**, assuming there is no charge there.

Solution

In this case it's the A_l 's that must be zero (or else V would not go to zero at ∞), so

$$V(r, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta).$$

At the surface of the sphere, we require that

$$V(R, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{R^{l+1}} P_l(\cos \theta) = V_0(\theta).$$

Multiplying by $P_{l'}(\cos \theta) \sin \theta$ and integrating—exploiting, again, the orthogonality relation 3.68—we have

$$\frac{B_{l'}}{R^{l'+1}} \frac{2}{2l'+1} = \int_0^\pi \pi V_0(\theta) P_{l'}(\cos \theta) \sin \theta d\theta,$$

or

$$B_l = \frac{2l+1}{2} R^{l+1} \int_0^\pi \pi V_0(\theta) P_l(\cos \theta) \sin \theta d\theta.$$

Equation 3.72, with the coefficients given by Eq. 3.73, is the solution to our problem. ■

4.3.2 Case Study: 3D Rectangular Pipe

Let's turn the problem up a bit and study a 3D case. For this, imagine, an infinitely long rectangular metal pipe (sides a and b) is grounded, but one end, at $x = 0$, is maintained at a specified potential $V_0(y, z)$, as indicated in Fig. 3.22.

We will try to find the potential inside the pipe. To start, we can say that this is a 3D problem with the following Laplace equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0,$$

which are subject to the following boundary conditions:

Deriving the ODEs

As usual, we are looking for solutions that are products:

$$V(x, y, z) = X(x) Y(y) Z(z).$$

Putting this into the Laplace equation presents us:

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0.$$

It follows that:

$$\frac{1}{X} \frac{d^2 X}{dx^2} = C_1, \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2, \quad \frac{1}{Z} \frac{d^2 Z}{dz^2} = C_3, \quad \text{where } C_1 + C_2 + C_3 = 0.$$



For simplicity, we will assume C_1 must be positive and C_2, C_3 are negative. Setting $C_2 = -k^2$ and $C_3 = -l^2$, we have $C_1 = k^2 + l^2$, and therefore:

$$\frac{d^2 X}{dx^2} = (k^2 + l^2) X, \quad \frac{d^2 Y}{dy^2} = -k^2 Y, \quad \frac{d^2 Z}{dz^2} = -l^2 Z$$

We have yet again turned a PDE into a series of ODE. The solutions are:

$$\begin{aligned} X(x) &= A \exp(\sqrt{k^2 + l^2} x) + B \exp(-\sqrt{k^2 + l^2} x), \\ Y(y) &= C \sin ky + D \cos ky, \\ Z(z) &= E \sin lz + F \cos lz. \end{aligned}$$

Applying the Boundary Conditions

Boundary condition **(v)** implies $A = 0$, **(i)** gives $D = 0$ and **(iii)** gives $F = 0$, whereas **(ii)** and **(iv)** require $k = n\pi/a$ and $l = m\pi/b$, where n and m are positive integers.

Combining the remaining constants, we are presented with:

$$V(x, y, z) = C \exp\left(-\pi\sqrt{(n/a)^2 + (m/b)^2} x\right) \sin(n\pi y/a) \sin(m\pi z/b).$$

This solution meets all the boundary condition except **(vi)**. It contains two (2) unspecified integers (m and n) and the most general linear combination is a **double sum**:

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{n,m} \exp\left(-\pi\sqrt{(n/a)^2 + (m/b)^2} x\right) \sin(n\pi y/a) \sin(m\pi z/b).$$

Deriving the Coefficients

We hope to fit the remaining boundary condition:

$$V(0, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{n,m} \sin(n\pi y/a) \sin(m\pi z/b) = V_0(y, z).$$

by choosing appropriate coefficients of $C_{n,m}$. To determine these constants, we multiply by $\sin(n'\pi y/a) \sin(m'\pi z/b)$, where n' and m' are arbitrary positive integers, and integrate:

$$\begin{aligned} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{n,m} \int_0^a \sin(n\pi y/a) \sin(n'\pi y/a) dy \int_0^b \sin(m\pi z/b) \sin(m'\pi z/b) dz \\ = \int_0^a \int_0^b V_0(y, z) \sin(n'\pi y/a) \sin(m'\pi z/b) dy dz. \end{aligned}$$

So:

$$C_{n,m} = \frac{4}{ab} \int_0^a \int_0^b V_0(y, z) \sin(n\pi y/a) \sin(m\pi z/b) dy dz.$$



We are almost at the end of our problem. For example, if the end of the tube is a conductor at constant period V_0 :

$$C_{n,m} = \frac{4V_0}{ab} \int_0^a \sin(n\pi y/a) dy \int_0^b \sin(m\pi z/b) dz$$

$$= \begin{cases} 0, & \text{if } n \text{ or } m \text{ is even,} \\ \frac{16V_0}{\pi^2 nm}, & \text{if } n \text{ and } m \text{ are odd.} \end{cases}$$

In this case:

$$V(0, y, z) = \frac{16V_0}{\pi^2} \sum_{n,m=1,3,5,\dots}^{\infty} \frac{1}{nm} \exp\left(-\pi\sqrt{(n/a)^2 + (m/b)^2}x\right) \sin(n\pi y/a) \sin(m\pi z/b)$$

As the successive terms decrease rapidly, a reasonable approximation would be obtained by keeping only the first few terms ■

4.3.3 Spherical Coordinates

In the examples considered thus far, Cartesian coordinates were clearly appropriate, given the boundaries were planes. For round objects, however, spherical coordinates are more natural. In the spherical system, Laplace's equation becomes:

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0. \quad (4.8)$$

We shall assume the problem has **azimuthal symmetry**, so that V is **independent** of ϕ in that case, Eq. (4.8) reduces to:

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial^2 V}{\partial r^2} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) = 0.$$

Similar to Cartesian coordinates, we look for solutions that are products:

$$V(r, \theta) = R(r) \Theta(\theta). \quad (4.9)$$

As before, we look for solutions that are products:

$$V(r, \theta) = R(r) \Theta(\theta). \quad (4.10)$$

Putting this into Eq. (4.9), and dividing by V , we get the following:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0. \quad (4.11)$$

Since the first term is dependent only on r , and the second only on θ , it follows that each must be a constant:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = l(l+1), \quad \text{and} \quad \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -l(l+1). \quad (4.12)$$

Table 4.1: Legendre Polynomials

Polynomial	Value
$P_0(x)$	1
$P_1(x)$	x
$P_2(x)$	$(3x^2 - 1)/2$
$P_3(x)$	$(5x^3 - 3x)/2$
$P_4(x)$	$(35x^4 - 30x^2 + 3)/8$
$P_5(x)$	$(63x^5 - 70x^3 + 15x)/8$

Here $l(l+1)$ is just a prettier way of writing the separation constant.³¹ As always, separation of variables has converted a PDE in Eq. (4.9) into pair of ODEs Eq. (4.12).

³¹We will see its usefulness in just a minute.

To solve, let's start with the radial equation:

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = l(l+1)R, \quad (4.13)$$

has the following general solution:

$$R(r) = Ar^l + \frac{B}{r^{l+1}}, \quad (4.14)$$

as we can see, A and B are the two (2) arbitrary constants to be expected in the solution of a 2nd-order ODE. But the angular equation,

$$\frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -l(l+1) \sin \theta \Theta, \quad (4.15)$$

is unfortunately, **NOT** so simple. The solutions are **Legendre polynomials** in the variable $\cos \theta$:

$$\Theta(\theta) = P_l(\cdot) \cos \theta. \quad (4.16)$$

$P_l(x)$ is most conveniently defined by the **Rodrigues formula**:

$$P_l(x) \equiv \frac{1}{2^l l!} \left(\frac{d}{dx} \right)^l (x^2 - 1)^l. \quad (4.17)$$

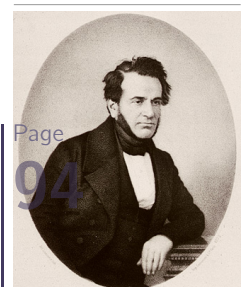
The first few Legendre polynomials are listed in **Tbl. 4.1** and the rest of the values can easily be generated using Eq. (4.17).

Notice that $P_l(x)$ is³² an l^{th} order polynomial in x ; it contains only even powers, if l is even, and odd powers, if l is odd. The factor in front $(1/2^l l!)$ was chosen in order that

³²As the name obviously suggests.

$$P_l(1) = 1. \quad (4.18)$$

The Rodrigues formula³³ obviously works only for non-negative integer values of l .



Moreover, it provides us with only one (1) solution. But Eq. (4.15) is 2nd- order, and it should possess two (2) independent solutions, for every value of l . It turns out that these “other solutions” blow up at $\theta = 0$ and/or $\theta = \pi$, and are therefore unacceptable on physical grounds.

For instance, the second solution for $l = 0$ is

$$\Theta(\theta) = \ln \left(\tan \frac{\theta}{2} \right). \quad (4.19)$$

Which as we all now is **undefined**. We can of course check for our self that this satisfies Eq. (4.15). In the case of azimuthal symmetry, then, the most general separable solution to Laplace’s equation, consistent with minimal physical requirements, is:³⁴

$$V(r, \theta) = \left(Ar^l + \frac{B}{r^{l+1}} \right) P_l(\cos \theta).$$

³⁴There was no need to include an overall constant in Eq. (4.16) as it can be absorbed into A and B at this stage.

As before, separation of variables gives an infinite set of solutions, one for each l . The **general solution** is the linear combination of separable solutions:

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta). \quad (4.20)$$

The following examples illustrate the power of this important result.

Example 3.9. A specified charge density $\sigma_0(\theta)$ is glued over the surface of a spherical shell of radius R . Find the resulting potential inside and outside the sphere. Solution You could, of course, do this by direct integration: $V = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma_0}{\lambda} da$, but separation of variables is often easier. For the interior region, we have

$$V(r, \theta) = \sum_{i=1}^{\infty} A_i r^i P_i(\cos \theta) \quad (r \leq R) \quad (4.21)$$

(no B_l terms—they blow up at the origin); in the exterior region

$$V(r, \theta) = \sum_{i=0}^{\infty} \frac{B_i}{r^{i+1}} P_i(\cos \theta) \quad (r \geq R) \quad (4.22)$$

(no A_l terms—they don’t go to zero at infinity). These two functions must be joined together by the appropriate boundary conditions at the surface itself. First, the potential is **continuous** at $r = R$ (Eq. 2.34):

$$\sum_{l=0}^{\infty} A_l R^l P_l(\cos \theta) = \sum_{l=0}^{\infty} \frac{B_l}{R^{l+1}} P_l(\cos \theta). \quad (4.23)$$



It follows that the coefficients of like Legendre polynomials are equal:

$$B_l = A_l R^{2l+1}. \quad (4.24)$$

(To prove that formally, multiply both sides of Eq. 3.80 by $P_l(\cos \theta) \sin \theta$ and integrate from 0 to π , using the orthogonality relation 3.68.) Second, the radial derivative of V suffers a discontinuity at the surface (Eq. 2.36):

$$\left(\frac{\partial V_{\text{out}}}{\partial r} - \frac{\partial V_{\text{in}}}{\partial r} \right) \bigg|_{r=R} = -\frac{1}{\epsilon_0} \sigma_0(\theta). \quad (4.25)$$

Thus $-\sum_{l=0}^{\infty} (l+1) \frac{B_l}{R^{l+2}} P_l(\cos \theta) - \sum_{l=0}^{\infty} l A_l R^{l-1} P_l(\cos \theta) = -\frac{1}{\epsilon_0} \sigma_0(\theta)$, or, using Eq. 3.81,

$$\sum_{l=0}^{\infty} (2l+1) A_l R^{l-1} P_l(\cos \theta) = \frac{1}{\epsilon_0} \sigma_0(\theta). \quad (4.26)$$

From here, the coefficients can be determined using Fourier's trick:

$$A_l = \frac{1}{2\epsilon_0 R^{l-1}} \int_0^\pi \sigma_0(\theta) P_l(\cos \theta) \sin \theta d\theta. \quad (4.27)$$

Equations 3.78 and 3.79 constitute the solution to our problem, with the coefficients given by Eqs. 3.81 and 3.84. For instance, if

$$\sigma_0(\theta) = k \cos \theta = k P_1(\cos \theta), \quad (4.28)$$

for some constant k , then all the A_l 's are zero except for $l = 1$, and $A_1 = \frac{k}{2\epsilon_0} \int_0^\pi [P_1(\cos \theta)]^2 \sin \theta d\theta = \frac{k}{3\epsilon_0}$. The potential inside the sphere is therefore

$$V(r, \theta) = \frac{k}{3\epsilon_0} r \cos \theta \quad (r \leq R), \quad (4.29)$$

whereas outside the sphere

$$V(r, \theta) = \frac{kR^3}{3\epsilon_0} \frac{1}{r^2} \cos \theta \quad (r \geq R). \quad (4.30)$$

In particular, if $\sigma_0(\theta)$ is the induced charge on a metal sphere in an external field $E_0 \hat{z}$, so that $k = 3\epsilon_0 E_0$ (Eq. 3.77), then the potential inside is $E_0 r \cos \theta = E_0 z$, and the field is $-E_0 \hat{z}$ —exactly right to cancel off the external field, as of course it should be. Outside the sphere the potential due to this surface charge is $E_0 \frac{R^3}{r^2} \cos \theta$, consistent with our conclusion in Ex. 3.8.

Exercise 4.2 A Long Rectangular Tube

An infinitely long rectangular metal pipe (sides a and b) is grounded, but one end at $x = 0$, is maintained at a specific potential $V_0(y, z)$ as indicated above. Find the potential inside the pipe.

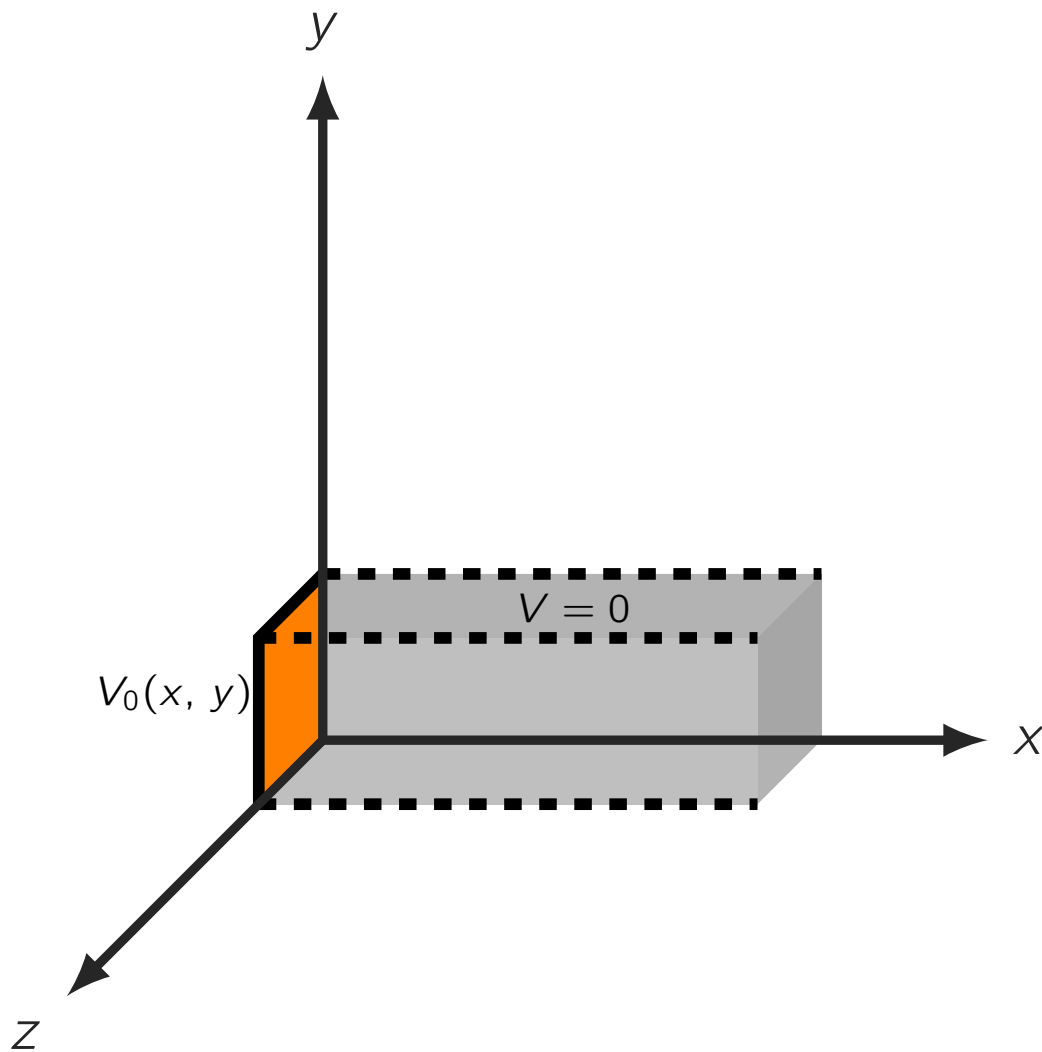


Figure 4.9

SOLUTION This is a 3D problem with the following **Laplace** equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0,$$

which are subject to the following boundary conditions given by the question either explicitly or implicitly:

- | | |
|-----------------------|-----------------------------|
| (i) $V = 0$ | when $y = 0$, |
| (ii) $V = 0$ | when $y = a$, |
| (iii) $V = 0$ | when $z = 0$, |
| (iv) $V = 0$ | when $z = b$, |
| (v) $V \rightarrow 0$ | as $x \rightarrow \infty$, |
| (vi) $V = V_0(x, y)$ | when $x = 0$. |

As usual, we are looking for solutions that are products using **separation of variables**:

$$V(x, y, z) = X(x) Y(y) Z(z)$$

Putting this into the Laplace equation presents to us:

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0.$$

Where once we isolate the variables, we can get the following relation:

$$\begin{aligned} \frac{1}{X} \frac{d^2 X}{dx^2} &= C_1, \\ \frac{1}{Y} \frac{d^2 Y}{dy^2} &= C_2, \\ \frac{1}{Z} \frac{d^2 Z}{dz^2} &= C_3, \quad \text{with} \\ C_1 + C_2 + C_3 &= 0. \end{aligned}$$

To keep this solution relatively short, We will assume C_1 **must** be positive and C_2, C_3 are negative. Setting $C_2 = -k^2$ and $C_3 = -l^2$, we have $C_1 = k^2 + l^2$, and therefore:

$$\begin{aligned} \frac{d^2 X}{dx^2} &= (k^2 + l^2) X, \\ \frac{d^2 Y}{dy^2} &= -k^2 Y, \\ \frac{d^2 Z}{dz^2} &= -l^2 Z \end{aligned}$$

We have yet again turned a PDE into a series of ODEs. The solutions are:

$$\begin{aligned} X(x) &= A \exp\left(\sqrt{k^2 + l^2} x\right) + B \left(-\sqrt{k^2 + l^2} x\right), \\ Y(y) &= C \sin ky + D \cos ky, \\ Z(z) &= E \sin lz + F \cos lz. \end{aligned}$$

Boundary condition (v) implies $A = 0$, (i) gives $D = 0$ and (iii) yields $F = 0$, whereas (ii) and (iv) require $k = n\pi/a$ and $l = m\pi/b$, where n and m are positive integers. Combining the remaining constants, we are presented with:

$$\begin{aligned} V(x, y, z) &= \\ &C \exp\left(-\pi \sqrt{(n/a)^2 + (m/b)^2} x\right) \\ &\times \sin(n\pi y/a) \sin(m\pi z/b). \end{aligned}$$

4.4 Multipole Expansion

4.4.1 Approximating Potentials at Large Distances

To start let's imagine we are at a point in space and we are trying to spot some charges. If we are very far away from a localised charge distribution, it may "look" like a point charge, and the potential is, to good approximation, would be:³⁵

$$\left(\frac{1}{4\pi\epsilon_0} \right) \frac{Q}{r} \quad [V]$$

where Q is, of course, the total charge (C). We have often used this as a check on formulas for V .

But what if Q is zero?

We might think the potential is then approximately zero, and of course, we would be right, in a sense.³⁶ But we're looking for something a bit more informative than that.

³⁵If we were to look at this a slightly humorous way, we basically state; *If it looks to small to me, It must be a tiny dot.*

³⁶indeed, the potential at large r is pretty small even if Q is not zero

Case Study: Estimating Field From Far Away

To get an idea of this method let's work on the following problem:

Let's have a system where a physical electric dipole consists of two (2) equal and opposite charges ($\pm q$) separated by a distance d . Let's find the approximate potential ($V(r)$) at points far from the dipole.

To start let's define our environment. Let z_- be the distance from q and z_+ the distance from $+q$, which we can see in **Fig. 4.10**. Then

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{z_-} - \frac{q}{z_+} \right), \quad [V]$$

and using the the law of cosines we derive the following:

$$z_{\pm}^2 = r^2 + \left(\frac{d}{2} \right)^2 \mp rd \cos \theta = r^2 \left(1 \mp \frac{d}{r} \cos \theta + \frac{d^2}{4r^2} \right).$$

We're interested in the scenario of $r \gg d$, so the third term can be assumed negligible, and the binomial expansion gives us:

$$\frac{1}{r_{\pm}} \cong \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta \right)^{-1/2} \cong \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right).$$

Therefore:

$$\frac{1}{z_1} - \frac{1}{z_2} \cong \frac{d}{r^2} \cos \theta, \quad \therefore \quad V(\mathbf{r}) \cong \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2}. \quad (4.31)$$

figures.pdf

Figure 4.10

The setup in question. Here we have two (2) charges and we are trying to estimate its value from far away.

As we can see from the aforementioned equation, the potential of a dipole decreases with a factor of r^2 at large r ; as we might have anticipated, it falls off more rapidly than the potential of a point charge.

If we put together a pair of equal and opposite dipoles to make a quadrupole,³⁷ the potential goes with a factor of r^3 ; for back-to-back **quadrupoles**³⁸, it goes with a factor of r^4 ; and so on. Figure 3.27 summarizes this hierarchy; for completeness let's include the electric monopole as well which decreases with a factor of r .

³⁷A major application of quadrupole configuration is its use as a *mass spectrometer*, which is the gold standard when it comes to chemical analysis and identification.

This short example we have conducted describes a very special charge configuration. We will now try to develop a systematic expansion for the potential of any localized charge distribution, in powers of $1/r$. Figure 3.28 defines the relevant variables; the potential at r is given by

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{\lambda} \rho(\mathbf{r}') d\tau'. \quad (4.32)$$

Using the law of cosines,

$$z^2 = r^2 + (r')^2 - 2rr' \cos \alpha = r^2 \left[1 + \left(\frac{r'}{r} \right)^2 - 2 \left(\frac{r'}{r} \right) \cos \alpha \right],$$

where α is the angle between $\mathbf{r}$ and $\mathbf{r}'$. Therefore we can write:

$$z = r\sqrt{1 + \epsilon}, \quad \text{where} \quad \epsilon \equiv \left(\frac{r'}{r} \right) \left(\frac{r'}{r} - 2 \cos \alpha \right).$$

«FIGURE»

For points **well outside** the charge distribution, ϵ is much less than 1, and this invites a binomial expansion:

$$\frac{1}{z} = \frac{1}{r} (1 + \epsilon)^{-1/2} = \frac{1}{r} \left(1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \frac{5}{16}\epsilon^3 + \dots \right), \quad (4.33)$$

or, in terms of r , r' , and α :

$$\begin{aligned} \frac{1}{z} &= \frac{1}{r} \left[1 - \frac{1}{2} \left(\frac{r'}{r} \right) \left(\frac{r'}{r} - 2 \cos \alpha \right) + \frac{3}{8} \left(\frac{r'}{r} \right)^2 \left(\frac{r'}{r} - 2 \cos \alpha \right)^2 \right. \\ &\quad \left. - \frac{5}{16} \left(\frac{r'}{r} \right)^3 \left(\frac{r'}{r} - 2 \cos \alpha \right)^3 + \dots \right] \\ &= \frac{1}{r} \left[1 + \left(\frac{r'}{r} \right) (\cos \alpha) + \left(\frac{r'}{r} \right)^2 \left(\frac{3 \cos^2 \alpha - 1}{2} \right) \right. \\ &\quad \left. + \left(\frac{r'}{r} \right)^3 \left(\frac{5 \cos^3 \alpha - 3 \cos \alpha}{2} \right) + \dots \right]. \end{aligned}$$

In the last step, we have collected together like powers of (r'/r) ; surprisingly, their coefficients³⁹ are **Legendre polynomials**. The remarkable result⁴⁰ is that

$$\frac{1}{z} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \alpha). \quad (4.34)$$

³⁹which are the terms inside round brackets.

⁴⁰This suggests a second way of defining the Legendre polynomials, the first being Rodrigues' formula and $1/z$ is called the **generating function** for Legendre polynomials.



Substituting this back into Eq. (4.32), and noting that r is a **constant**, as far as the integration is concerned, we can conclude the following expression

$$V(r) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{(n+1)}} \int (r')^n P_n(\cos \alpha) \rho(r') d\tau' \quad (4.35)$$

or, written explicitly,

$$V(r) = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} \int \rho(r') d\tau' + \frac{1}{r^2} \int r' \cos \alpha \rho(r') d\tau' \right. \quad (4.36)$$

$$\left. + \frac{1}{r^3} \int (r')^2 \left(\frac{3}{2} \cos^2 \alpha - \frac{1}{2} \right) \rho(r') d\tau' + \dots \right]. \quad (4.37)$$

This is the desired result—the **multipole expansion** of V in powers of $1/r$. The first term ($n = 0$) is the monopole contribution (it goes like $1/r$); the second ($n = 1$) is the dipole (it goes like $1/r^2$); the third is quadrupole; the fourth octopole; and so on. Remember that α is the angle between $\mathbf{r}$ and $\mathbf{r}'$, so the integrals depend on the direction to the field point. If you are interested in the potential along the z' axis (or—putting it the other way around—if you orient your $\mathbf{r}'$ coordinates so the z' axis lies along $\mathbf{r}$), then α is the usual polar angle θ' . As it stands, Eq. 3.95 is exact, but it is useful primarily as an approximation scheme: the lowest nonzero term in the expansion provides the approximate potential at large r , and the successive terms tell us how to improve the approximation if greater precision is required.

4.4.2 Monopole and Dipole Terms

To put it simply, the multipole expansion is dominated⁴¹ by the monopole term:

$$V_{\text{mon}}(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}, \quad (4.38)$$

where $Q = \int \rho d\tau$ is the total charge of the configuration (C). This is just what we expect for the approximate potential at large distances from the charge. For a point charge placed at the origin, $V_{\text{mon}}(r)$ is the **exact** potential and **NOT** merely a first approximation at large r .

In this case, all the higher multi-poles vanish. If the total charge is zero, the dominant term in the potential will be the dipole (unless, of course, it **also** vanishes):

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \int r' \cos \alpha \rho(r') d\tau'.$$

Since α is the angle between $\mathbf{r}'$ and $\mathbf{r}$ (Fig. 3.28),

$$r' \cos \alpha = \hat{\mathbf{r}} \cdot \mathbf{r}'$$

and the dipole potential can be written more succinctly:

$$V_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{r}} \cdot \int \mathbf{r}' \rho(r') d\tau'.$$

⁴¹when we assume r to be large



This integral (which does not depend on $\mathbf{r}$) is called the **dipole moment** of the distribution:

$$\mathbf{p} \equiv \int \mathbf{r}' \rho(\mathbf{r}') d\tau', \quad (4.39)$$

and the dipole contribution to the potential simplifies to

$$V_{dip}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}. \quad (4.40)$$

The dipole moment is determined by the geometry (size, shape, and density) of the charge distribution. Eq. (4.39) translates in the usual way for point, line, and surface charges. Therefore, the dipole moment of a collection of **point** charges is

$$\mathbf{p} = \sum_{i=1}^n q_i \mathbf{r}'_i. \quad (4.41)$$

For a **physical dipole**,⁴² we write:

$$\mathbf{p} = q\mathbf{r}'_+ - q\mathbf{r}'_- = q(\mathbf{r}'_+ - \mathbf{r}'_-) = q\mathbf{d}, \quad (4.42)$$

⁴²equal and opposite charges, $\pm q$.

where $\mathbf{d}$ is the vector from the negative charge to the positive one (Fig. 3.29).

Is this consistent with what we got in Ex. 3.10? Yes: If you put Eq. 3.101 into Eq. 3.99, you recover Eq. 3.90. Notice, however, that this is only the **approximate** potential of the physical dipole—evidently there are higher multipole contributions. Of course, as you go farther and farther away, V_{dip} becomes a better and better approximation, since the higher terms die off more rapidly with increasing r . By the same token, at a fixed r the dipole approximation improves as you shrink the separation d . To construct a **perfect** (point) **dipole** whose potential is given exactly by Eq. 3.99, you'd have to let d approach zero. Unfortunately, you then lose the dipole term **too**, unless you simultaneously arrange for q to go to infinity! A **physical** dipole becomes a **pure** dipole, then, in the rather artificial limit $d \rightarrow 0$, $q \rightarrow \infty$, with the product $qd = p$ held fixed. When someone uses the word "dipole," you can't always tell whether they mean a physical dipole (with finite separation between the charges) or an **ideal** (point) dipole. If in doubt, assume that d is small enough (compared to r) that you can safely apply Eq. 3.99. Dipole moments are vectors, and they add accordingly: if you have two dipoles, $\mathbf{p}_1$ and $\mathbf{p}_2$, the total dipole moment is $\mathbf{p}_1 + \mathbf{p}_2$. For instance, with four charges at the corners of a square, as shown in Fig. 3.30, the net dipole moment is zero. You can see this by combining the charges in pairs (vertically, $\downarrow + \uparrow = 0$, or horizontally, $\rightarrow + \leftarrow = 0$ or by adding up the four contributions individually, using Eq. 3.100. This is a **quadrupole**, as I indicated earlier, and its potential is dominated by the quadrupole term in the multipole expansion.

4.4.3 Origin of Coordinates in Multipole Expansions

We discussed previously, that a **point charge** at the origin constitutes a "pure" monopole.

If it is **NOT** at the origin, it's no longer a pure monopole.



For instance, the charge in Fig. 3.32 has a dipole moment $\mathbf{p} = qd\hat{y}$, and a corresponding dipole term in its potential. The monopole potential $(1/4\pi\epsilon_0)q/r$ is not quite correct for this configuration; rather, the exact potential is $(1/4\pi\epsilon_0)q/\lambda$. The multipole expansion is, remember, a series in inverse powers of r (the distance to the **origin**), and when we expand $1/\nu$, we get **all** powers, not just the first.

So moving the origin⁴³ can radically alter a multipole expansion. The **monopole moment** Q does not change, since the total charge is obviously independent of the coordinate system. (In Fig. 3.32, the monopole term was unaffected when we moved q away from the origin—it's just that it was no longer the whole story: a dipole term—and for that matter all higher poles—appeared as well.) Ordinarily, the dipole moment does change when you shift the origin, but there is an important exception: If the total charge is zero, then the dipole moment is independent of the choice of origin. For suppose we displace the origin by an amount $\mathbf{a}$ (Fig. 3.33). The new dipole moment is then

$$\begin{aligned}\bar{\mathbf{p}} &= \int \bar{\mathbf{p}}' \rho(\mathbf{r}') d\tau' = \int (\mathbf{r}' - \mathbf{a}) \rho(\mathbf{r}') d\tau' \\ &= \int \mathbf{r}' \rho(\mathbf{r}') d\tau' - \mathbf{a} \int \rho(\mathbf{r}') d\tau' = \mathbf{p} - Q\mathbf{a}.\end{aligned}$$

In particular, if $Q = 0$, then $\bar{\mathbf{p}} = \mathbf{p}$. So if someone were for the dipole moment in Fig. 3.34(a),⁴⁴ we can answer with confidence “ $q\mathbf{d}$ ”, but if we're asked for the dipole moment in Fig. 3.34(b), the appropriate response would be:

With respect to what origin?

So far we have worked only with potentials. Now let's try to calculate the electric field of a **perfect** dipole. If we choose coordinates so that $\mathbf{p}$ is at the origin and points in the $\hat{z}$ direction (Fig. 3.36), then using Eq. (4.40), the potential at r, θ is:

$$V_{\text{dip}}(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}} \cdot \mathbf{p}}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}. \quad (4.43)$$

To get the field, we take the **negative gradient** of V :

$$E_r = -\frac{\partial V}{\partial r} = \frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{r^3}, \quad E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta} = \frac{1}{4\pi\epsilon_0} \frac{p \sin \theta}{r^3}, \quad E_\phi = -\frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} = 0.$$

Which gives us the result we have been looking for:

$$\mathbf{E}_{\text{dip}}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}). \quad (4.44)$$

This formula makes explicit reference to a particular coordinate system⁴⁵ and assumes a particular orientation for $\hat{\mathbf{p}}$ which is along $\mathbf{z}$ axis. It can be recast in a coordinate-free form, analogous to the potential in Eq. (4.40).

Notice that the dipole field falls off as the inverse **cube** of r ; the **monopole** field $(Q/4\pi\epsilon_0 r^2)\hat{\mathbf{r}}$ goes as the inverse square, of course. Quadrupole fields go like $1/r^4$, octopole like $1/r^5$, and so on. (This merely reflects the fact that monopole **potentials** fall off like $1/r$, dipole like $1/r^2$, quadrupole like

⁴³or, what amounts to the same thing, moving the **charge**

⁴⁴Perhaps while you are commiserating while having lunch between lectures, perhaps?

⁴⁵In which case it is spherical



Figure 4.11

A visual comparison between (a) physical description of a dipole and (b) a **pure** dipole.

figures.pdf

$1/r^3$, and so on-the gradient introduces another factor of $1/r$.) Figure **Fig. 4.11a** shows the field lines of a "pure" dipole as described in Eq. (4.44).

For comparison, Let's also compare the pure representation with the field lines for a "physical" dipole, also shown in **Fig. 4.11b**.

Please observe, how similar the two (2) pictures become if we blot out the central region; up close, however, they are entirely different. Only for points $r \gg d$ does Eq. 3.103 represent a valid approximation to the field of a physical dipole.

As discussed earlier, this behaviour can be reached either by going to large r or by squeezing the charges very close together.⁴⁶

⁴⁶Even in the limit, there remains an infinitesimal region at the origin where the field of a physical dipole points in the "wrong" direction.



Part III

Magnetic Fields

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The experimental investigation by which Ampère established the law of the mechanical action between electric currents is one of the most brilliant achievements in science.

The whole, theory and experiment, seems as if it had leaped, full grown and full armed, from the brain of the “Newton of electricity”. It is perfect in form, and unassailable in accuracy, and it is summed up in a formula from which all the phenomena may be deduced, and which must always remain the cardinal formula of electro-dynamics.

(James Clerk Maxwell in A Treatise on Electricity and Magnetism (1873) quoted from 3rd edition (1892) Vol. 2, Ch. 3, p. 175.)

Chapter 5

Magnetostatics

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5.1 Lorentz Force Law

5.1.1 Magnetic Fields

Let's recall the key question classical electrodynamics tries to understand:

We have a collection of charges $q_1, q_2, \dots$ ¹ and we want to calculate the force they exert on some other charge Q .²

¹called the "source" charges,
²Which are, as we recall are called the "test charge".

If we were to use the principle of **superposition**, it is sufficient to find the force of a **single** source charge.

The total is then **the vector sum** of all the individual forces.

Till now, we have confined our attention to the simplest case, **electrostatics**, in which the source charge is at rest.³ The time has come to consider the forces between charges in **motion**.

³This restriction does **NOT** necessary apply to the test charge.

To give some sense of what is in store, imagine we've set up the following demonstration:

There are two (2) wires hanging from the ceiling, a few cm apart and we have wired in a

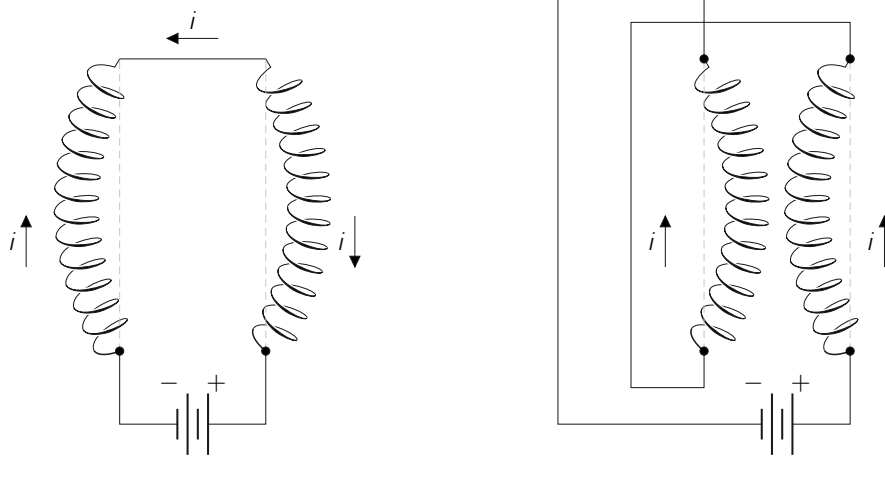


Figure 5.1

A simple coil experiment
(a) Here we see the currents in opposite direction repel and (b) Current in the same directions attract.

way that when we turn on a current, it passes up one wire and back down the other as seen in **Fig. 5.1a**. We observe the wires jump apart. They evidently repel one another.

How do we explain this?

We might think the battery⁴ is charging up the wire, and that the force is simply due to the electrical repulsion of same polarity charges. But this statement would be **incorrect**, as we could hold up a test charge near these wires, and there would be **no** force acting on it,⁵ given the wires are in fact electrically neutral.⁶ Moreover, if we were to hook up the demonstration so as to make the current flow up **both** wires **Fig. 5.1b**, they are found to **attract** each other.

Whatever force accounts for the attraction of parallel currents and the repulsion of antiparallel ones is **NOT** electrostatic in nature but of magnetic forces. Whereas a **stationary** charge produces only an electric field ($\mathbf{E}$) in the space around it, a **moving** charge generates, in addition, a magnetic field ($\mathbf{B}$).

In fact, magnetic fields are a lot easier to detect, in practice as all we need really need is a compass.⁷

Ordinarily, this means **north**, in response to the earth's magnetic field, but in the laboratory, where typical fields may be hundreds of times stronger than that, the compass indicates the direction of whatever magnetic field is present.

Now, if we were to hold up a tiny compass in the vicinity of a current-carrying wire, we would quickly discover a very peculiar thing: The field does **NOT** point **toward** the wire, nor **away** from it, but rather it **circles around the wire**. In fact, if we were to grab the wire with our right hand with the thumb in the direction of the current, our fingers curl **around in the direction of the magnetic field** (Fig. 5.3).

How can such a field lead to a force of attraction on a nearby parallel current?

⁴or whatever drives the current.

⁵This is **NOT** necessarily the correct description but to keep it simple, let's assume as such.

⁶It's true that electrons are flowing down the line as that's the definition of current, but there are just as many stationary plus charges as moving minus charges on any given segment.

⁷How these devices work is irrelevant at the moment. It is enough to know that the needle points in the direction of the local magnetic field.

At the second wire, the magnetic field points **into the page** Fig. 5.1b, the current is **upward**, and yet the resulting force is **to the left**.

It's going to take a strange law to account for these directions.

5.1.2 Magnetic Forces

In fact, this combination of directions is just right for a **cross product**:

the magnetic force ($\mathbf{F}$) on a charge (Q), moving with velocity ($\mathbf{v}$) under the effect of a magnetic field ($\mathbf{B}$), is⁸

$$\mathbf{F}_{\text{mag}} = Q(\mathbf{v} \times \mathbf{B}) \quad (5.1)$$

This is known as the **Lorentz force law**. In the presence of both electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields, the net force on Q would be:

$$\mathbf{F} = Q[\mathbf{E} + (\mathbf{v} \times \mathbf{B})]. \quad (5.2)$$

Our main job from now on is to calculate the magnetic field ($\mathbf{B}$). But before we proceed, it is worthwhile to take a closer look at the *Lorentz force law* itself. It is an important law, and it leads to some truly bizarre particle trajectories.

⁸While we shall treat $\mathbf{B}$ as a vector it is in fact what is called a pseudovector. A pseudovector, a vector-like quantity that behaves like a normal (polar) vector under rotations but flips sign under a mirror reflection (parity inversion). Unlike true vectors (like velocity, which flips in a mirror), pseudovectors (like angular velocity, torque, magnetic field) retain their direction even when mirrored, often appearing to obey the left-hand rule after reflection.

Example 5.1: Motion inside a Cyclotron

The principle motion of a charged particle in a magnetic field is **circular**, with the magnetic force ($\mathbf{F}$) providing the centripetal acceleration. In Fig. 5.5, a uniform magnetic field points **into the page**. If the charge Q moves counterclockwise, with speed v , around a circle of radius R , the magnetic force points **inward**, and has a fixed magnitude QvB —just right to sustain uniform circular motion:

$$QvB = m \frac{v^2}{R}, \quad \text{or} \quad p = QBR \quad (5.3)$$

where m is the particle's mass and $p = mv$ is its momentum. Eq. (5.3) is known as the **cyclotron formula** because it describes the motion of a particle in a cyclotron—the first of the modern particle accelerators.

It also suggests a simple experimental technique for finding the momentum of a charged particle:

send it through a region of known magnetic field, and measure the radius of its trajectory.

This is in fact the standard means for determining the momenta of elementary particles.

Note — Here we assume the particle has no $\hat{z}$ -component.

A Small Digression

An implication of the Lorentz force law given in Eq. (5.1) deserves special attention:

Magnetic forces do **NOT** work.

For if Q moves an amount $d\mathbf{l} = \mathbf{v} dt$, the work done is

$$dW_{\text{mag}} = \mathbf{F}_{\text{mag}} \cdot d\mathbf{l} = Q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = 0. \quad (5.4)$$

This follows as $(\mathbf{v} \times \mathbf{B})$ is perpendicular to $\mathbf{v}$, so $(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} = 0$. Magnetic forces may alter the **direction** in which a particle moves, but they cannot speed it up or slow it down. The fact that magnetic forces do no work is an elementary and direct consequence of the Lorentz force law, but there are many situations in which it **appears** so manifestly false that one's confidence is bound to waver. When a magnetic crane lifts the carcass of a junked car, for instance, **something** is obviously doing work, and it seems perverse to deny that the magnetic force is responsible. Well, perverse or not, deny it we must, and it can be a very subtle matter to figure out who **does** deserve the credit in such circumstances.

5.1.3 Currents

The current in a wire is the charge per unit time passing a given point. By definition, negative charges moving to the left count the same as positive ones to the right. This conveniently reflects the physical fact that almost all phenomena involving moving charges depend on the product of charge and their velocity.

Information: Sign Independence of Current

If we were to reverse the signs of q and $\mathbf{v}$, we would get the same answer, so it doesn't really matter which you have.⁹

⁹The Lorentz force law is a case in point with the Hall effect is a significant exception.

In practice, it is ordinarily the negatively charged electrons which actually does the moving, that is in the direction **opposite to the electric current**. To avoid the complications this entails, let us pretend it's the positive charges which move.

Current is measured in coulombs-per-second, or amperes (A):

$$1 \text{ A} = 1 \text{ C} \cdot \text{s}^{-1} \quad (5.5)$$

A line charge (λ) traveling down a wire at speed $\mathbf{v}$ constitutes a current:

$$I = \lambda v, \quad (5.6)$$

because a segment of length $v\Delta t$, carrying charge $\lambda v\Delta t$, passes point P in a time interval Δt . Current is actually a **vector**:

$$\mathbf{I} = \lambda \mathbf{v} \quad (5.7)$$



Because the path of the flow is dictated by the shape of the wire, one doesn't ordinarily bother to display the direction of I explicitly,¹⁰ but when it comes to surface and volume currents we cannot afford to be so casual, and for the sake of notational consistency it is a good idea to acknowledge the vectorial character of currents right from the start.

A neutral wire, of course, contains as many stationary positive charges as mobile negative ones. The former do not contribute to the current—the charge density λ in Eq. (5.6) refers only to the **moving** charges. In the unusual situation where **both** types move, $I = \lambda[+]v_+ + \lambda_-v_-$.

The magnetic force on a segment of current-carrying wire is

$$\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B}) dq = \int (\mathbf{v} \times \mathbf{B}) \lambda dl = \int (I \times \mathbf{B}) dl. \quad (5.8)$$

Given I and dl both point in the same direction, we can just as well write this as

$$\mathbf{F}_{\text{mag}} = \int I (d\mathbf{l} \times \mathbf{B}). \quad (5.9)$$

Typically, the current is constant¹¹ along the wire, and in that case I can come **outside the integral**:

$$\mathbf{F}_{\text{mag}} = I \int (d\mathbf{l} \times \mathbf{B}). \quad (5.10)$$

When charge flows over a **surface**, we describe it by the surface current density ($\mathbf{K}$) defined as follows: let us consider a “ribbon” of infinitesimal width $dl_{\perp}$, running parallel to the flow. If the current in this ribbon is dI , the surface current density is

$$\mathbf{K} \equiv \frac{dI}{dl_{\perp}}. \quad (5.11)$$

In words, $\mathbf{K}$ is the **current per unit width**. In particular, if the surface charge density is σ and its velocity is $\mathbf{v}$, then

$$\mathbf{K} = \sigma \mathbf{v}. \quad (5.12)$$

In general, $\mathbf{K}$ will vary from point to point over the surface, reflecting variations in σ and/or $\mathbf{v}$. The magnetic force on the surface current is:

$$\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B}) \sigma da = \int (\mathbf{K} \times \mathbf{B}) da. \quad (5.13)$$

Just as $\mathbf{E}$ suffers a discontinuity at a surface **charge**, so $\mathbf{B}$ is discontinuous at a surface current. In Eq. (5.13), we must be careful to use the average field.

When the flow of charge is distributed throughout a three-dimensional region, we describe it by the volume current density ($\mathbf{J}$). To image this consider a “tube” of infinitesimal cross section $da_{\perp}$, running parallel to the flow (Fig. 5.14). If the current in this tube is dI , the volume current density is

$$\mathbf{J} \equiv \frac{dI}{da_{\perp}} \quad (5.14)$$

¹⁰For the same reason, if we are describing a locomotive constrained to move along a specified track, we would probably speak of its speed, rather than its velocity.

¹¹in magnitude.

In words, J is the **current per unit area**. If the (mobile) volume charge density is ρ and the velocity is $\mathbf{v}$, then

$$\mathbf{J} = \rho \mathbf{v} \quad (5.15)$$

The magnetic force on a volume current is therefore

$$\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B}) \rho d\tau = \int (\mathbf{J} \times \mathbf{B}) d\tau. \quad (5.16)$$

According to Eq. (5.14), the total current crossing a surface S can be written as¹²

$$I = \int_c \mathbf{J} \cdot d\mathbf{a}_\perp = \int_c \mathbf{J} \cdot d\mathbf{a}. \quad (5.17)$$

¹²The dot product serves neatly to pick out the appropriate component of $d\mathbf{a}$.

In particular, the charge per unit time leaving a volume V is

$$\oint_c \mathbf{J} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{J}) d\tau.$$

Because charge is conserved, whatever flows out through the surface must come at the expense of what remains inside:

$$\int_V (\nabla \cdot \mathbf{J}) d\tau = -\frac{d}{dt} \int_V \rho d\tau = -\int_V \left(\frac{\partial \rho}{\partial t} \right) d\tau.$$

(The minus sign reflects the fact that an outward flow decreases the charge left in V .) Since this applies to **any** volume, we conclude that

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}. \quad (5.18)$$

This is the precise mathematical statement of local charge conservation; it is called the **continuity equation**.

For future reference, let me summarize the "dictionary" we have implicitly developed for translating equations into the forms appropriate to point, line, surface, and volume currents:

$$\sum_{i=1}^n (-) q_i \mathbf{v}_i \sim \int_{\text{line}} (-) I d\mathbf{l} \sim \int_{\text{surface}} (-) \mathbf{K} d\mathbf{a} \sim \int_{\text{volume}} (-) \mathbf{J} d\tau. \quad (5.19)$$

This correspondence, which is analogous to $q \sim \lambda dl \sim \sigma da \sim \rho d\tau$ for the various charge distributions, generates Eqs. 5.15, 5.24, and 5.27 from the original Lorentz force law (5.1).



5.2 The Biot-Savart Law

5.2.1 Steady Currents

Stationary charges produce electric fields that are constant in time; hence the term electrostatics.¹³ Steady currents produce magnetic fields that are constant in time; the theory of steady currents is called magnetostatics.

¹³Actually, it is not necessary that the charges be stationary, but only that the charge density at each point be constant.

Stationary charges constant electric fields: electrostatics. $\Rightarrow$ Steady currents constant magnetic fields: magnetostatics. $\Rightarrow$

By steady current we of course mean a continuous flow which has been going on forever, without change and without charge piling up anywhere. Formally, both electro and magnetostatics follow the following behaviours:

$$\frac{\partial \rho}{\partial t} = 0, \quad \frac{\partial \mathbf{J}}{\partial t} = 0 \quad (5.20)$$

at all places and all times. Of course, there's no such thing in practice as a **truly** steady current, any more than there is a **truly** stationary charge. In this sense, both electrostatics and magnetostatics describe artificial worlds that exist only in textbooks. However, they represent suitable **approximations** as long as the actual fluctuations are remote, or gradual-in fact, for most purposes magnetostatics applies very well to household currents, which alternate 120 times a second!

Notice that a moving **point** charge **cannot possibly constitute a steady current**. If it's here one instant, it's gone the next. This may seem like a minor thing to you, but it's a major headache for me. I developed each topic in electrostatics by starting out with the simple case of a point charge at rest; then I generalized to an arbitrary charge distribution by invoking the superposition principle. This approach is not open to us in magnetostatics because a moving point charge does not produce a static field in the first place. We are forced to deal with extended current distributions right from the start, and, as a result, the arguments are bound to be more cumbersome.

When a steady current flows in a wire, its magnitude I must be the same all along the line; otherwise, charge would be piling up somewhere, and it wouldn't be a steady current. More generally, since $\partial \rho / \partial t = 0$ in magnetostatics, the continuity equation (5.29) becomes

$$\nabla \cdot \mathbf{J} = 0. \quad (5.21)$$

5.2.2 The Magnetic Field of a Steady Current

The magnetic field of a steady line current is given by the Biot-Savart law:



$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{z}}}{r^2} d\ell' = \frac{\mu_0}{4\pi} I \int \frac{d\ell' \times \hat{\mathbf{z}}}{r^2}. \quad (5.22)$$

The integration is along the current path, in the direction of the flow; $d\ell'$ is an element of length along the wire, and λ , as always, is the vector from the source to the point $\mathbf{r}$ (Fig. 5.17). The constant μ_0 is called the permeability of free space¹⁴

$$\mu_0 = 4\pi \times 10^{-7} \quad (5.23)$$

These units are such that B itself comes out in newtons per ampere-meter (as required by the Lorentz force law), or teslas (T):

$$1T = 1N/(A \cdot m) \quad (5.24)$$

As the starting point for magnetostatics, the Biot-Savart law plays a role analogous to Coulomb's law in electrostatics. Indeed, the $1/r^2$ dependence is common to both laws.

For surface and volume currents, the Biot-Savart law becomes

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(\mathbf{r}') \times \hat{\mathbf{v}}\ell}{r^2} da' \quad \text{and} \quad \mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{v}}\ell}{r^2} d\tau', \quad (5.42)$$

respectively. You might be tempted to write down the corresponding formula for a moving point charge, using the "dictionary" (Eq. (5.19)):

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \hat{\mathbf{z}}}{r^2}, \quad (5.25)$$

but this is simply wrong. As I mentioned earlier, a point charge does not constitute a steady current, and the Biot-Savart law, which only holds for steady currents, does **not** correctly determine its field.

The superposition principle applies to magnetic fields just as it does to electric fields: if you have a collection of source currents, the net field is the (vector) sum of the fields due to each of them taken separately.

¹⁴This is an exact number, not an empirical constant. It serves (via Eq. (5.23)) to define the ampere, and the ampere in turn defines the coulomb.



5.3 Divergence and Curl of B

5.3.1 Straight Line Currents

The magnetic field of an infinite straight wire is shown in **Fig. 5.2** where the current is coming **out** of the page. At a glance, it is clear that this field has a nonzero curl¹⁵ let's calculate it. According to Eq. (??), the integral of **B** around a circular path of radius s , centered at the wire, is

$$\oint \mathbf{B} \cdot d\mathbf{l} = \oint \frac{\mu_0 I}{2\pi s} d\mathbf{l} = \frac{\mu_0 I}{2\pi s} \oint d\mathbf{l} = \mu_0 I.$$

Notice that the answer is independent of s ; that's because B decreases at the same rate as the circumference **increases**. In fact, it doesn't have to be a circle; **any** old loop that encloses the wire would give the same answer.

For if we use cylindrical coordinates (s, ϕ, z) , with the current flowing along the z axis, $\mathbf{B} = (\mu_0 I / 2\pi s) \hat{\phi}$ and $d\mathbf{l} = ds \hat{s} + s d\phi \hat{\phi} + dz \hat{z}$, so

$$\oint \mathbf{B} \cdot d\mathbf{l} = \frac{\mu_0 I}{2\pi} \oint \frac{1}{s} s d\phi = \frac{\mu_0 I}{2\pi} \int_0^{2\pi} d\phi = \mu_0 I.$$

This assumes the loop encircles the wire exactly once; if it went around twice, then ϕ would run from 0 to 4π , and if it didn't enclose the wire at all, then ϕ would go from ϕ_1 to ϕ_2 and back again, with $\int d\phi = 0$ (Fig. 5.28).

Now suppose we have a **bundle** of straight wires. Each wire that passes through our loop contributes $\mu_0 I$, and those outside contribute nothing (Fig. 5.29). The line integral will then be

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}, \quad (5.26)$$

where I_{enc} stands for the total current enclosed by the integration path. If the flow of charge is represented by a volume current density $\mathbf{J}$, the enclosed current is

$$I_{\text{enc}} = \int \mathbf{J} \cdot d\mathbf{a}, \quad (5.27)$$

with the integral taken over any surface bounded by the loop. Applying Stokes' theorem to Eq. (5.26), then, $\int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a}$, and hence

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (5.28)$$

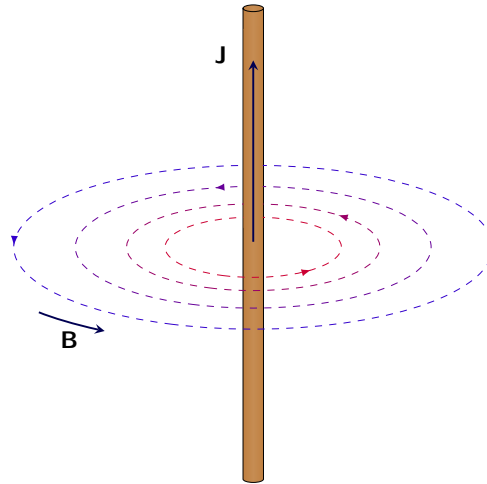


Figure 5.2

The magnetic field of an infinite straight wire is shown.

¹⁵something you'll never see in an **electrostatic** field.

With minimal labor, we have actually obtained the general formula for the curl of $\mathbf{B}$. But our derivation is seriously flawed by the restriction to infinite straight line currents. Most current configurations **cannot** be constructed out of infinite straight wires, and we have no right to assume that Eq. (5.28) applies to them. So the next section is devoted to the formal derivation of the divergence and curl of $\mathbf{B}$, starting from the Biot-Savart law itself.

5.3.2 Amperés Law

The equation for the curl of $\mathbf{B}$,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \quad (5.29)$$

is called Ampère's law, in differential form. It can be converted to integral form by the usual device of applying one of the fundamental theorems.¹⁶

¹⁶In this case it would be Stokes' theorem.

$$\int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a}$$

Now, $\int \mathbf{J} \cdot d\mathbf{a}$ is the total current passing through the surface, which we call I_{enc} (the current enclosed by the Amperian loop). Thus

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}. \quad (5.30)$$

This is the integral version of Ampère's law; it generalizes Eq. (5.26) to arbitrary steady currents. Notice that Eq. (5.30) inherits the sign ambiguity of Stokes' theorem (Sect. 1.3.5):

Which way around the loop am I supposed to go? and which direction through the surface corresponds to a "positive" current?

And The resolution, as always, is the right-hand rule: If the fingers of your right hand indicate the direction of integration around the boundary, then your thumb defines the direction of a positive current.

Just as the Biot-Savart law plays a role in magnetostatics that Coulomb's law assumed in electrostatics, so Ampère's plays the part of Gauss's: In particular, for currents with appropriate symmetry, Ampère's law in integral form offers a lovely and extraordinarily efficient way of calculating the magnetic field.

Like Gauss's law, Ampère's law is always true (for steady currents), but it is not always **useful**. Only when the symmetry of the problem enables you to pull $\mathbf{B}$ outside the integral $\oint \mathbf{B} \cdot d\mathbf{l}$ can you calculate the magnetic field from Ampère's law. When it **does** work, it's by far the fastest method; when it doesn't, you have to fall back on the Biot-Savart law. The current configurations that can be handled by Ampère's law are 1. Infinite straight lines 2. Infinite planes 3. Infinite solenoids 4.

Toroids The last of these is a surprising and elegant application of Ampère's law. As in Exs. 5.8 and 5.9, the hard part is figuring out the **direction** of the field (which we will now have done, once and for all, for each of the four geometries); the actual application of Ampère's law takes only one line.

Chapter 6

Magnetic Fields in Matter

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6.1 Magnetisation

6.1.1 Diamagnets, Paramagnets, and Ferromagnets

If the topic of **magnetism** ever comes to play during lunch break,¹ you may think about refrigerator decorations, compass needles, and the North Pole which initially seems to have no obvious connection with moving charges or current-carrying wires.

¹Perhaps when you are discussing the current events or why the elevator is again under maintenance at MCI IV.

However, it is important to state, all magnetic phenomena are caused by electric charges in motion. If we could examine a piece of magnetic material on an atomic scale you would find tiny currents:

electrons orbiting around nuclei and spinning about their axes.

For macroscopic domain,² these current loops are **minuscule** so we treat them as magnetic dipoles. Ordinarily, they cancel each other out due to the **random orientation of the atoms**. But when a magnetic field is applied, a **net alignment** of these magnetic dipoles occurs, and the medium becomes magnetically polarised, or **magnetised**.

²The domain in which most day-to-day events occur.

Unlike electric polarisation we have looked previously, which is **almost** always in the same direction

as the $\mathbf{E}$ -field, some materials acquire a magnetisation **parallel** to $\mathbf{B}$, called **paramagnets** and some **opposite** to $\mathbf{B}$, called **diamagnets**. A few substances³ retain their magnetization even after the external field has been removed—for these, the magnetisation is **NOT** determined by the **present** field but by the whole magnetic “history” of the object.

³called **ferromagnets**, named after to the most common example, iron (Fe).

Information: Lode Stones

Lodestones are naturally magnetized pieces of the mineral magnetite [2, 3]. They are naturally occurring magnets, which can attract iron. The property of magnetism was first discovered in antiquity through lodestones [4]. Pieces of lodestone, suspended so they could turn, were the first magnetic compasses, [4] and their importance to early navigation is indicated by the name lodestone, which in Middle English means “course stone” or “leading stone”, from the now-obsolete meaning of lode as “journey, way”.

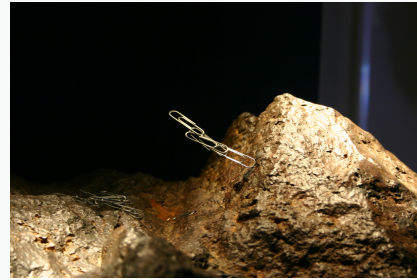


Figure 6.1: Lodestone in the Hall of Gems of the Smithsonian [1].

Lodestone is one of only a very few minerals that is found naturally magnetised.

Permanent Magnet (PM)s made of iron are the most familiar examples of magnetism, but from a theoretical point of view they are the most complicated.

We will have a look at ferromagnetism for the end of the chapter, and begin with qualitative models of paramagnetism and diamagnetism.

6.1.2 Torques and Forces on Magnetic Dipoles

A magnetic dipole experiences a torque under the effect of a magnetic field ($\mathbf{B}$), similar to an electric dipole in an electric field.

Let's calculate the torque on a rectangular current loop in a uniform field $\mathbf{B}$ ⁴. To begin, centre the loop at the origin, and tilt it an angle θ from the $\hat{z}$ -axis towards the $\hat{y}$ -axis (Fig. 6.2).

Let $\mathbf{B}$ point in the $\hat{z}$ direction. The forces on the two (2) sloping sides cancel.⁵ The forces on the “horizontal” sides are likewise equal and opposite, which makes the net **force** on the loop is zero, but regardless they generate a torque:

$$\mathbf{N} = aF \sin \theta \hat{x}. \quad (6.1)$$

The magnitude of the force on each of these segments is

$$F = IbB, \quad (6.2)$$

and therefore

⁴Since any current loop could be built up from infinitesimal rectangles, with all the “internal” sides canceling, as indicated in Fig. 6.1, there is no real loss of generality here.

⁵they tend to **stretch** the loop, but they don't **rotate** it

$$\mathbf{N} = IabB \sin \theta \hat{\mathbf{x}} = mB \sin \theta \hat{\mathbf{x}} \quad \text{or} \quad \mathbf{N} = \mathbf{m} \times \mathbf{B} \quad (6.3)$$

where $m = Iab$ is the magnetic dipole moment of the loop with Eq. (6.3) giving the torque on any localized current distribution, in the presence of a **uniform** field; in a nonuniform field it is the exact torque⁶ for a perfect dipole of infinitesimal size.

⁶about the center.

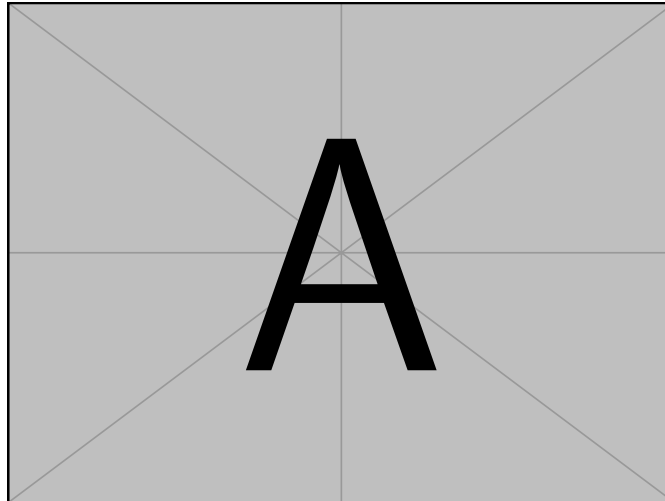


Figure 6.2: Figure 6.1

Here, Eq. (6.3) is **identical** in form to the electrical analog, Eq. 4.4: $\mathbf{N} = \mathbf{p} \times \mathbf{E}$. In particular, the torque is again in such a direction as to line the dipole up **parallel** to the field. It is this torque which accounts for **paramagnetism** effect.

Given every electron constitutes a magnetic dipole,⁷ we might expect paramagnetism to be a **universal phenomenon**.

⁷picture it, if you wish, as a tiny spinning sphere of charge

Actually, quantum mechanics⁸ tends to lock the electrons within a given atom together in pairs with opposing spins,⁹ and this effectively **neutralises** the torque on the combination.

⁸specifically, the Pauli exclusion principle.

As a result, paramagnetism most often occurs in atoms or molecules with an **odd number** of electrons, where the “extra” unpaired member is subject to the magnetic torque.

⁹This is **NOT** always true for the outermost electrons in unfilled shells.

Even here, the alignment is far from complete, given random thermal collisions tend to destroy the order.

In a uniform field, the net **force** on a current loop is zero:

$$\mathbf{F} = I \oint (d\mathbf{l} \times \mathbf{B}) = I \left(\oint d\mathbf{l} \right) \times \mathbf{B} = 0; \quad (6.4)$$

Here, the constant $\mathbf{B}$ comes **outside the integral**, and the net displacement $(\oint d\mathbf{l})$ around a closed loop vanishes. In a **non-uniform** field this is no longer the case.

As an example, assume a circular wire ring of radius of R , carrying a current of I , is suspended above a short solenoid in the “fringing” region (Fig. 6.3). Here $\mathbf{B}$ has a radial component, and there is a net downward force on the loop (Fig. 6.4):

$$F = 2\pi IRB \cos \theta. \quad (6.5)$$

For an **infinitesimal** loop, with dipole moment $\mathbf{m}$, in a field $\mathbf{B}$, the force is:

$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B}) \quad (6.6)$$

Once again the magnetic formula is identical to its electrical “twin,” if we write the latter in the form of:

$$\mathbf{F} = \nabla(\mathbf{p} \cdot \mathbf{E})$$

Information: An Unfortunate Analogy

If you're starting to get a sense of déjà vu, perhaps you will have more respect for those early physicists who thought magnetic dipoles consisted of positive and negative magnetic “charges” (north and south “poles,” they called them), separated by a small distance, just like electric dipoles (Fig. 6.5(a)). They wrote down a “Coulomb’s law” for the attraction and repulsion of these poles, and developed the whole of magnetostatics in exact analogy to electrostatics. It’s not a bad model, for many purposes—it gives the correct field of a dipole (at least, away from the origin), the right torque on a dipole (at least, on a stationary dipole), and the proper force on a dipole (at least, in the absence of external currents). But it’s bad physics, because there’s no such thing as a single magnetic north pole or south pole. If you break a bar magnet in half, you don’t get a north pole in one hand and a south pole in the other; you get two complete magnets. Magnetism is not due to magnetic monopoles, but rather to moving electric charges; magnetic dipoles are tiny current loops (Fig. 6.5(c)), and it’s an extraordinary thing, really, that the formulas involving $\mathbf{m}$ bear any resemblance to the corresponding formulas for $\mathbf{p}$. Sometimes it is easier to think in terms of the “Gilbert” model of a magnetic dipole (separated monopoles), instead of the physically correct “Ampere” model (current loop).

Indeed, this picture occasionally offers a quick and clever solution to an otherwise cumbersome problem

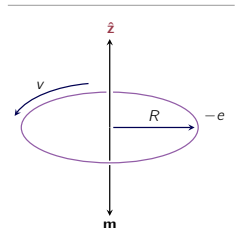
We just copy the corresponding result from electrostatics, changing:

$$p \rightarrow m \quad \frac{1}{\epsilon_0} \rightarrow \mu_0 \quad \mathbf{E} \rightarrow \mathbf{B}.$$

However, if we want to have a closer look at the features of the dipoles, the two models can produce significant results.

6.1.3 Effect of a Magnetic Field on Atomic Orbits

Electrons **NOT** only **spin** as they also **revolve** around the nucleus. To give a better understanding of the aforementioned statement, let’s assume the orbit is a circle¹⁰ of radius R .



¹⁰A visualisation of the electron orbit.



While, under the strictest technical definitions, this orbital motion does **NOT** constitute a steady current, in practice the period:

$$T = 2\pi R/v$$

is imperceptibly short that we might as well assume it is **steady-current**.¹¹

$$I = \frac{-e}{T} = -\frac{ev}{2\pi R}.$$

Following this chain of thought, the orbital dipole moment ($I\pi R^2$) is:

$$\mathbf{m} = -\frac{1}{2}evR\hat{\mathbf{z}}. \quad (6.7)$$

Similar to any other magnetic dipole, this one is subject to a torque ($\mathbf{m} \times \mathbf{B}$) when we turn on a magnetic field.

But it's a lot harder to tilt the entire orbit than it is the spin, so the orbital contribution to paramagnetism is small.

There is, however, a more significant effect on the orbital motion which is worth mention and it is that the electron speeds up or slows down, depending on the orientation of $\mathbf{B}$. For whereas the centripetal acceleration v^2/R is ordinarily sustained by electrical forces alone,

To avoid confusion with the magnetic dipole moment m , we will write the electron mass with subscript: m_e .

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} = m_e \frac{v^2}{R}, \quad (6.8)$$

in the presence of a magnetic field there is an additional force, described as:

$$-e(\mathbf{v} \times \mathbf{B})$$

For the sake of argument, let's say that $\mathbf{B}$ is perpendicular to the plane of the orbit, as shown in Fig. 6.10; then

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} + e\bar{v}B = m_e \frac{\bar{v}^2}{R}. \quad (6.9)$$

Under these conditions, the new speed $\bar{v}$ is **greater** than v :

$$e\bar{v}B = \frac{m_e}{R}(\bar{v}^2 - v^2) = \frac{m_e}{R}(\bar{v} + v)(\bar{v} - v), \quad (6.10)$$

or, assuming the change $\Delta v = \bar{v} - v$ is small,

$$\Delta v = \frac{eRB}{2m_e}. \quad (6.11)$$

When $\mathbf{B}$ is turned on, then, the electron speeds up.¹² A change in orbital speed means a change in the dipole moment (Eq. (6.7)):

$$\Delta \mathbf{m} = -\frac{1}{2}e(\Delta v)R\hat{\mathbf{z}} = -\frac{e^2 R^2}{4m_e} \mathbf{B}. \quad (6.12)$$

¹¹The minus sign accounts for the negative charge of the electron.

¹²I said (Eq. 5.11) that magnetic fields do no work, and are incapable of speeding a particle up. I stand by that. However, as we shall see in Chapter 7, a changing magnetic field induces an electric field, and it is the latter that accelerates the electrons in this instance.

Notice that the change in $\mathbf{m}$ is opposite to the direction of $\mathbf{B}$. (An electron circling the other way would have a dipole moment pointing upward, but such an orbit would be slowed down by the field, so the **change** is still opposite to $\mathbf{B}$.) Ordinarily, the electron orbits are randomly oriented, and the orbital dipole moments cancel out. But in the presence of a magnetic field, each atom picks up a little "extra" dipole moment, and these increments are all **antiparallel** to the field. This is the mechanism responsible for diamagnetism. It is a universal phenomenon, affecting all atoms. However, it is typically much weaker than paramagnetism, and is therefore observed mainly in atoms with even numbers of electrons, where paramagnetism is usually absent.

In deriving Eq. (6.12), I assumed that the orbit remains circular, with its original radius R . I cannot offer a justification for this at the present stage. If the atom is stationary while the field is turned on, then my assumption can be proved—this is not magnetostatics, however, and the details will have to await Chapter 7 (see Prob. 7.52). If the atom is moved into the field, the situation is enormously more complicated. But never mind—I'm only trying to give you a qualitative account of diamagnetism. Assume, if you prefer, that the velocity remains the same while the **radius** changes—the formula (Eq. (6.12)) is altered (by a factor of 2), but the qualitative conclusion is unaffected. The truth is that this classical model is fundamentally flawed,¹³ so there's not much point in refining the details. What is important is the empirical fact that in diamagnetic materials the induced dipole moments point opposite to the magnetic field.

¹³ diamagnetism is really a quantum phenomenon.

6.1.4 Magnetization

In the presence of a magnetic field, matter becomes **magnetized**; that is, upon microscopic examination, it will be found to contain many tiny dipoles, with a net alignment along some direction. We have discussed two mechanisms that account for this magnetic polarization:

1. paramagnetism (the dipoles associated with the spins of unpaired electrons experience a torque tending to line them up parallel to the field), and
2. diamagnetism (the orbital speed of the electrons is altered in such a way as to change the orbital dipole moment in a direction opposite to the field)

Whatever the cause, we describe the state of magnetic polarization by the vector quantity

$$\mathbf{M} \equiv \text{magnetic dipole moment per unit volume} \quad (6.13)$$

where $\mathbf{M}$ is called the magnetization and plays a role analogous to the polarization $\mathbf{P}$ in electrostatics. In the following section, we will not worry about how the magnetization **got** there—it could be paramagnetism, diamagnetism, or even ferromagnetism—we shall take $\mathbf{M}$ as given, and calculate the field this magnetization itself produces.

Incidentally, it may have surprised you to learn that materials other than the famous ferromagnetic trio (iron, nickel, and cobalt) are affected by a magnetic field at all. You cannot, of course, pick up a piece of wood or aluminum with a magnet. The reason is that diamagnetism and paramagnetism are extremely weak:

It takes a delicate experiment and a powerful magnet to detect them at all.

If you were to suspend a piece of paramagnetic material above a solenoid, as in Fig. 6.3, the induced magnetization would be upward, and hence the force downward. By contrast, the magnetization of a diamagnetic object would be downward and the force upward. In general, when a sample is placed in a region of nonuniform field, the paramagnet is attracted into the field, whereas the diamagnet is repelled away. But the actual forces are pitifully weak—in a typical experimental arrangement the force on a comparable sample of iron would be 10^4 or 10^5 times as great. That's why it was reasonable for us to calculate the field inside a piece of copper wire, say, in Chapter 5, without worrying about the effects of magnetization.

6.2 The Field of a Magnetised Object

6.2.1 Bound Currents

Let us assume we have a piece of magnetized material and the magnetic dipole moment per unit volume ($\mathbf{M}$) is given.

What field does this object produce?

Well, the vector potential of a single dipole $\mathbf{m}$ is given by Eq. 5.85:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{z}}}{r^2}. \quad (6.14)$$

In the magnetized object, each volume element $d\tau'$ carries a dipole moment $\mathbf{M}d\tau'$, so the total vector potential is (Fig. 6.11)

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{M}(\mathbf{r}') \times \hat{\mathbf{z}}}{r^2} d\tau'. \quad (6.15)$$

That **does** it, in principle. But, similar to the electrical case, the integral can be cast in a more illuminating form by exploiting the following identity:

$$\nabla' \frac{1}{r} = \frac{\hat{\mathbf{z}}}{r^2}.$$

Using this, we can re-express the equation in Eq. (6.15) as:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \left[\mathbf{M}(\mathbf{r}') \times \left(\nabla' \frac{1}{r} \right) \right] d\tau'. \quad (6.16)$$

Integrating by parts, using product rule 7, gives

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \left\{ \int \frac{1}{r} [\nabla' \times \mathbf{M}(\mathbf{r}')] d\tau' - \int \nabla' \times \left[\frac{\mathbf{M}(\mathbf{r}')}{r} \right] d\tau' \right\}.$$

Let's express the latter as a surface integral,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{1}{r} [\nabla' \times \mathbf{M}(\mathbf{r}')] d\tau' + \frac{\mu_0}{4\pi} \oint \frac{1}{r} [\mathbf{M}(\mathbf{r}') \times d\mathbf{a}']. \quad (6.17)$$

The first term looks just like the potential of a volume current,

$$\mathbf{J}_b = \nabla \times \mathbf{M}, \quad (6.18)$$

while the second looks like the potential of a surface current,

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} \quad (6.19)$$



where $\hat{n}$ is the normal unit vector. With these definitions,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}_b(\mathbf{r}')}{r} d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{\mathbf{K}_b(\mathbf{r}')}{r} da'. \quad (6.20)$$

What this means is that the potential¹⁴ of a magnetized object is the same as would be produced by a volume current $\mathbf{J}_b = \nabla \times \mathbf{M}$ throughout the material, plus a surface current $\mathbf{K}_b = \mathbf{M} \times \hat{n}$, on the boundary. Instead of integrating the contributions of all the infinitesimal dipoles, using Eq. (6.15), we first determine the **bound currents**, and then find the field **they** produce, in the same way we would calculate the field of any other volume and surface currents. Notice the striking parallel with the electrical case: there the field of a polarized object was the same as that of a bound volume charge $\rho_b = -\nabla \cdot \mathbf{P}$ plus a bound surface charge $\sigma_b = \mathbf{P} \cdot \hat{n}$.

¹⁴and hence also the field.

It is no accident that the same factors appear in the "contact" term for the fields of electric and magnetic dipoles (Eqs. 3.106 and 5.94). In fact, one good way to model a perfect dipole is to take the limit ($R \rightarrow 0$) of a polarized/magnetized sphere.

6.2.2 Physical Interpretation of Bound Currents

In the last section, we found that the field of a magnetized object is identical to the field that would be produced by a certain distribution of "bound" currents, $\mathbf{J}_b$ and $\mathbf{K}_b$. I want to show you how these bound currents arise physically. This will be a heuristic argument-the rigorous derivation has already been given. Figure 6.15 depicts a thin slab of uniformly magnetized material, with the dipoles represented by tiny current loops. Notice that all the "internal" currents cancel: every time there is one going to the right, a contiguous one is going to the left. However, at the edge there is no adjacent loop to do the canceling. The whole thing, then, is equivalent to a single ribbon of current I flowing around the boundary (Fig. 6.16). What is this current, in terms of M ? Say that each of the tiny loops has area a and thickness t (Fig. 6.17). In terms of the magnetization M , its dipole moment is $m = Mat$. In terms of the circulating current I , however, $m = Ia$. Therefore $I = Mt$, so the surface current is $\mathbf{K}_b = I/t = M$. Using the outward-drawn unit vector $\hat{n}$ (Fig. 6.16), the direction of $\mathbf{K}_b$ is conveniently indicated by the cross product:¹⁵

$$\mathbf{K}_b = \mathbf{M} \times \hat{n}$$

This bound surface current is exactly what we obtained in Sect. 6.2.1. It is a peculiar **kind** of current, in the sense that no single charge makes the whole trip- on the contrary, each charge moves only in a tiny little loop within a single atom. Nevertheless, the net effect is a macroscopic current flowing over the surface of the magnetized object. We call it a "bound" current to remind ourselves that every charge is attached to a particular atom, but it's a perfectly genuine current, and it produces a magnetic field in the same way any other current does.

When the magnetization is **nonuniform**, the internal currents no longer cancel. Figure 6.18(a) shows two adjacent chunks of magnetized material, with a larger arrow on the one to the right suggesting greater magnetization at that point. On the surface where they join, there is a net

¹⁵This expression also records the fact that there is no current on the top or bottom surface of the slab; here M is parallel to $\hat{n}$, so the cross product vanishes.

current in the x direction, given by

$$I_x = [M_z(y + dy) - M_z(y)]dz = \frac{\partial M_z}{\partial y} dy dz. \quad (6.21)$$

The corresponding volume current density is therefore

$$(J_b)_x = \frac{\partial M_z}{\partial y}$$

By the same token, a nonuniform magnetization in the y direction would contribute an amount $-\partial M_y/\partial z$ (Fig. 6.18(b)), so

$$(J_b)_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}. \quad (6.22)$$

In general, then, $J_b = \nabla \times M$, consistent, again, with the result of Sect. 6.2.1.

Incidentally, like any other steady current, J_b should obey the conservation law 5.33:

$$\nabla \cdot J_b = 0$$

Does it? Yes, for the divergence of a curl is always zero.

Information: The Magnetic Field Inside Matter

Similar to the electric field, the actual **microscopic** magnetic field inside matter fluctuates wildly from point to point and instant to instant. When we speak of magnetic field in matter, we mean the **macroscopic** field: the average over regions large enough to contain many atoms. (The magnetization M is "smoothed out" in the same sense.) It is this macroscopic field that one obtains when the methods of Sect. 6.2.1 are applied to points inside magnetized material.

6.3 The Auxiliary H Field

6.3.1 Ampère's Law in Magnetized Materials

In the previous section, we found that the effect of magnetization is to establish bound currents $\mathbf{J}_b = \nabla \times \mathbf{M}$ within the material and $\mathbf{K}_b = \mathbf{M} \times \hat{n}$ on the surface. The field due to magnetization of the medium is just the field produced by these bound currents. We are now ready to put everything together: the field attributable to bound currents, plus the field due to everything else—which we shall call the **free current**.

The free current might flow through wires imbedded in the magnetized substance or, if the latter is a conductor, through the material itself. In any event, the total current can be written as

$$\mathbf{J} = \mathbf{J}_b + \mathbf{J}_f. \quad (6.23)$$

There is no new physics in Eq. (6.23); it is simply a **convenience** to separate the current into these two parts, as they got there by quite different means: the free current is there because somebody hooked up a wire to a battery—it involves actual transport of charge; the bound current is there because of magnetization—it results from the conspiracy of many aligned atomic dipoles.

In view of Eq. (6.18) and Eq. (6.23), Ampère's law can be written as:

$$\frac{1}{\mu_0} (\nabla \times \mathbf{B}) = \mathbf{J} = \mathbf{J}_f + \mathbf{J}_b = \mathbf{J}_f + \nabla \times \mathbf{M}$$

or, collecting together the two (2) curls:

$$\nabla \times \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right) = \mathbf{J}_f. \quad (6.24)$$

The quantity in parentheses is designated by the letter $\mathbf{H}$:

$$\mathbf{H} \equiv \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}. \quad (6.25)$$

In terms of $\mathbf{H}$, then, Ampère's law reads

$$\nabla \times \mathbf{H} = \mathbf{J}_f, \quad \text{in integral form} \quad \oint \mathbf{H} \cdot d\mathbf{l} = I_{f_{\text{enc}}}, \quad (6.26)$$

where $I_{f_{\text{enc}}}$ is the total **free** current passing through the Amperian loop. $\mathbf{H}$ plays a role in magnetostatics similar to $\mathbf{D}$ in electrostatics: Just as $\mathbf{D}$ allowed us to write **Gauss's** law in terms of the free **charge** alone, $\mathbf{H}$ permits us to express **Ampère's** law in terms of the free **current** alone—and free current is what we control directly. Bound current, like bound charge, comes along for the ride—the material gets magnetized, and this results in bound currents; we cannot turn them on or off independently, as we can free currents. In applying Eq. (6.26), all we need to worry about is the

free current, which we know about because we **put** it there. In particular, when symmetry permits, we can calculate H immediately from Eq. (??) by the usual Ampère's law methods. (For example, Probs. 6.7 and 6.8 can be done in one line by noting that $H = 0$.)

As it turns out, H is a more useful quantity than D . In the laboratory, you will frequently hear people talking about H (more often even than B), but you will never hear anyone speak of D (only E). The reason is this: To build an electromagnet you run a certain (free) current through a coil. The **current** is the thing you read on the dial, and this determines H (or at any rate, the line integral of H); B depends on the specific materials you used and even, if iron is present, on the history of your magnet. On the other hand, if you want to set up an electric field, you do not plaster a known free charge on the plates of a parallel plate capacitor; rather, you connect them to a battery of known **voltage**. It's the **potential difference** you read on your dial, and that determines E (or rather, the line integral of E); D depends on the details of the dielectric you're using. If it were easy to measure charge, and hard to measure potential, then you'd find experimentalists talking about D instead of E . So the relative familiarity of H , as contrasted with D , derives from purely practical considerations; theoretically, they're on an equal footing. Many authors call H , not B , the "magnetic field." Then they have to invent a new word for B : the "flux density," or magnetic "induction" (an absurd choice, since that term already has at least two other meanings in electrodynamics). Anyway, B is indisputably the fundamental quantity, so I shall continue to call it the "magnetic field," as everyone does in the spoken language. H has no sensible name: just call it " H ."

6.3.2 A Deceptive Parallel

Equation 6.19 looks just like Ampère's original law (Eq. 5.56), except that the **total** current is replaced by the **free** current, and B is replaced by $\mu_0 H$. As in the case of D , however, I must warn you against reading too much into this correspondence. It does **not** say that $\mu_0 H$ is "just like B , only its source is J_f instead of J ." For the curl alone does not determine a vector field—you must also know the divergence. And whereas $\nabla \cdot B = 0$, the divergence of H is **not**, in general, zero. In fact, from Eq. (6.25)

$$\nabla \cdot H = -\nabla \cdot M. \quad (6.27)$$

Only when the divergence of M vanishes is the parallel between B and $\mu_0 H$ faithful. If you think I'm being pedantic, consider the example of the bar magnet—a short cylinder of iron that carries a permanent uniform magnetization M parallel to its axis. (See Probs. 6.9 and 6.14.) In this case there is no free current anywhere, and a naïve application of Eq. (??) might lead you to suppose that $H = 0$, and hence that $B = \mu_0 M$ inside the magnet and $B = 0$ outside, which is nonsense. It is quite true that the **curl** of H vanishes everywhere, but the divergence does not. (Can you see where $\nabla \cdot M \neq 0$?) **Advice:** When you are asked to find B or H in a problem involving magnetic materials, first look for symmetry. If the problem exhibits cylindrical, plane, solenoidal, or toroidal symmetry, then you can get H directly from Eq. (??) by the usual Ampère's law methods. (Evidently, in such cases $\nabla \cdot M$ is automatically zero, since the free current alone determines the answer.) If the requisite symmetry is absent, you'll have to think of another approach, and in particular you must **not** assume that H is zero just because there is no free current in sight.

6.3.3 Boundary Conditions

The magnetostatic boundary conditions of Sect. 5.4.2 can be rewritten in terms of $\mathbf{H}$ and the **free** current. From Eq. (6.27) it follows that

$$H_{\text{above}}^{\perp} - H_{\text{below}}^{\perp} = -(M_{\text{above}}^{\perp} - M_{\text{below}}^{\perp}), \quad (6.28)$$

while Eq. (??) says

$$H_{\text{above}}^{\parallel} - H_{\text{below}}^{\parallel} = K_f \times \hat{n}. \quad (6.29)$$

In the presence of materials, these are sometimes more useful than the corresponding boundary conditions on $\mathbf{B}$ (Eqs. 5.74 and 5.76):

$$B_{\text{above}}^{\perp} - B_{\text{below}}^{\perp} = 0, \quad (6.30)$$

and

$$B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0(K \times \hat{n}). \quad (6.31)$$

You might want to check them, for Ex. 6.2 or Prob. 6.14.

6.4 Linear and Non-Linear Media

6.4.1 Magnetic Susceptibility and Permeability

For both paramagnetic and diamagnetic materials, the magnetisation (M) is sustained by the field (B).

when B is removed, M disappears.

In fact, for most substances the magnetization is **proportional** to the field, provided the field is **NOT** too strong. To keep consistency with the electrical case (Eq. 4.30), we will express the proportionality as follows:

$$M = \frac{1}{\mu_0} \chi_m B.$$

While the above expression is mathematically correct, scientific tradition prefers it be written in terms of H , instead of B :

$$M = \chi_m H. \quad (6.32)$$

The constant of proportionality (χ_m) is called the **magnetic susceptibility** and is a dimensionless quantity which varies from one substance to another.

This is positive for **paramagnets** and **negative** for diamagnets.

Table 6.1: Magnetic Susceptibilities (unless otherwise specified, values are for 1 atm, 20°C). Data from Handbook of Chemistry and Physics, 91st ed. (Boca Raton: CRC Press, Inc., 2010) and other references.

Diamagnetic		Paramagnetic	
Material	Susceptibility	Susceptibility	Material
Bismuth	$-1,7 \times 10^{-4}$	Oxygen (O_2)	$1,7 \times 10^{-6}$
Gold	$-3,4 \times 10^{-5}$	Sodium	$8,5 \times 10^{-6}$
Silver	$-2,4 \times 10^{-5}$	Aluminum	$2,2 \times 10^{-5}$
Copper	$-9,7 \times 10^{-6}$	Tungsten	$7,0 \times 10^{-5}$
Water	$-9,0 \times 10^{-6}$	Platinum	$2,7 \times 10^{-4}$
Carbon Dioxide	$-1,1 \times 10^{-8}$	Liquid Oxygen	$3,9 \times 10^{-3}$
Hydrogen (H_2)	$-2,1 \times 10^{-9}$	Gadolinium	0,48

Typical values are around 10^{-5} (see Table 6.1). Materials that obey Eq. (6.32) are called linear media. In view of Eq. (6.25), $B = \mu_0(H + M) = \mu_0(1 + \chi_m)H$, (6.30) for linear media. Thus B is also proportional to H :¹⁶ $B = \mu H$, (6.31) where

$$\mu \equiv \mu_0(1 + \chi_m). \quad (6.33)$$

μ is called the **permeability** of the material.¹⁷ In a vacuum, where there is no matter to magnetize, the susceptibility χ_m vanishes, and the permeability is μ_0 . That's why μ_0 is called the **permeability of free space**.

We might assume that linear media escape the defect in the parallel between B and H . given M and H are now **proportional** to B .

If this is true, doesn't this mean, their divergence, similar to B 's, must always vanish? Unfortunately, it does **NOT** at the boundary between two (2) materials of different permeability, the divergence of M can actually be infinite.

Information: Divergence of Auxiliary-Field

Formally:

$$\nabla \cdot H = \nabla \cdot \left(\frac{1}{\mu} B \right) = \frac{1}{\mu} \nabla \cdot B + B \cdot \nabla \left(\frac{1}{\mu} \right) = B \cdot \nabla \left(\frac{1}{\mu} \right),$$

so, in general, H is **NOT** divergenceless at points where μ is changing.

For instance, at the end of a cylinder of linear paramagnetic material, M is zero on one side but **NOT** on the other. For the "Gaussian pillbox" shown in Fig. 6.23, $\oint M \cdot da \neq 0$, and hence, by the divergence theorem, $\nabla \cdot M$ cannot vanish everywhere within it.

Incidentally, the volume bound current density in a homogeneous linear material is proportional to the free current density:

$$J_b = \nabla \times M = \nabla \times (\chi_m H) = \chi_m J_f. \quad (6.34)$$

Particularly, unless free current actually flows through the material, all bound current will be at the surface.

6.4.2 Ferromagnetism

In a linear medium, the alignment of atomic dipoles is maintained by a magnetic field imposed from the outside. Ferromagnets, which are emphatically **NOT** linear¹⁸, require no external fields to sustain the magnetization; the alignment is "frozen in." Like paramagnetism, ferromagnetism involves the magnetic dipoles associated with the spins of unpaired electrons. The new feature, which makes ferromagnetism so different from paramagnetism, is the interaction between nearby dipoles:

In a ferromagnet, each dipole "wants" to point in the same direction as its **neighbours**.

The reason for this preference is essentially quantum mechanical and therefore significantly beyond the scope of an engineers need. It is enough to know that the correlation is so strong as to align

¹⁶Physically, therefore, Eq. (??) would say exactly the same as Eq. (6.32), only the constant χ_m would have a different value. Equation 6.29 is a little more convenient, because experimentalists find it handier to work with H than B .

¹⁷If you factor out μ_0 , what's left is called the **relative permeability**: $\mu_r \equiv 1 + \chi_m = \mu/\mu_0$. By the way, formulas for H in terms of B (Eq. (??)), in the case of linear media) are called constitutive relations, just like those for D in terms of E .

¹⁸In this sense, it is misleading to speak of the susceptibility or permeability of a ferromagnet. The terms are used for such materials, but they refer to the proportionality factor between a differential increase in H and the resulting differential change in M (or B); moreover, they are **NOT** constants, but functions of H .

virtually 100% of the unpaired electron spins. If you could somehow magnify a piece of iron and "see" the individual dipoles as tiny arrows, it would look something like Fig. 6.25, with all the spins pointing the same way.

But if that is true, why isn't every wrench and nail a powerful magnet?

The answer is that the alignment occurs in relatively small patches, called **domains**. Each domain contains billions of dipoles, all lined up,¹⁹ but the domains **themselves** are randomly oriented. The a standard wrench contains an enormous number of domains, and their magnetic fields cancel, so the wrench as a whole is not magnetized.²⁰

How, then, would you produce a **permanent magnet**?

If we were to put a piece of iron into a strong magnetic field, the torque $N = m \times B$ tends to align the dipoles parallel to the field. Since they like to stay parallel to their neighbors, most of the dipoles will resist this torque. However, at the **boundary** between two domains, there are **competing** neighbors, and the torque will throw its weight on the side of the domain most nearly parallel to the field; this domain will win some converts, at the expense of the less favorably oriented one. The net effect of the magnetic field, then, is to move the domain boundaries. Domains parallel to the field grow, and the others shrink. If the field is strong enough, one domain takes over entirely, and the iron is said to be saturated. It turns out that this process²¹ is not entirely reversible: When the field is switched off, there will be **some** return to randomly oriented domains, but it is far from complete as there remains a preponderance of domains in the original direction. You now have a permanent magnet.

A simple way to accomplish this, in practice, is to wrap a coil of wire around the object to be magnetized (Fig. 6.27). Run a current I through the coil; this provides the external magnetic field (pointing to the left in the diagram). As you increase the current, the field increases, the domain boundaries move, and the magnetization grows. Eventually, you reach the saturation point, with all the dipoles aligned, and a further increase in current has no effect on M (Fig. 6.28, point b). Now suppose you **reduce** the current. Instead of retracing the path back to $M = 0$, there is only a **partial** return to randomly oriented domains; M decreases, but even with the current off there is some residual magnetization (point c). The wrench is now a permanent magnet. If you want to eliminate the remaining magnetization, you'll have to run a current backwards through the coil (a negative I). Now the external field points to the right, and as you increase I (negatively),

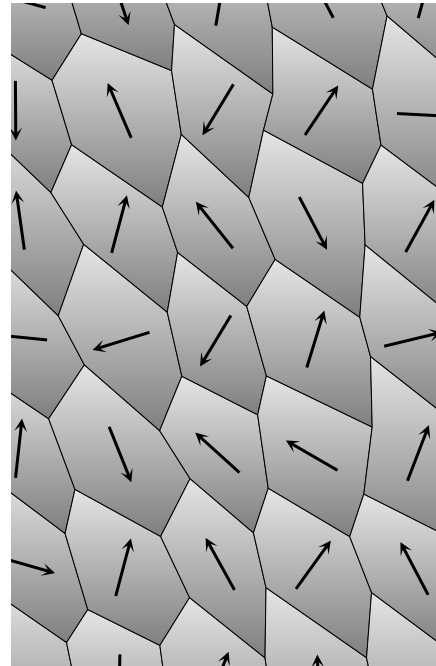


Figure 6.3

¹⁹these domains are actually **visible** under a microscope, using suitable etching techniques, as an example shown in Fig. 6.4.

²⁰Actually, the orientation of domains is not **completely** random; within a given crystal, there may be some preferential alignment along the crystal axes. But there will be just as many domains pointing one way as the other, so there is still no large-scale magnetization. Moreover, the crystals themselves are randomly oriented within any sizable chunk of metal.

²¹the shifting of domain boundaries in response to an external field



M drops down to zero (point d). If you turn I still higher, you soon reach saturation in the other direction—all the dipoles now pointing to the right (e). At this stage, switching off the current will leave the wrench with a permanent magnetization to the right (point f). To complete the story, turn I on again in the positive sense: M returns to zero (point g), and eventually to the forward saturation point (b). The path we have traced out is called a **hysteresis loop**. Notice that the magnetization of the wrench depends not only on the applied field (that is, on I), but also on its previous magnetic "history".²² For instance, at three different times in our experiment the current was zero (a , c , and f), yet the magnetization was different for each of them. Actually, it is customary to draw hysteresis loops as plots of B against H , rather than M against I . (If our coil is approximated by a long solenoid, with n turns per unit length, then $H = nI$, so H and I are proportional. Meanwhile, $B = \mu_0(H + M)$, but in practice M is huge compared to H , so to all intents and purposes B is proportional to M .) To make the units consistent (teslas), I have plotted $(\mu_0 H)$ horizontally (Fig. 6.29); notice, however, that the vertical scale is 104 times greater than the horizontal one. Roughly speaking, $\mu_0 H$ is the field our coil **would** have produced in the absence of any iron; B is what we **actually** got, and compared to $\mu_0 H$, it is gigantic. A little current goes a long way, when you have ferromagnetic materials around. That's why anyone who wants to make a powerful electromagnet will wrap the coil around an iron core. It doesn't take much of an external field to move the domain boundaries, and when you do that, you have all the dipoles in the iron working with you.

²²Etymologically, the word hysteresis has nothing to do with the word history—nor with the word hysteria. It derives from a Greek verb meaning "lag behind."

One final point about ferromagnetism: It all follows, remember, from the fact that the dipoles within a given domain line up parallel to one another. Random thermal motions compete with this ordering, but as long as the temperature doesn't get too high, they cannot budge the dipoles out of line. It's not surprising, though, that very high temperatures do destroy the alignment. What is surprising is that this occurs at a precise temperature (770°C , for iron). Below this temperature (called the **Curie point**), iron is ferromagnetic; above, it is paramagnetic. The Curie point is rather like the boiling point or the freezing point in that there is no gradual transition from ferro- to para-magnetic behavior, any more than there is between water and ice. These abrupt changes in the properties of a substance, occurring at sharply defined temperatures, are known in statistical mechanics as phase

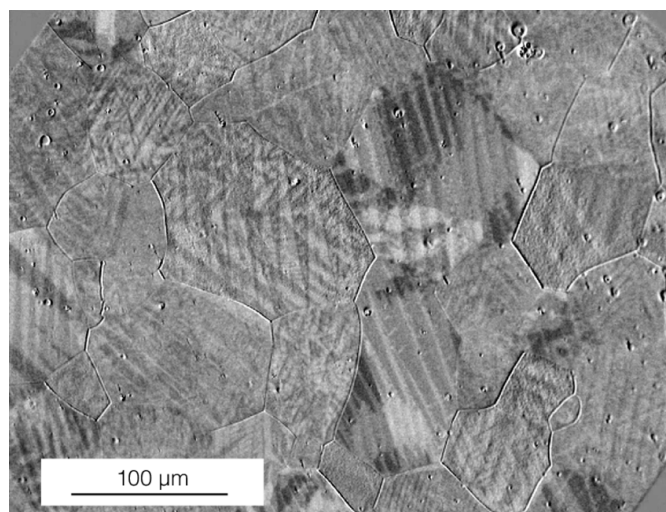


Figure 6.4: Illustration of the magnetic domains structure at the surface of a Fe-27%Co polycrystal [5] -Kerr microscopy.

transitions.



Part IV

Electromagnetic Fields

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This velocity is so nearly that of light, that it seems we have strong reason to conclude that light itself is an electromagnetic disturbance in the form of waves propagated through the electromagnetic field according to electromagnetic laws

*(James Clerk Maxwell, in A Dynamical Theory of the Electromagnetic Field (1864)
Introduction, p. 466.)*

Chapter 7

Electrodynamics

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7.1 Electromotive Force

Before we start with thing that change with time¹ it is worthwhile to discuss some essential terminologies for understanding electrical circuits, electrical machines, and wide variety of electrical phenomena:

¹i.e., where time actually becomes a variable.

the Electro-motive Force (EMF).

7.1.1 Ohm's Law

To make a current flow, we have to *push* on the charges. How *fast* they move, in response to a given push, depends on the nature of the material. For most substances, the current density $\mathbf{J}$ is proportional to the *force per unit charge*, $\mathbf{f}$:

$$\mathbf{J} = \sigma \mathbf{f} \quad (7.1)$$

The proportionality factor σ , which should **NOT** to be confused with surface charge, is an empirical constant which varies from one material to another; it's called the *conductivity* of the medium. Actually, most resources usually list the *reciprocal* of σ , called the *resistivity*: $\rho = 1/\sigma$.² Some typical

²Which shouldn't be confused with charge density-I'm sorry, but we're running out of Greek letters, and this is the standard notation

values are listed in Table 7.1. Looking at the table we can see even *insulators* conduct slightly, though the conductivity of a metal is significantly greater.

In fact, for most purposes metals can be regarded as **perfect conductors**, with $\sigma = \infty$, while for insulators we can pretend $\sigma = 0$.

In principle, the force that drives the charges to produce the current could be anything:

chemical, gravitational, or pressure

For our purposes, though, it's usually an electromagnetic force that does the job. In this case Eq. (7.1) becomes:³

$$\mathbf{J} = \sigma (\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.2)$$

³This equation should remind you of the Lorentz force law.

Ordinarily, the velocity of the charges is sufficiently small that the second term can be ignored:⁴

$$\mathbf{J} = \sigma \mathbf{E}. \quad (7.3)$$

⁴However, in plasmas, for instance, the magnetic contribution to $\mathbf{f}$ can be significant and cannot be ignored.

Eq. (7.3) is called Ohm's law⁵, though the physics behind it is really contained in Eq. (7.1), of which Eq. (7.3) is considered to be just a special case.

If we recall from electrostatics, we said $\mathbf{E} = 0$ inside a conductor, which here we are contradicting. But that's for *stationary* charges $\mathbf{J} = 0$.

Moreover, for *perfect* conductors $\mathbf{E} = \mathbf{J}/\sigma = 0$ even if current is flowing. In practice, metals are such good conductors that the electric field ($\mathbf{E}$) required to drive current in them is assumed **negligible**. Therefore we routinely treat the connecting wires in electric circuits as **equipotentials**. Resistors, by contrast, are made from *poorly* conducting materials.

As these aforementioned examples show, the total current flowing from one electrode to the other is proportional to the potential difference between them:

$$V = IR \quad (7.4)$$

This, of course, is the more familiar, and more practical, version of *Ohm's law*. The constant of proportionality R is called the **resistance** and it's a function of the geometry of the arrangement and the conductivity of the medium between the electrodes.

Resistance is measured in ohms (Ω) where an ohm is a volt per ampere ($\text{V} \cdot \text{A}^{-1}$). Notice that the proportionality between V and I is a direct consequence of Eq. (7.3):

if we want to double V , we simply double the charge on the electrodes. This in turn doubles E , which, for an ohmic material, doubles $\mathbf{J}$, which doubles I .



⁵a German mathematician and physicist. As a school teacher, Ohm began his research with the new electrochemical cell, invented by Italian scientist *Alessandro Volta*. Using equipment of his own creation, Ohm found that there is a direct proportionality between the potential difference (voltage) applied across a conductor and the resultant electric current. This relation is known as Ohm's law.



For steady currents and *uniform* conductivity,

$$\nabla \cdot \mathbf{E} = \frac{1}{\sigma} \nabla \cdot \mathbf{J} = 0, \quad (7.5)$$

and therefore the charge density is zero (0) and any unbalanced charge resides on the surface.⁶

It follows, in particular, that Laplace's equation holds within a homogeneous ohmic material carrying a steady current, so all the tools and tricks of Chapter 3 are available for calculating the potential.

⁶We proved this long ago, for the case of *stationary* charges, using the fact that $\mathbf{E} = 0$; evidently, it is still true when the charges are allowed to move.

7.1.2 Determining the Current Density

It's a little surprising that Ohm's law ever holds.

After all, a given field $\mathbf{E}$ produces a force $q\mathbf{E}$, acting on a charge q , and according to Newton's second law, the charge will accelerate.

But if the charges are accelerating, why doesn't the current increase with time, growing larger and larger the longer you leave the field on?

Ohm's law implies, on the contrary, a constant field produces a constant current, which suggests a constant velocity.

Isn't that a contradiction to Newton's law?

No, for we are forgetting the **frequent collisions** electrons make as they pass down the wire. If the length of the wire is λ and electron acceleration is a , the time it takes to go through a wire:

$$t = \sqrt{\frac{2\lambda}{a}} \quad \text{which makes the average speed} \quad v_{\text{ave}} = \frac{1}{2}at = \sqrt{\frac{\lambda a}{2}}$$

While this is a better description we are still not finished. The aforementioned equation says that the velocity is proportional to the square root of the acceleration ($v \propto \sqrt{1/a}$), and therefore the current should be proportional to the square root of the field.

In practice, the charges are already moving very fast due to of their thermal energy. But the thermal velocities have **random directions, and average to zero**. The drift velocity we are concerned with is a tiny extra bit. So the time between collisions is actually much shorter than we supposed.

If we assume for the sake of argument that all charges travel the same distance λ between collisions, then:

$$t = \frac{\lambda}{v_{\text{thermal}}} \quad \text{therefore} \quad v_{\text{ave}} = \frac{1}{2}at = \frac{a\lambda}{2v_{\text{thermal}}}$$

If there are n molecules per unit volume, and f free electrons per molecule, each with charge q and mass m , the current density ($\mathbf{J}$) is:⁷

$$\mathbf{J} = nfqv_{\text{ave}} = \frac{nfq\lambda}{2v_{\text{thermal}}} \frac{\mathbf{F}}{m} = \left(\frac{nf\lambda q^2}{2mv_{\text{thermal}}} \right) \mathbf{E}. \quad (7.6)$$

⁷This model is also known as the **Drude model** which was proposed in 1900.

While **NOT** perfectly accurate to modern standards, it does indicate the basic ingredients, and it correctly predicts that conductivity is proportional to the density of the moving charges and decreases with increasing temperature.

As a result of all the collisions, the work done by the electrical force is converted into heat in the resistor. Since the work done per unit charge is V and the charge flowing per unit time is I , the power delivered is

$$P = VI = I^2 R. \quad (7.7)$$

This is the **Joule heating law**. With I in amperes (A) and R in ohms (Ω), P comes out in watts (W or $\text{J} \cdot \text{s}^{-1}$).

7.1.3 Electromotive Force

Think of a typical electric circuit:

a battery hooked up to a light bulb.

An interesting question arises: In practice, the *current* is the same all *the way around the loop*; why is this the case, when the only obvious driving force is inside the battery? Off hand, you might expect a large current in the battery and none at all in the lamp. Who's doing the pushing, in the rest of the circuit, and how does it happen that this push is exactly right to produce the same current in each segment? What's more, given that the charges in a typical wire move (literally) at a *snail's* pace (see Prob. 5.20), why doesn't it take half an hour for the current to reach the light bulb? How do all the charges know to start moving at the same instant?

Answer: If the current were *not* the same all the way around (for instance, during the first split second after the switch is closed), then charge would be piling up somewhere, and here's the crucial point—the electric field of this accumulating charge is in such a direction as to even out the flow. Suppose, for instance, that the current *into* the bend in Fig. 7.8 is greater than the current *out*. Then charge piles up at the "knee," and this produces a field aiming *away* from the kink. This field *opposes* the current flowing in (slowing it down) and *promotes* the current flowing out (speeding it up) until these currents are equal, at which point there is no further accumulation of charge, and equilibrium is established. It's a beautiful system, automatically self-correcting to keep the current uniform, and it does it all so quickly that, in practice, you can safely assume the current is the same all around the circuit, even in systems that oscillate at radio frequencies.

There are really two (2) forces involved in driving current around a circuit:

- i. the source, $\mathcal{E}$, which is ordinarily confined to one portion of the loop (a battery, say), and
- ii. an *electrostatic* force, which serves to smooth out the flow and communicate the influence of

the source to distant parts of the circuit:

$$\mathbf{f} = \mathbf{f}_s + \mathbf{E}. \quad (7.8)$$

The physical agency responsible for $\mathbf{f}_s$ can be of different things:

in a battery it's a chemical force; in a piezoelectric crystal mechanical pressure is converted into an electrical impulse; in a thermocouple it's a temperature gradient that does the job; in a photoelectric cell it's light; and in a Van de Graaff generator the electrons are literally loaded onto a conveyer belt and swept along.

Whatever the *mechanism*, its net effect is determined by the line integral of $\mathbf{f}$ around the circuit:

$$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l}. \quad (7.9)$$

As we know $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for electrostatic fields, it doesn't matter whether we use $\mathbf{f}$ or $\mathbf{f}_s$.

Here, $\mathcal{E}$ is called the **electromotive force**,⁸ or **emf**, of the circuit. Within an ideal source of emf (a resistanceless battery,⁴ for instance), the *net* force on the charges is *zero* (Eq. 7.1 with $\sigma = \infty$), so $\mathbf{E} = -\mathbf{f}_s$. The potential difference between the terminals (a and b) is therefore⁹

$$V = - \int_a^b \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E}. \quad (7.10)$$

The function of a battery, then, is to establish and maintain a voltage difference equal to the electromotive force.¹⁰ The resulting electrostatic field drives current around the rest of the circuit (notice, however, that *inside* the battery $\mathbf{f}_s$ drives current in the direction *opposite* to $\mathbf{E}$). Because it's the line integral of $\mathbf{f}_s$, $\mathcal{E}$ can be interpreted as the *work done per unit charge*, by the source—indeed, in some books electromotive force is *defined* this way. However, as you'll see in the next section, there is some subtlety involved in this interpretation, so I prefer Eq. 7.9.

7.1.4 Motional EMF

In the last section, we have listed several possible sources of EMF with batteries being the most familiar. But we did **NOT** mention the commonest one of all:

the electric generator.¹¹

Generators exploit **motional** EMFs, which arise when you *move a wire* through a magnetic field. Figure 7.10 suggests a primitive model for a generator. In the shaded region there is a uniform magnetic field ($\mathbf{B}$), pointing **into the page**, and the resistor R represents whatever it is¹² we're trying to drive current through. If the entire loop is pulled to the right with speed v , the charges in

⁸Some textbooks and resources use the term *electromance*.

⁹We can extend the integral to the entire loop because $\mathbf{f}_s = 0$ outside the source anyway.

¹⁰A 6 V battery, for example, holds the positive terminal 6 V above the negative terminal.

¹¹A significant application of a synchronous generator is covered by the lecturer of this course on **B.Sc Drive Technologies**.

¹²It basically represents anything which consumes power.

segment ab experience a magnetic force whose vertical component qvB drives current around the loop, in the clockwise direction. The EMF is

$$\mathcal{E} = \oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh, \quad (7.11)$$

where h is the width of the loop.¹³ Notice that the integral you perform to calculate $\mathcal{E}$ (Eq. 7.9 or 7.11) is carried out at one instant of time—take a "snapshot" of the loop, if you like, and work from that. Therefore $d\mathbf{l}$, for the segment ab in Fig. 7.10, points straight up, even though the loop is moving to the right. You can't quarrel with this—it's simply the way emf is *defined*—but it is important to be clear about it. In particular, although the magnetic force is responsible for establishing the emf, it is not doing any work—magnetic forces never do work. Who, then, is supplying the energy that heats the resistor? Answer: The person who's pulling on the loop. With the current flowing, the free charges in segment ab have a vertical velocity (call it $\mathbf{u}$) in addition to the horizontal velocity $\mathbf{v}$ they inherit from the motion of the loop. Accordingly, the magnetic force has a component quB to the left. To counteract this, the person pulling on the wire must exert a force per unit charge

$$\mathbf{f}_{\text{null}} = uB$$

to the *right* (Fig. 7.11). This force is transmitted to the charge by the structure of the wire. Meanwhile, the particle is actually *moving* in the direction of the resultant velocity $\mathbf{w}$, and the distance it goes is $(h/\cos\theta)$. The work done per unit charge is therefore:¹⁴

$$\int \mathbf{f}_{\text{pull}} \cdot d\mathbf{l} = (uB) \left(\frac{h}{\cos\theta} \right) \sin\theta = vBh = \mathcal{E}$$

As it turns out, then, the work done per unit charge is *exactly equal to the emf, though the integrals are taken along entirely different paths* (Fig. 7.12), and completely different forces are involved. To calculate the emf, you integrate around the loop at one instant, but to calculate the work done you follow a charge in its journey around the loop; $\mathbf{f}_{\text{pull}}$ contributes nothing to the emf, because it is perpendicular to the wire, whereas $\mathbf{f}_{\text{mag}}$ contributes nothing to work because it is perpendicular to the motion of the charge. There is a particularly nice way of expressing the emf generated in a moving loop. Let Φ be the flux of $\mathbf{B}$ through the loop:

$$\Phi \equiv \int_{\mathbf{f}_{\text{null}}} \mathbf{B} \cdot d\mathbf{a}. \quad (7.12)$$

For the rectangular loop in Fig. 7.10,

$$\Phi = Bhx.$$

As the loop moves, the flux decreases:

$$\frac{d\Phi}{dt} = Bh \frac{dx}{dt} = -Bhv.$$

(The minus sign accounts for the fact that dx/dt is negative.) But this is precisely the emf (Eq. 7.11); evidently the emf generated in the loop is minus the rate of change of flux through the loop:

$$\mathcal{E} = -\frac{d\Phi}{dt}. \quad (7.13)$$

¹³The horizontal segments bc and ad contribute nothing, since the force there is perpendicular to the wire.

¹⁴In the equation below, $\sin\theta$ is coming from the dot product.

This is the flux rule for motional EMF. Apart from its delightful simplicity, the flux rule has the virtue of applying to nonrectangular loops moving in arbitrary directions through nonuniform magnetic fields; in fact, the loop need not even maintain a fixed shape. **Proof.** Figure 7.13 shows a loop of wire at time t , and also a short time dt later. Suppose we compute the flux at time t , using surface S , and the flux at time $t + dt$, using the surface consisting of S plus the "ribbon" that connects the new position of the loop to the old. The *change* in flux, then, is

$$d\Phi = \Phi(t + dt) - \Phi(t) = \Phi_{\text{ribbon}} = \int_{\text{ribbon}} \mathbf{B} \cdot d\mathbf{a}.$$

Focus your attention on point P : in time dt , it moves to P' . Let $\mathbf{v}$ be the velocity of the *wire*, and $\mathbf{u}$ the velocity of a charge *down* the wire; $\mathbf{w} = \mathbf{v} + \mathbf{u}$ is the resultant

velocity of a charge at P . The infinitesimal element of area on the ribbon can be written as

$$d\mathbf{a} = (\mathbf{v} \times d\mathbf{l}) dt$$

(see inset in Fig. 7.13). Therefore

$$\frac{d\Phi}{dt} = \oint \mathbf{B} \cdot (\mathbf{v} \times d\mathbf{l}).$$

Since $\mathbf{w} = (\mathbf{v} + \mathbf{u})$ and $\mathbf{u}$ is parallel to $d\mathbf{l}$, we can just as well write this as

$$\frac{d\Phi}{dt} = \oint \mathbf{B} \cdot (\mathbf{w} \times d\mathbf{l}).$$

Now, the scalar triple-product can be rewritten:

$$\mathbf{B} \cdot (\mathbf{w} \times d\mathbf{l}) = -(\mathbf{w} \times \mathbf{B}) \cdot d\mathbf{l},$$

so

$$d\Phi = - \oint (\mathbf{w} \times \mathbf{B}) \cdot d\mathbf{l}.$$

But $(\mathbf{w} \times \mathbf{B})$ is the magnetic force per unit charge, $\mathbf{f}_{\text{mag}}$, so

$$\frac{d\Phi}{dt} = - \oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l},$$

and the integral of $\mathbf{f}_{\text{mag}}$ is the emf:

$$\mathcal{E} = - \frac{d\Phi}{dt}.$$

□

There is a sign ambiguity in the definition of emf (Eq. 7.9): Which way around the loop are you supposed to integrate? There is a compensatory ambiguity in the definition of *flux* (Eq. 7.12): Which is the positive direction for $d\mathbf{a}$? In applying

the flux rule, sign consistency is governed (as always) by your right hand: If your fingers define the positive direction around the loop, then your thumb indicates the direction of $d\mathbf{a}$. Should the emf come out negative, it means the current will flow in the negative direction around the circuit. The flux rule is a nifty short-cut for calculating motional emfs. It does not contain any new physics—just the Lorentz force law. But it can lead to error or ambiguity if you're not careful. The flux rule

assumes you have a single wire loop—it can move, rotate, stretch, or distort (continuously), but beware of switches, sliding contacts, or extended conductors allowing a variety of current paths. A standard "flux rule paradox" involves the circuit in Figure 7.14. When the switch is thrown (from *a* to *b*) the flux through the circuit doubles, but there's no motional emf (no conductor moving through a magnetic field), and the ammeter (*A*) records no current.

Example 7.4 (the Faraday disk, or Faraday dynamo) involves a motional emf that you can't calculate (at least, not directly) from the flux rule. The flux rule assumes the current flows along a well-defined path, whereas in this example the current spreads out over the whole disk. It's not even clear what the "flux through the circuit" would *mean* in this context.

Even more tricky is the case of **eddy currents**. Take a chunk of aluminum (say), and shake it around in a nonuniform magnetic field. Currents will be generated in the material, and you will feel a kind of "viscous drag"—as though you were pulling the block through molasses (this is the force I called f_{pull} in the discussion of motional emf). Eddy currents are notoriously difficult to calculate, but easy and dramatic to demonstrate. You may have witnessed the classic experiment in which an aluminum disk mounted as a pendulum on a horizontal axis swings down and passes between the poles of a magnet (Fig. 7.16a). When it enters the field region it suddenly slows way down. To confirm that eddy currents are responsible, one repeats the demonstration using a disk that has many slots cut in it, to prevent the flow of large-scale currents (Fig. 7.16b). This time the disk swings freely, unimpeded by the field.

7.2 Electromagnetic Induction

7.2.1 Faraday's Law

In 1831 Michael Faraday¹⁵ reported on a series of experiments, including three (3) which can be characterized as follows:

Experiment 1

He pulled a loop of wire to the right through a magnetic field (Fig. 7.21a). A current flowed in the loop.

Experiment 2

He moved the *magnet* to the *left*, holding the loop still (Fig. 7.21b). Again, a current flowed in the loop.

Experiment 3

With both the loop and the magnet at rest (Fig. 7.21c), he changed the strength of the field (he used an electromagnet, and varied the current in the coil). Once again, current flowed in the loop.

The first experiment, of course, is a straightforward case of motional emf; according to the flux rule:

$$\mathcal{E} = -\frac{d\Phi}{dt}. \quad (7.14)$$

I don't think it will surprise you to learn that exactly the same emf arises in Experiment 2—all that really matters is the *relative* motion of the magnet and the loop. Indeed, in the light of special relativity it has to be so. But Faraday knew nothing of relativity, and in classical electrodynamics this simple reciprocity is a remarkable coincidence. For if the *loop* moves, it's a *magnetic* force that sets up the emf, but if the loop is *stationary*, the force *cannot* be magnetic—stationary charges experience no magnetic forces. In that case, what is responsible? What sort of field exerts a force on charges at rest? Well, *electric* fields do, of course, but in this case there doesn't seem to be any electric field in sight. Faraday had an ingenious inspiration:

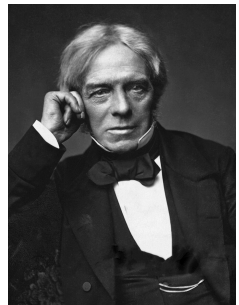
A changing magnetic field induces an electric field.

It is this induced⁸ electric field that accounts for the emf in Experiment 2.⁹ Indeed, if (as Faraday found empirically) the emf is again equal to the rate of change of the flux,

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}, \quad (7.15)$$

then $\mathbf{E}$ is related to the change in $\mathbf{B}$ by the equation

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a}. \quad (7.16)$$



¹⁵Michael Faraday
(1791 - 1867)

An English chemist and physicist who contributed to the study of electro-chemistry and electromagnetism. His main discoveries include the principles underlying electromagnetic induction, diamagnetism, and electrolysis. Although Faraday received little formal education, as a self-made man, he was one of the most influential scientists in history [1]. It was by his research on the magnetic field around a conductor carrying a direct current Faraday established the concept of the electromagnetic field in physics. Faraday also established magnetism could affect rays of light and that there was an underlying relationship between the two phenomena. He similarly discovered the principles of electromagnetic induction, diamagnetism, and the laws of electrolysis. His inventions of electromagnetic rotary devices formed the foundation of electric motor technology, and it was largely due to his efforts that electricity became practical for use in technology. The SI unit of capacitance, the farad, is named after him.

This is Faraday's law, in integral form. We can convert it to differential form by applying Stokes' theorem:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (7.17)$$

Note that Faraday's law reduces to the old rule $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ (or, in differential form, $\nabla \times \mathbf{E} = 0$ in the static case (constant $\mathbf{B}$) as, of course, it should. In Experiment 3, the magnetic field changes for entirely different reasons, but according to Faraday's law an electric field will again be induced, giving rise to an emf $-d\Phi/dt$.

Indeed, one can subsume all three cases¹⁶ into a kind of universal flux rule: Whenever (and for whatever reason) the magnetic flux through a loop changes, an emf

¹⁶and for that matter any combination of them.

$$\mathcal{E} = -\frac{d\Phi}{dt} \quad (7.18)$$

will appear in the loop. Many people call this "Faraday's law." Maybe I'm overly fastidious, but I find this confusing. There are really two totally different mechanisms underlying Eq. 7.17, and to identify them both as "Faraday's law" is a little like saying that because identical twins look alike we ought to call them by the same name. In Faraday's first experiment it's the Lorentz force law at work; the emf is *magnetic*. But in the other two it's an *electric* field (induced by the changing magnetic field) that does the job. Viewed in this light, it is quite astonishing that all three processes yield the same formula for the emf. In fact, it was precisely this "coincidence" that led Einstein to the special theory of relativity—he sought a deeper understanding of what is, in classical electrodynamics, a peculiar accident. But that's a story for Chapter 12. In the meantime, I shall reserve the term "Faraday's law" for electric fields induced by changing magnetic fields, and I do *not* regard Experiment 1 as an instance of Faraday's law.

Keeping track of the *signs* in Faraday's law can be a real headache. For instance, in Ex. 7.5 we would like to know which way around the ring the induced current flows. In principle, the right-hand rule does the job (we called Φ positive to the left, in Fig. 7.22, so the positive direction for current in the ring is counter-clockwise, as viewed from the left; since the first spike in Fig. 7.23b is negative, the first current pulse flows *clockwise*, and the second counterclockwise). But there's a handy rule, called Lenz's law, whose sole purpose is to help you get the directions right:

Nature abhors a change in flux. The induced current will flow in such a direction that the flux it produces tends to cancel the change. (As the front end of the magnet in Ex. 7.5 enters the ring, the flux increases, so the current in the ring must generate a field to the *right*—it therefore flows *clockwise*.) Notice that it is the *change* in flux, not the flux itself, that nature abhors (when the tail end of the magnet exits the ring, the flux drops, so the induced current flows *counterclockwise*, in an effort to restore it). Faraday induction is a kind of "inertial" phenomenon: A conducting loop "likes" to maintain a constant flux through it; if you try to *change* the flux, the loop responds by sending a current around in such a direction as to frustrate your efforts. (It doesn't succeed completely; the flux produced by the induced current is typically only a tiny fraction of the original. All Lenz's law tells you is the *direction* of the flow.)

7.2.2 The Induced Electric Field

Faraday's law generalises the electrostatic rule, we have derived in previous chapters.¹⁷ If $\mathbf{E}$ is a pure Faraday field,¹⁸ then:

$$\nabla \cdot \mathbf{E} = 0, \quad \text{and} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

This is mathematically identical to magnetostatics,

$$\nabla \cdot \mathbf{B} = 0, \quad \text{and} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

Conclusion Faraday-induced electric fields are determined by $-\partial \mathbf{B} / \partial t$ in exactly the same way as magnetostatic fields are determined by $\mu_0 \mathbf{J}$. The analog to Biot-Savart is¹⁹

$$\mathbf{E} = -\frac{1}{4\pi} \int \frac{(\partial \mathbf{B} / \partial t) \times \hat{\mathbf{z}}}{z^2} d\tau = -\frac{1}{4\pi} \frac{\partial}{\partial t} \int \frac{\mathbf{B} \times \hat{\mathbf{z}}}{z^2} d\tau, \quad (7.19)$$

and if symmetry permits, we can use all the methods and simplifications associated with *Ampère's law* in integral form ($\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$), only now it's *Faraday's law* in integral form:

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}. \quad (7.20)$$

The rate of change of magnetic flux through the Amperian loop plays the role formerly assigned to $\mu_0 I_{\text{enc}}$.

Information: The Importance of Time

I must warn you, now, of a small limitation which **restricts** applications of Faraday's law: Electro-magnetic induction, of course, occurs only when the magnetic fields are **changing**, and yet we would like to use the apparatus of magnetostatics²⁰ to calculate those magnetic fields.

Technically, any result derived in this way is only approximately correct. But in practice the error is usually negligible, unless the field fluctuates extremely rapidly, or if we are interested in points very far from the source. Even the case of a wire snipped by a pair of scissors is considered *static enough* for Ampère's law to apply. This régime, in which magnetostatic rules can be used to calculate the magnetic field on the right hand side of Faraday's law, is called quasistatic.

Generally speaking, it is only when we come to electromagnetic waves and radiation that we must worry seriously about the breakdown of magnetostatics itself.

¹⁷ $\nabla \times \mathbf{E} = 0$ to the time-dependent régime. The *divergence* of $\mathbf{E}$ is still given by Gauss's law ($\nabla \cdot \mathbf{E} = \rho/\epsilon_0$).

¹⁸ due exclusively to a changing $\mathbf{B}$, with $\rho = 0$

¹⁹ Magnetostatics holds only for time-independent currents, but there is no such restriction on $\partial \mathbf{B} / \partial t$.

²⁰ Ampère's law, the Biot-Savart law, and the rest.

7.2.3 Inductance

Suppose we have two (2) loops of wire, at rest (Fig. 7.30). If were to run a steady current I_1 around loop 1, it produces a magnetic field $\mathbf{B}_1$. Some of the field lines pass through loop 2; let Φ_2 be the



flux of $\mathbf{B}_1$ through 2. You might have a tough time actually *calculating* B_1 , but a glance at the Biot-Savart law,

$$\mathbf{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\mathbf{l}_1 \times \hat{\mathbf{z}}}{r^2}$$

reveals one significant fact about this field: It is proportional to the current I_1 . Therefore, so too is the flux through loop 2:

$$\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2 \quad \text{therefore} \quad \Phi_2 = M_{21} I_1 \quad (7.21)$$

where M_{21} is the constant of proportionality; it is known as the **mutual inductance** of the two loops. There is a cute formula for the mutual inductance, which you can derive by expressing the flux in terms of the vector potential, and invoking Stokes' theorem:

$$\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2 = \int (\nabla \times \mathbf{A}_1) \cdot d\mathbf{a}_2 = \oint \mathbf{A}_1 \cdot d\mathbf{l}_2.$$

Now, according to Eq. 5.66,

$$\mathbf{A}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{d\mathbf{l}_1}{r},$$

We can use this equation to derive:

$$\Phi_2 = \frac{\mu_0 I_1}{4\pi} \oint \left(\oint \frac{d\mathbf{l}_1}{r} \right) \cdot d\mathbf{l}_2. \quad \text{therefore} \quad M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{r}. \quad (7.22)$$

This is the Neumann formula; it involves a double line integral—

one integration around loop 1, the other around loop 2

(Fig. 7.31). It's not very useful for practical calculations, but it does reveal two important things about mutual inductance:

- i. M_{21} is a purely geometrical quantity, having to do with the sizes, shapes, and relative positions of the two loops.
- ii. The integral in Eq. (7.22) is unchanged if we switch the roles of loops 1 and 2: it follows that

$$M_{21} = M_{12}. \quad (7.23)$$

This is an astonishing conclusion: Whatever the shapes and positions of the loops, the flux through 2 when we run a current I around 1 is identical to the flux through 1 when we send the same current I around 2. We may as well drop the subscripts and call them both M .

Suppose, now, we *vary* the current in loop 1. The flux through loop 2 will vary accordingly, and Faraday's law says this changing flux will induce an emf in loop 2:

$$\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -M \frac{dI_1}{dt}. \quad (7.24)$$

(In quoting Eq. 7.22—which was based on the Biot-Savart law—I am tacitly assuming that the currents change slowly enough for the system to be considered quasistatic.) What a remarkable thing: Every time you change the current in loop 1, an induced current flows in loop 2—even though there

are no wires connecting them! Come to think of it, a changing current not only induces an emf in any nearby loops, it also induces an emf in the source loop *itself* (Fig 7.33). Once again, the field (and therefore also the flux) is proportional to the current:

$$\Phi = LI. \quad (7.25)$$

The constant of proportionality L is called the self inductance (or simply the **inductance**) of the loop. As with M , it depends on the geometry (size and shape) of the loop. If the current changes, the emf induced in the loop is

$$\mathcal{E} = -L \frac{dI}{dt}. \quad (7.26)$$

Inductance is measured in henries (H); a henry is a volt-second per ampere. **FIGURE 7.33**

Inductance (like capacitance) is an intrinsically *positive* quantity. Lenz's law, which is enforced by the minus sign in Eq. 7.27, dictates that the emf is in such a direction as to *oppose* any *change in current*. For this reason, it is called a back emf. Whenever you try to alter the current in a wire, you must fight against this back emf. Inductance plays somewhat the same role in electric circuits that *mass* plays in mechanical systems: The greater L is, the harder it is to change the current, just as the larger the mass, the harder it is to change an object's velocity.

7.2.4 Energy in Magnetic Fields

It takes a certain amount of energy to start a current flowing in a circuit. Here, we are **NOT** talking about the energy delivered to the resistors and converted into heat.²¹ What we are concerned with, rather, is the work you must do **against the back emf** to get the current going. This is a fixed amount, and it is *recoverable*:

We get it back when the current is turned off.

In the meantime, it represents energy present in the circuit, and as we'll see in a moment, it can be regarded as energy stored in the magnetic field. The work done on a unit charge, against the back emf, in one trip around the circuit is $-\mathcal{E}$.²² The amount of charge per unit time passing down the wire is I .

So the total work done per unit time is

$$\frac{dW}{dt} = -\mathcal{E}I = LI \frac{dI}{dt}.$$

If we start with zero current and build it up to a final value I , the work done²³ is:

$$W = \frac{1}{2}LI^2 \quad (7.27)$$

It does not depend on how *long* we take to crank up the current, only on the geometry of the loop (in the form of L) and the final current I . There is a cleaner way to write W , which has the

²¹that is irretrievably lost, as far as the circuit is concerned, and can either be large or small, depending on how long the current is left to run.

²²The minus sign records the fact that this is the work done by you *against* the emf, not the work done by the emf

²³integrating the last equation over time



advantage that it is readily generalized to surface and volume currents. Remember that the flux Φ through the loop is equal to LI shown in Eq. (7.25). On the other hand,

$$\Phi = \int \mathbf{B} \cdot d\mathbf{a} = \int (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint \mathbf{A} \cdot d\mathbf{l}$$

where the line integral is around the perimeter of the loop. We can then make the conclusion that:

$$LI = \oint \mathbf{A} \cdot d\mathbf{l} \quad \text{therefore} \quad W = \frac{1}{2} I \oint \mathbf{A} \cdot d\mathbf{l} = \frac{1}{2} \oint (\mathbf{A} \cdot \mathbf{I}) d\mathbf{l}$$

In this form, the generalization to volume currents is obvious:

$$W = \frac{1}{2} \int_{\Omega} (\mathbf{A} \cdot \mathbf{J}) d\tau. \quad (7.28)$$

But we can do even better, and express W entirely in terms of the magnetic field: Ampère's law, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$, lets us eliminate $\mathbf{J}$:

$$W = \frac{1}{2\mu_0} \int \mathbf{A} \cdot (\nabla \times \mathbf{B}) d\tau. \quad (7.29)$$

Integration by parts transfers the derivative from $\mathbf{B}$ to $\mathbf{A}$; specifically, product rule 6 states that

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}).$$

so

$$\mathbf{A} \cdot (\nabla \times \mathbf{B}) = \mathbf{B} \cdot \nabla - \nabla \cdot (\mathbf{A} \times \mathbf{B}).$$

Consequently,

$$\begin{aligned} W &= \frac{1}{2\mu_0} \left[\int B^2 d\tau - \int \nabla \cdot (\mathbf{A} \times \mathbf{B}) d\tau \right] \\ &= \frac{1}{2\mu_0} \left[\int_V B^2 d\tau - \oint_S (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a} \right], \end{aligned} \quad (7.30)$$

where S is the surface bounding the volume V . Now, the integration in Eq. 7.32 is to be taken over the *entire volume occupied* by the current. But any region larger than this will do just as well, for $\mathbf{J}$ is zero out there anyway. In Eq. 7.34, the larger the region we pick the greater is the contribution from the volume integral, and therefore the smaller is that of the surface integral (this makes sense: as the surface gets farther from the current, both $\mathbf{A}$ and $\mathbf{B}$ decrease). In particular, if we agree to integrate over all space, then the surface integral goes to zero, and we are left with

$$W = \frac{1}{2\mu_0} \int_{\text{all space}} B^2 d\tau. \quad (7.31)$$

In view of this result, we say the energy is "stored in the magnetic field," in the amount $(B^2/2\mu_0)$ per unit volume. This is a nice way to think of it, though someone looking at Eq. 7.32 might prefer to say that the energy is stored in the *current distribution*, in the amount $\frac{1}{2}(\mathbf{A} \cdot \mathbf{J})$ per unit volume. The distinction is one of bookkeeping; the important quantity is the total energy W , and we need not worry about where (if anywhere) the energy is "located."

You might find it strange that it takes energy to set up a magnetic field—after all, magnetic fields *themselves* do no work. The point is that producing a magnetic field, where previously there was none, requires *changing* the field, and a changing B-field, according to Faraday, induces an *electric*

field. The latter, of course, can do work. In the beginning, there is no E , and at the end there is no E ; but in between, while $\mathbf{B}$ is building up, there is an $\mathbf{E}$, and it is against *this* that the work is done. (You see why I could not calculate the energy stored in a magnetostatic field back in Chapter 5.) In the light of this, it is extraordinary how similar the magnetic energy formulas are to their electrostatic counterparts:

$$W_{\text{elec}} = \frac{1}{2} \int (V\rho) d\tau = \frac{\epsilon_0}{2} \int E^2 d\tau,$$

$$W_{\text{mag}} = \frac{1}{2} \int (\mathbf{A} \cdot \mathbf{J}) d\tau = \frac{1}{2\mu_0} \int B^2 d\tau.$$



7.3 Maxwell's Equations

7.3.1 Electrodynamics Before Maxwell

So far, we have encountered the following laws, specifying the divergence and curl of electric and magnetic fields:

- (i) $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$ Gauss's law
- (ii) $\nabla \cdot \mathbf{B} = 0$ -
- (iii) $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ Faraday's law
- (iv) $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ Ampere's law

These equations represent the state of electromagnetic theory in the mid 19th century, when *Maxwell* began his work. They were **NOT** written in the compact form [2] we know today, but their physical content was familiar.²⁴ Now, it happens there is a major inconsistency in these formulas. It has to do with the old rule that **divergence of curl is always zero**. If we were to apply the divergence operation ($\nabla \cdot$) to number (iii), everything works out:

$$\nabla \cdot (\nabla \times \mathbf{E}) = \nabla \cdot \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \cdot \mathbf{B})$$

The left side is zero given divergence of curl is zero and the right side is zero by virtue of equation (ii). But when we do the same thing to number (iv), we seem to hit a problem:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 (\nabla \cdot \mathbf{J}) \quad (7.32)$$

the left side must be zero, but the right side, in general, is **NOT**. For steady currents, the divergence of $\mathbf{J}$ is zero, which we have seen in **Magnetostatics**, but when we go beyond magnetostatics Ampère's law cannot be right.

There's another way to see that Ampère's law is bound to fail for nonsteady currents. Suppose we're in the process of charging up a capacitor, shown in **Fig. ??**. To recall, in integral form, Ampère's law reads;

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}.$$

Let's apply it to the Amperian loop shown in the diagram. Before we do that, we need to ask the following question:

How do I determine I_{enc} ?

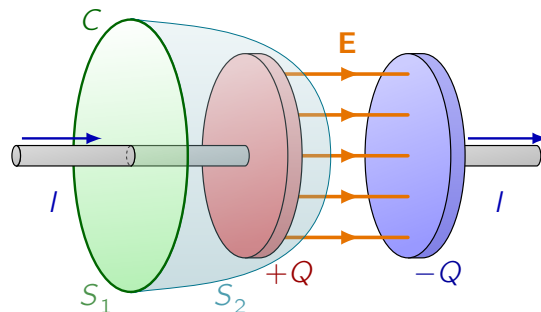


Figure 7.1

The charging of the capacitor was one of the problem which the current knowledge of electrodynamics couldn't explain and was a pivotal point in developing the final element of electrodynamics.

Well, it's the total current passing through the loop, or, more precisely, the current piercing a surface that has the loop for its boundary. In this case, the **simplest** surface lies in the plane of the loop—the wire punctures this surface, therefore $I_{\text{enc}} = I$.

But what if we draw instead the balloon-shaped surface in **Fig. 7.1**?

No current passes through *this* surface, and I conclude that $I_{\text{enc}} = 0$! We never had this problem in magnetostatics because the conflict arises only when charge is piling up somewhere.²⁵ But for *non-steady* currents²⁶ “the current enclosed by the loop” is an ill-defined notion as it depends entirely on what surface you use.

²⁵in this case, on the capacitor plates.

²⁶such as the charging of a capacitor.

This may sound like an unnecessary argument as one could say to use the plane surface, but remember, the Amperian loop could be some contorted shape that doesn't even lie in a plane.

Of course, we had no right to *expect* Ampère's law to hold outside of magnetostatics given we derived it from the *Biot-Savart law*. However, if we put all these knowledge in the time of *Maxwell's*, there was no *experimental* reason to doubt that *Ampère's law* was of wider validity. The flaw was a purely theoretical one, and *Maxwell* fixed it by purely theoretical arguments.

7.3.2 How Maxwell Fixed Ampère's Law

The problem is on the right side of Eq. (7.32), which should be zero, but isn't. Applying the continuity equation and Gauss's law, the problematic term can be rephrased as:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial t} (\epsilon_0 \nabla \cdot \mathbf{E}) = -\nabla \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right).$$

If we were to combine:

$$\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad \text{with} \quad \mathbf{J},$$

in *Ampère's law*, it would be just right to kill off the extra divergence:²⁷

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (7.33)$$

Such a modification changes nothing, as far as magnetostatics is concerned:

when $\mathbf{E}$ is constant, we still have $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$.

In fact, Maxwell's term is hard to detect in ordinary electromagnetic experiments, that's why Faraday and the others never discovered it in the laboratory.

However, it plays a crucial role in the propagation of electromagnetic waves.

Apart from curing the defect in Ampère's law, Maxwell's term has a certain aesthetic appeal:

²⁷Maxwell himself had other reasons for wanting to add this quantity to Ampère's law. To him, the rescue of the continuity equation was a happy dividend rather than a primary motive. But today we recognize this argument as a far more compelling one than Maxwell's, which was based on a now-discredited model of the *ether* as at that time it was thought luminiferous aether (or ether), was a theorized medium for the propagation of light [3].

Just as a changing *magnetic* field induces an *electric* field,²⁸ so is a changing electric field induces a magnetic field.

²⁸Faraday's law.

Of course, theoretical convenience and aesthetic consistency are only suggestive as there might be other ways to rephrase Ampère's law. The real confirmation of Maxwell's theory came in 1888 with Hertz's experiments on electro-magnetic waves [4].

Maxwell called his extra term the **displacement current** [2]:²⁹

$$\mathbf{J}_d \equiv \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (7.34)$$

²⁹It's a misleading name given $\epsilon_0(\partial \mathbf{E}/\partial t)$ has nothing to do with current, except that it adds to $\mathbf{J}$ in Ampère's law.

Let's see now how displacement current resolves the paradox of the charging capacitor in **Fig. 7.1**. If the capacitor plates are very close together,³⁰ then the electric field between them is:

³⁰The calculation is simpler if we assume this.

$$E = \frac{1}{\epsilon_0} \sigma = \frac{1}{\epsilon_0} \frac{Q}{A},$$

where Q is the charge on the plate (C) and A is its area (m^2). Therefore, between the plates:

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} \frac{dQ}{dt} = \frac{1}{\epsilon_0 A} I.$$

Now, Eq. (7.33) reads, in its integral form as:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \int \left(\frac{\partial \mathbf{E}}{\partial t} \right) \cdot d\mathbf{a}. \quad (7.35)$$

If we were choose the *flat* surface, then $E = 0$ and $I_{\text{enc}} = I$. If, on the other hand, we use the balloon-shaped surface, then $I_{\text{enc}} = 0$, but:

$$\int \left(\frac{\partial \mathbf{E}}{\partial t} \right) \cdot d\mathbf{a} = \frac{I}{\epsilon_0}$$

So we get the same answer for either surface, though in the first case it comes from the conduction current, and in the second from the displacement current.

7.3.3 Maxwell's Equations

In the last section we put the finishing touches on Maxwell's equations:

$$\begin{array}{ll} \text{(i)} & \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \text{Gauss's law} \\ \text{(ii)} & \nabla \cdot \mathbf{B} = 0 \quad - \\ \text{(ii)} & \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \text{Faraday's law} \\ \text{(iv)} & \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad \text{Ampere's law} \end{array} \quad (7.36)$$

Together with the force law,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.37)$$

summarise the entire theoretical content of classical electrodynamics.³¹ Even the continuity equation,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} \quad (7.38)$$

³¹Except for some special properties of matter, which we have looked in the previous chapters.

which is the mathematical expression of conservation of charge, can be derived from Maxwell's equations by applying the divergence to number (iv).

In the aforementioned set of equations, we have written Maxwell's equations in the traditional way, which emphasizes that they specify the divergence and curl of $\mathbf{E}$ and $\mathbf{B}$. In this form, they reinforce the notion that electric fields can be produced *either* by charges (ρ) or by changing magnetic fields ($\partial \mathbf{B} / \partial t$), and magnetic fields can be produced *either* by currents ($\mathbf{J}$) or by changing electric fields ($\partial \mathbf{E} / \partial t$).

Actually, this is misleading, as $\partial \mathbf{B} / \partial t$ and $\partial \mathbf{E} / \partial t$ are *themselves* due to charges and currents, therefore it may be better to write these equations as:

$$\begin{aligned} \text{(i)} \quad \nabla \cdot \mathbf{E} &= \frac{1}{\epsilon_0} \rho, & \text{(iii)} \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0, \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} &= 0, & \text{(iv)} \quad \nabla \times \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \mathbf{J} \end{aligned} \quad (7.39)$$

with the fields ($\mathbf{E}$ and $\mathbf{B}$) on the left and the sources (ρ and $\mathbf{J}$) on the right. This notation emphasizes that all electromagnetic fields are ultimately attributable to charges and currents. Maxwell's equations tell you how *charges* produce *fields*; reciprocally, the force law tells you how *fields* affect *charges*.

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Acronyms

E

EMF An energy transfer to an electric circuit per unit of electric charge, measured in volts. Devices called electrical transducers provide an EMF by converting other forms of energy into electrical energy. Other types of electrical equipment also produce an emf, such as batteries, which convert chemical energy, and generators, which convert mechanical energy. This energy conversion is achieved by physical forces applying physical work on electric charges. However, electromotive force itself is not a physical force

L

LHS Left Hand Side

O

ODE Ordinary Differential Equation

P

PDE Partial Differential Equation

PM An object made from a material that is magnetized and creates its own persistent magnetic field

